

2024, Mar. 2024 Release LOC:Acute Adult**General Medical**

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview**Select Day**

Initial review ^(1, 2)

Episode Day 1 ⁽¹⁾

Episode Day 2 ⁽²⁴⁸⁾

Episode Day 3-X ⁽²⁶⁹⁾

Discharge Screens ⁽²⁸⁸⁾

Utilization Benchmarks**Notes**

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Overview

Instruction: This subset is intended to be used when a patient's condition, symptom, or finding is not included in a condition-specific subset. Links to condition-specific subsets have been added where applicable. Surgery, transplant, and trauma are excluded and can be found as separate subsets. The exceptions to this are as follows:

- Use this subset for the organ transplant candidate who is listed for a transplant and admitted for end-stage organ disease.
- Use this subset for the bone marrow transplant (BMT), stem cell transplant (SCT), or organ transplant recipient who was discharged and is readmitted for the medical management of a complication related to the transplant.

This subset may be used on admission or when the reviewer is changing subsets based on a change in the patient's clinical condition. For example:

- If a patient is admitted to the Acute level of care for management of pulmonary arterial hypertension, the reviewer should refer to Episode Day 1 within this subset and conduct a review.
- If a patient originally admitted with epilepsy develops abdominal pain of unknown etiology, and that is the primary factor driving continued stay, the reviewer should refer to Episode Day 1 within this subset and conduct a review.
- When a patient has initially met criteria in the General Medical subset, but the condition changes and criteria for this condition is located in the General Medical subset, conduct an Episode Day 1 review for the new condition.

InterQual® content contains references to patient sex or gender. Depending on the context, these references may refer to either genotype or phenotype. At the individual patient level, a variety of factors, including but not limited to gender identity and gender affirmation via surgery or hormonal treatment, may affect the applicability of some InterQual criteria. This occurs most often with genetic testing and some procedures that require the presence of specific anatomy. For the purpose of these criteria, references to "Male" and "Female" are based on sex assigned at birth, unless otherwise defined. InterQual users should carefully consider patient genotype and anatomy when appropriate.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The

Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽³⁾● **Gastrointestinal or biliary, One:**● Abdominal pain, unknown etiology (excludes trauma) and, **Both:** ^(4, 5)● Finding, ≥ **One:**● Systemic finding, **Both:**– Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁾● Finding, ≥ **One:**– WBC ≥ 12,000/cu.mm(12x10⁹/L)

– Bands > 10%(0.10)

– Heart rate 100-120/min, sustained ⁽⁷⁾– History of abdominal surgery ⁽⁸⁾– Mental status changes (excludes coma, stupor, or obtundation) ⁽⁹⁾– Vomiting, persistent and unresponsive to ≥ 2 doses antiemetic ⁽¹⁰⁾– Imaging performed ^(11, 12)● **ACUTE, One:** ⁽¹³⁾● **Gastrointestinal or biliary, ≥ One:**● Ascites and, **One:** ⁽¹⁴⁾

– Initial episode

– Refractory and symptoms interfering with ADLs ⁽¹⁵⁾● Hepatic encephalopathy Grade II and, ≥ **One:** ^(16, 17)– Disorientation to time ⁽¹⁸⁾

– Lethargy

– Perceived change in behavior or personality

– Ileus, confirmed by CT or x-ray ⁽¹⁹⁾● **INTERMEDIATE, One:** ⁽²⁰⁾● **Gastrointestinal or biliary, One:**● Hepatic encephalopathy Grade III and, ≥ **One:** ^(16, 17)– Unusual or bizarre behavior ⁽²¹⁾

– Confused

– Gross disorientation ⁽²²⁾

– Responsive to noxious stimuli only

– Somnolence

● **CRITICAL, One:** ⁽²³⁾● **Gastrointestinal or biliary, One:**– Hepatic encephalopathy Grade IV and coma ^(16, 17)● **Episode Day 1, One:** ⁽¹⁾

(Symptom or finding within 24h)

● **OBSERVATION, ≥ One:** ⁽³⁾● **General, ≥ One:**● Danger to self or others, **Both:** ⁽²⁴⁾● Condition, ≥ **One:**● Neurocognitive disorder, intellectual disability, or autism and, **Both:** ⁽²⁵⁾– Agitation or escalating self-injurious behavior ⁽²⁶⁾– Support system unavailable or inadequate ^(27, 28)– Suicidal or homicidal ideation ⁽²⁹⁾● Intervention, ≥ **One:**– Antipsychotic ≥ 2 doses ⁽³⁰⁾– Psychiatric crisis intervention or stabilization and observation every 15 min ⁽³¹⁾

- Pain, severe, **Both**:
 - Unresponsive to ≥ 2 doses of analgesic prior to admit decision
- Intervention, **Both**:
 - Requiring additional analgesic ≥ 2 doses
 - Pain assessment at least every 4h ⁽³²⁾
- Spinal headache and, $\geq$ **One**:
 - IV caffeine and fluid
 - Epidural blood patch performed within last 24h
- **Cardiovascular, One**:
 - Diagnosis, actual or suspected, **One**:
 - Atrial arrhythmia (**use Arrhythmia: Atrial subset**)
 - Chest pain and negative troponin (**use Acute Coronary Syndrome subset**)
 - Deep vein thrombosis (**use Deep Vein Thrombosis subset**)
 - Heart failure (**use Heart Failure subset**)
 - Hypertension (**use Hypertension subset**)
 - Syncope (**use Syncope subset**)
 - Pulmonary arterial hypertension and, **Both**: ^(33, 34)
 - Epoprostenol, treprostinil, or iloprost (includes inhaled) dose adjustment ^(35, 36, 37)
 - Previous symptomatic vital sign instability during pulmonary arterial hypertension medication adjustment
- **Gastrointestinal or biliary, One**:
 - Appendicitis, suspected (**use Infection: GI/GYN subset**)
 - Cholecystitis (**use Cholecystitis subset**)
 - Dehydration or gastroenteritis (**use Dehydration or Gastroenteritis subset**)
 - Diverticulitis (**use Infection: GI/GYN subset**)
 - Pancreatitis (**use Pancreatits subset**)
 - Surgical intervention (**use General Surgical subset**)
- Abdominal pain, unknown etiology (excludes trauma) and, **Both**: ^(4, 5)
 - Finding, $\geq$ **One**:
 - Systemic finding, **Both**:
 - Temperature $\geq 100.4^{\circ}\text{F}(38.0^{\circ}\text{C})$ ⁽⁶⁾
 - Finding, $\geq$ **One**:
 - WBC $\geq 12,000/\text{cu.mm}(12 \times 10^9/\text{L})$
 - Bands $> 10\%(0.10)$
 - Heart rate 100-120/min, sustained ⁽⁷⁾
 - History of abdominal surgery ⁽⁸⁾
 - Mental status changes (excludes coma, stupor, or obtundation) ⁽⁹⁾
 - Vomiting, persistent and unresponsive to ≥ 2 doses antiemetic ⁽¹⁰⁾
 - Intervention, **All**:
 - Imaging performed ^(11, 12)
 - NPO and IV fluid ⁽³⁸⁾
 - Pain uncontrolled requiring change in medication, dose, or frequency ^(32, 39, 40)
- **Genitourinary, $\geq$ One**:
 - Acute kidney injury (AKI) (**use Acute Kidney Injury subset**)
- Renal calculus without evidence of obstruction by imaging and, **Both**:
 - Analgesic ≥ 2 doses
 - IV fluid ⁽³⁸⁾
- **Neurological, One**:
 - Diagnosis, actual or suspected, **One**:
 - Epilepsy (**use Epilepsy subset**)
 - Stroke (**use Stroke subset**)
 - TIA (**use TIA subset**)
 - Other neurological diagnosis, actual or suspected, $\geq$ **One**: ^(41, 42)

- Acute exacerbation of known neurological disorder and, ≥ **One:** ^(43, 44, 45)
 - CNS imaging scheduled or performed within 24h ⁽⁴⁶⁾
 - EEG scheduled or performed within 24h ⁽⁴⁷⁾
 - Functional impairment (new) requiring rehabilitation initiation within 24h ^(48, 49)
 - Medication initiation or dose adjustment (includes PO) ⁽³⁹⁾
- New onset of neurological disorder and, **Both:**
 - Finding, ≥ **One:**
 - Ataxia
 - Seizure (excludes epilepsy), resolved within the last 24h ^(50, 51, 52)
 - Vertigo ⁽⁵³⁾
 - Intervention, ≥ **One:**
 - CNS imaging scheduled or performed within 24h ⁽⁴⁶⁾
 - EEG scheduled or performed within 24h ⁽⁴⁷⁾
 - Functional impairment (new) requiring rehabilitation initiation within 24h ^(48, 49)
 - Medication initiation (includes PO)
- Mental status change (excludes coma, stupor, or obtundation) and, ≥ **One:** ⁽⁹⁾
 - CNS imaging scheduled or performed within 24h ⁽⁴⁶⁾
 - Culture results pending
 - Medication initiation or dose adjustment (includes PO) ⁽³⁹⁾
- Migraine and, **Both:**
 - Finding, ≥ **One:**
 - Failed outpatient treatment (includes PO) ⁽⁵⁴⁾
 - Focal neurological finding ⁽⁵⁵⁾
 - Intractable or incapacitating ⁽⁵⁶⁾
 - Intervention, ≥ **One:**
 - Analgesic or anti-migraine agent ≥ 2x/24h ⁽⁵⁷⁾
 - Dihydroergotamine (DHE) and antiemetic ⁽⁵⁸⁾
- **Respiratory, One:**
 - Diagnosis, actual or suspected, **One:**
 - Asthma (**use Asthma subset**)
 - COPD (**use COPD subset**)
 - Other respiratory diagnosis, actual or suspected, ≥ **One:** ^(41, 59)
 - Anaphylaxis and, **Both:** ⁽⁶⁰⁾
 - Airway patent and hemodynamic stability following epinephrine administration and, ≥ **One:** ^(61, 62)
 - ≥ 2 doses epinephrine required for symptom control
 - History of biphasic reaction ⁽⁶³⁾
 - Continuous oximetry and, ≥ **One:** ⁽⁶⁴⁾
 - Antihistamine (includes PO) ⁽⁶⁵⁾
 - Corticosteroid (includes PO) ⁽⁶⁶⁾
 - Angioedema, actual or suspected and, **All:** ⁽⁶⁷⁾
 - Airway patent and hemodynamic stability following medication administration and, ≥ **One:** ^(61, 62, 68)
 - Antihistamine (includes PO)
 - Blood product transfusion ⁽⁶⁹⁾
 - Corticosteroid (includes PO) ⁽⁷⁰⁾
 - Immunomodulator
 - Finding, ≥ **One:** ⁽⁷¹⁾
 - Persistent symptoms following treatment
 - Severe presenting symptoms ⁽⁷²⁾
 - Continuous oximetry ⁽⁷³⁾
 - Dyspnea and, **Both:** ^(74, 75)
 - Oxygenation, ≥ **One:**
 - Arterial Po₂ 56-60 mmHg(7.4-8.0 kPa) and < baseline ⁽⁷⁵⁾

- O₂ sat 90-91%(0.90-0.91) and < baseline ⁽⁷⁵⁾
- **Intervention, Both:**
 - Oximetry
 - Requiring supplemental oxygen ⁽⁷⁶⁾
- **Pneumomediastinum and, Both:**
 - Assessment and monitoring at least every 4h
 - Requiring supplemental oxygen ⁽⁷⁶⁾
- **Pneumothorax confirmed by imaging and, ≥ One:** ^(77, 78, 79)
 - **Intervention, Both:**
 - Small bore chest tube placed ^(80, 81)
 - Oxygen requirement back to baseline post chest tube placement
 - Requiring oximetry and repeat chest x-ray (CXR), scheduled or performed within 24h
- **Toxic exposure or ingestion, ≥ One:** ⁽⁸²⁾
 - Acetaminophen overdose (**use Acetaminophen Overdose subset**)
 - Carbon monoxide poisoning (**use Carbon Monoxide Poisoning subset**)
- **Alcohol or drug intoxication and, All:**
 - **Finding, ≥ One:**
 - Agitation ⁽⁸³⁾
 - Confusion ⁽⁸⁴⁾
 - Disorientation ⁽⁸⁵⁾
 - Airway assessment
 - Aspiration precautions
 - Neurological assessment at least 6x/24h ⁽⁸⁶⁾
- **Ingestion of potentially toxic substance and, All:** ⁽⁸⁷⁾
 - Clinically stable
 - Unknown type or quantity
 - Assessment and monitoring at least every 4h ⁽⁸⁸⁾
- **ACUTE, ≥ One:** ⁽¹³⁾
 - **Cardiovascular or peripheral vascular, One:**
 - **Diagnosis, actual or suspected, One:**
 - Deep vein thrombosis (**use Deep Vein Thrombosis subset**)
 - Heart failure (**use Heart Failure subset**)
 - Pulmonary embolism (**use Pulmonary Embolism subset**)
 - Syncope (**use Syncope subset**)
 - **Other cardiovascular diagnosis, actual or suspected, One:** ⁽⁴¹⁾
 - **Acute limb ischemia, actual or suspected and, Both:**
 - **Finding, ≥ One:**
 - Confirmed by imaging
 - Coldness, mottling, pallor, or numbness
 - Limb swelling
 - Pain at rest
 - **Intervention, Both:**
 - **Anticoagulant, One:** ⁽⁸⁹⁾
 - Unfractionated heparin, continuous ≤ 5d ⁽⁹⁰⁾
 - HIT, suspected or by history and anticoagulation ⁽⁹¹⁾
 - Neurovascular assessment at least 6x/24h ⁽⁹²⁾
 - **Eating disorder and, All:**
 - Requiring medical stabilization ⁽⁹³⁾
 - **Finding, ≥ One:** ⁽⁹⁴⁾
 - Blood glucose < 60 mg/dL(3.3 mmol/L) ^(95, 96)
 - **Blood pressure, Both:**
 - Systolic < 90 mmHg ⁽⁹⁷⁾
 - Diastolic < 60 mmHg ⁽⁹⁷⁾

- **Cardiomyopathy and, One:** ^(98, 99)
 - Cardiomegaly by imaging
 - Tricuspid or mitral valve dysfunction by physical examination
 - EF < 40%(0.40)
- Co-occurring medical condition and unstable lab values requiring daily monitoring ⁽¹⁰⁰⁾
- **Dehydration and, ≥ One:**
 - BUN 25-45 mg/dL(8.9-16.1 mmol/L)
 - Creatinine 1.5-3.0 mg/dL(132.6-265.2 μmol/L) ⁽¹⁰¹⁾
 - Heart rate sustained > 100/min ^(7, 102)
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽⁹⁾
 - Urine specific gravity > 1.030
- **Electrolyte imbalance, ≥ One:** ^(103, 104)
 - Magnesium 1.0-1.4 mg/dL(0.41-0.58 mmol/L)
 - Phosphate 1.0-1.4 mg/dL(0.32-0.45 mmol/L)
 - Potassium < 3.0 mEq/L(3.0 mmol/L) without ECG changes
- **Sodium, One:**
 - 151-160 mEq/L(151-160 mmol/L)
 - 120-129 mEq/L(120-129 mmol/L)
- Lack of improvement with outpatient treatment ⁽¹⁰⁵⁾
- Heart rate < 50/min, sustained ^(7, 106)
- **Orthostatic hypotension, Both:**
 - Sustained drop in blood pressure within 3 min of sitting or standing
- **Blood pressure, ≥ One:**
 - Systolic drop ≥ 20 mmHg ⁽¹⁰⁷⁾
 - Diastolic drop ≥ 10 mmHg ⁽¹⁰⁸⁾
- Orthostatic changes in heart rate > 30/min ⁽¹⁰⁹⁾
- QTc interval > 450 ms ⁽¹¹⁰⁾
- Temperature < 96.8°F(36.0°C) ⁽¹¹¹⁾
- **Weight, ≥ One:**
 - BMI < 15 kg/m² ⁽¹¹²⁾
 - Weight loss > 10%(0.10) in last 6 months or > 20%(0.20) in last year ⁽¹¹³⁾
- **Intervention, Both:**
 - **Monitoring, All:**
 - Calorie count
 - Daily weight
 - Electrolytes monitored at least once daily ⁽¹¹⁴⁾
 - Observation and documentation of oral or tube feeding pattern ⁽¹¹⁵⁾
 - Vital signs at least 3x/24h
 - **Treatment, ≥ One:**
 - Blood glucose monitoring at least 4x/24h ⁽¹¹⁶⁾
 - Complete bed rest ⁽¹¹⁷⁾
 - Fluid, supplements, or medication to correct electrolyte abnormalities ⁽¹⁰⁴⁾
 - IV fluid ⁽³⁸⁾
 - Tube feeding or parenteral nutrition
 - **Weight-gain-specific treatment plan and, All:**
 - Accompanied or monitored bathroom visits ⁽¹¹⁸⁾
 - Nutritional assessment daily ⁽¹¹⁹⁾
 - Re-evaluation of plan effectiveness daily ⁽¹²⁰⁾
 - Supervised meals and snacks ⁽¹²¹⁾
- **Gastrointestinal or biliary, One:**
 - **Diagnosis, actual or suspected, One:**
 - Pancreatitis (use **Pancreatitis subset**)
 - Ascending cholangitis (use **Infection: GI/GYN subset**)

- Bowel obstruction (**use Bowel Obstruction subset**)
- Spontaneous bacterial peritonitis (**use Infection: GI/GYN subset**)
- Ulcerative colitis or Crohn's disease (**use Inflammatory Bowel Disease subset**)
- Other gastrointestinal diagnosis, actual or suspected, ≥ **One:** ⁽⁴¹⁾
 - Ascites ≤ 7d and, **Both:** ⁽¹⁴⁾
 - Finding, **One:**
 - Initial episode
 - Refractory and symptoms interfering with ADLs ⁽¹⁵⁾
 - Intervention, ≥ **One:**
 - Albumin
 - Diuretic ≥ 2x/24h
 - Sodium and fluid restriction ≤ 3d
 - Therapeutic paracentesis
 - Bilirubin > 3.0 mg/dL(51.3 μmol/L) (excludes uncomplicated viral hepatitis) and, **All:**
 - Finding, ≥ **One:**
 - Abdominal pain
 - Albumin level < 2.0 g/dL(20 g/L)
 - Blood urea nitrogen (BUN) > 45 mg/dL(16.1 mmol/L)
 - Creatinine > 3.0 mg/dL(265.2 μmol/L)
 - Hepatomegaly ⁽¹²²⁾
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽⁹⁾
 - PT ≥ 1.5x upper limit normal (ULN) or INR ≥ 2.0, without anticoagulation
 - Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁾
 - Vomiting, persistent and unresponsive to ≥ 2 doses antiemetic ^(10, 123)
 - Imaging scheduled or performed within last 24h ⁽¹²²⁾
 - Intervention, ≥ **One:**
 - Albumin
 - Antiemetic ≥ 3x/24h ⁽¹²³⁾
 - Serotonin antagonist ≥ 2x/24h ⁽¹²⁴⁾
 - Anti-infective
 - Endoscopic retrograde cholangiopancreatography (ERCP)
 - Magnetic resonance cholangiopancreatography (MRCP)
 - Fresh frozen plasma (FFP)
 - IV fluid ⁽³⁸⁾
 - Liver biopsy ≤ 24h
 - Gastroparesis, actual or suspected ≤ 4d since initiation of treatment and, **Both:** ^(125, 126)
 - Inadequate oral intake and IV fluid ^(38, 127, 128)
 - Intervention, ≥ **One:**
 - Medication discontinuation ⁽¹²⁹⁾
 - Medication ⁽¹³⁰⁾
 - Gastric scintigraphy, scheduled or performed ^(131, 132, 133)
 - Hepatic encephalopathy Grade II ≤ 7d and, **Both:** ^(16, 17, 134)
 - Finding, ≥ **One:**
 - Disorientation to time ⁽¹⁸⁾
 - Perceived change in behavior or personality
 - Lethargy
 - Medication, ≥ **One:**
 - Branched-chain amino acids (BCAA) (includes PO) ⁽¹³⁵⁾
 - Lactulose (includes PO or PR) ⁽¹³⁶⁾
 - L-ornithine L-aspartate (LOLA) (includes PO) ⁽¹³⁷⁾
 - Neomycin (includes PO) ⁽¹³⁸⁾
 - Rifaximin (includes PO) ⁽¹³⁹⁾
 - Ileus confirmed by imaging and IV fluid ≤ 5d ^(19, 38, 140)

- **Genitourinary, ≥ One:**
 - Acute kidney injury (AKI) (**use Acute Kidney Injury subset**)
- **Genitourinary obstruction, Both:**
 - **Condition, ≥ One:**
 - Obstruction of renal pelvis
 - Obstruction of ureter
 - Obstruction of urethra
 - **Intervention, ≥ One:**
 - Nephrostomy
 - **Creatinine > 1.8 mg/dL(159.1 μmol/L) and > baseline requiring urinary catheter and, One:**
 - Suprapubic
 - Transurethral
- **Hydronephrosis and, Both:** ⁽¹⁴¹⁾
 - **Finding, ≥ One:**
 - Hematuria
 - Pain
 - Renal failure
 - Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁾
 - **Intervention, ≥ One:**
 - Analgesic ≥ 3x/24h or continuous
 - Surgery planned within 24h
- **Nephrotic syndrome actual or suspected and, Both:**
 - **Finding, ≥ One:**
 - Generalized edema, severe
 - Proteinuria
 - Urine output < 0.5 mL/kg/h
 - **Intervention, ≥ One:**
 - ACE inhibitor (includes PO)
 - ARB (includes PO)
 - Albumin 25%(0.25)
 - Corticosteroid
 - Cytotoxic agent
 - Diuretic ≥ 2x/24h
- **Priapism and, Both:** ^(142, 143)
 - Alpha-adrenergic sympathomimetic
 - **Intervention, ≥ One:**
 - Intracavernous injection
 - Irrigation
- **Hematology or oncology, One:**
 - **Diagnosis, actual or suspected, One:**
 - Acute chest syndrome or vaso-occlusive crisis (**use Sickle Cell subset**)
 - **For anemia or bleeding, use the appropriate anemia or bleeding subset, One:**
 - Anemia (**use Anemia subset**)
 - Gastrointestinal (GI) Bleeding (**use Gastrointestinal (GI) Bleeding subset**)
 - Non-Traumatic Bleeding (**use Non-Traumatic Bleeding subset**)
 - **Other hematology or oncology diagnosis, actual or suspected, ≥ One:** ⁽⁴¹⁾
 - **Hyperviscosity syndrome and, Both:** ⁽¹⁴⁴⁾
 - **Finding, ≥ One:**
 - Mucosal bleeding
 - Neurological symptoms
 - Visual changes
 - **Intervention, All:**
 - Apheresis ⁽¹⁴⁵⁾

- Hydration or nutrition, ≥ **One**:
 - IV fluid ⁽³⁸⁾
 - Advancing diet as tolerated
 - Neurological assessment at least 3x/24h ⁽⁸⁶⁾
- Thrombocytopenia (excludes known Von Willebrand disease) and, **Both**: ⁽¹⁴⁶⁾
 - Finding, ≥ **One**:
 - Platelet count < 20,000/cu.mm(20x10⁹/L)
 - Purpura or petechiae, progressive
 - Intervention, ≥ **One**:
 - Blood product transfusion
 - Corticosteroid
 - Immunoglobulin ⁽¹⁴⁷⁾
- **Neurological, One**:
 - Diagnosis, actual or suspected, **One**:
 - Epilepsy (**use Epilepsy subset**)
 - Stroke (**use Stroke subset**)
 - Other neurological diagnosis, actual or suspected, **One**: ^(41, 42)
 - Acute exacerbation of known neurological disorder and, **Both**:
 - Finding, ≥ **One**:
 - Dysphagia ⁽¹⁴⁸⁾
 - Paresis or paralysis ⁽¹⁴⁹⁾
 - Seizure (excludes epilepsy) ≥ 2 within the last 24h ^(50, 51, 150)
 - Intervention, ≥ **One**:
 - Anticonvulsant (includes PO)
 - CNS imaging scheduled or performed within 24h ⁽⁴⁶⁾
 - Continuous EEG ≤ 72h since initiation ⁽¹⁵¹⁾
 - Corticosteroid ≤ 5d
 - Feeding tube, ≥ **One**:
 - Insertion performed within last 24h
 - Feedings initiated
 - Immunotherapy
 - Medication initiation or dose adjustment (includes PO) within 24h ⁽³⁹⁾
 - Plasmapheresis
 - New onset of neurological disorder and, **Both**:
 - Finding, ≥ **One**:
 - Abnormal reflexes ⁽¹⁵²⁾
 - Blindness, diplopia, or visual field loss ⁽¹⁵³⁾
 - Dysphagia ⁽¹⁴⁸⁾
 - Paresis or paralysis ⁽¹⁴⁹⁾
 - Seizure (excludes epilepsy) ≥ 2 within the last 24h ^(50, 51, 52, 150)
 - Intervention, ≥ **One**:
 - CNS imaging scheduled or performed within 24h ⁽⁴⁶⁾
 - Continuous EEG ≤ 72h since initiation ⁽¹⁵¹⁾
 - Feeding tube, ≥ **One**:
 - Insertion performed within last 24h
 - Feedings initiated
 - Pregnancy (excludes eclampsia and epilepsy) and, ≥ **One**:
 - ≥ 1 seizure within the last 24h
 - Postictal state within the last 24h ⁽¹⁵⁴⁾
 - Seizure evaluation (excludes epilepsy) ≤ 5d since initiation, scheduled admission ^(155, 156)
 - Unconscious and DNR, DNI, or CMO and discharge planning ⁽¹⁵⁷⁾
- **Respiratory, One**:
 - Diagnosis, actual or suspected, **One**:

- Asthma (**use Asthma subset**)
- COPD (**use Chronic Obstructive Pulmonary Disease subset**)
- Cystic fibrosis exacerbation (**use Cystic Fibrosis subset**)
- Other respiratory diagnosis, actual or suspected, **≥ One:** ⁽⁵⁹⁾
 - Chest tube placement and, **≥ One:** ^(77, 78, 79, 158)
 - Large bore chest tube placed ^(80, 81)
 - Suction, continuous
 - O₂ requirement > baseline post chest tube placement
 - Dyspnea and, **Both:** ⁽⁷⁵⁾
 - Oxygenation, **≥ One:**
 - Arterial Po₂ < 56 mmHg(7.4 kPa) and < baseline ⁽⁷⁵⁾
 - O₂ sat ≤ 89%(0.89) and < baseline ⁽⁷⁵⁾
 - Requiring supplemental oxygen ⁽⁷⁶⁾
 - Post thoracentesis and O₂ sat ≤ 89%(0.89)
 - Pulmonary infiltrate by imaging and, **Both:**
 - Non-infectious actual or suspected
 - O₂ sat ≤ 89%(0.89) and < baseline requiring supplemental oxygen and, **≥ One:** ^(75, 76)
 - Bronchoscopy scheduled or performed within 24h
 - Immunosuppressant ⁽¹⁵⁹⁾
 - Lung biopsy scheduled or performed within 24h
 - Upper airway mass and corticosteroid
- Skin, **≥ One:**
 - Dermatitis, **Both:**
 - Finding, **≥ One:**
 - Erythroderma
 - Exfoliative dermatitis
 - Inflammatory dermatitis and, **≥ One:** ⁽¹⁶⁰⁾
 - Failed outpatient treatment ≥ 5d (includes PO) ^(54, 161)
 - Progression despite ≥ 24h outpatient treatment (includes PO)
 - Skin involvement, **≥ One:**
 - > 10%(0.10) BSA
 - > 50%(0.50) of limb or torso
 - Immunosuppressant (includes PO) ≤ 3d ⁽¹⁵⁹⁾
 - Erythema multiforme major [Stevens-Johnson Syndrome or TEN] and, **≥ One:** ⁽¹⁶²⁾
 - Complex wound care ≥ 3x/24h ≤ 5d and, **≥ One:** ⁽¹⁶³⁾
 - Requiring parenteral analgesia
 - ≥ 15 min in duration
 - Immunoglobulin ≤ 3d ⁽¹⁴⁷⁾
 - Immunosuppressant (includes PO) ≤ 3d ⁽¹⁵⁹⁾
 - IV fluid ⁽³⁸⁾
 - Volume expander ≤ 2d ⁽¹⁶⁴⁾
 - Loss or damage of skin > 10%(0.10) BSA and, **≥ One:**
 - Analgesic ≥ 3x/24h or continuous
 - Complex wound care ≥ 3x/24h ≤ 5d and, **≥ One:** ⁽¹⁶³⁾
 - Requiring parenteral analgesia
 - ≥ 15 min in duration
 - Immunosuppressant (includes PO) ≤ 3d ⁽¹⁵⁹⁾
 - IV fluid ⁽³⁸⁾
 - Psoriasis ≥ 30%(0.30) BSA and, **Both:**
 - Failed outpatient treatment (includes PO) ≥ 5d ⁽¹⁶⁵⁾
 - Goeckerman regimen ≤ 7d ⁽¹⁶⁶⁾
 - Wound dehiscence, **≥ One:**
 - Complex wound care ≥ 3x/24h ≤ 5d and, **≥ One:** ⁽¹⁶³⁾

- Requiring parenteral analgesia
- ≥ 15 min in duration
- Return to OR within last 24h ⁽¹⁶⁷⁾
- **Transplant, ≥ One:**
 - Graft dysfunction and, **Both:** ^(168, 169)
 - Finding, ≥ **One:**
 - GVHD (Grade I or II) ⁽¹⁷⁰⁾
 - Heart and, ≥ **One:**
 - Echocardiogram finding, ≥ **One:**
 - Decreased EF and < baseline ⁽⁷⁵⁾
 - Decreased motility and < baseline ⁽⁷⁵⁾
 - Increased cardiac size and > baseline ⁽⁷⁵⁾
 - Sustained arrhythmia ⁽⁷⁾
 - Inflammatory cell infiltrate on graft biopsy
 - Kidney and, ≥ **One:**
 - Elevated or increasing blood urea nitrogen (BUN) or creatinine and > baseline ⁽⁷⁵⁾
 - Urine output < 1 mL/kg/h
 - Urinoma
 - Worsening proteinuria
 - Liver and, ≥ **One:**
 - Alkaline phosphatase > 4x upper limit normal (ULN)
 - Aspartate aminotransferase (AST) > 4x upper limit normal (ULN)
 - Alanine aminotransferase (ALT) > 4x upper limit normal (ULN)
 - Bilirubin > 2x upper limit normal (ULN)
 - Biloma
 - Hematoma
 - Hepatic abscess
 - Hepatic artery thrombosis
 - T-tube leak or obstruction
 - Lung and, ≥ **One:**
 - Pulmonary infiltrate and abnormal vital signs
 - Worsening pulmonary function, ≥ **One:**
 - Decreased exercise tolerance
 - Decreased spirometry ⁽¹⁷¹⁾
 - Increased dyspnea
 - Increased oxygen requirement
 - Primary graft non-function ⁽¹⁷²⁾
 - Small bowel and, ≥ **One:**
 - Increased ileal output or diarrhea
 - Temperature > 99.0°F(37.2°C)
- Intervention, ≥ **One:**
 - Surgical drainage (**use General Surgical subset**)
 - Percutaneous drainage
 - Immunosuppressant and, ≥ **One:** ^(159, 173)
 - New medication initiation ⁽¹⁷⁴⁾
 - Medication adjustment (includes PO) ≤ 3d ^(39, 174, 175)
- Post transplant complication and, ≥ **One:**
 - Kidney and, **Both:**
 - Finding, ≥ **One:**
 - Hematoma by imaging
 - Lymphocele by imaging and, ≥ **One:**
 - Compression of bladder
 - Compression of iliac vein and ipsilateral DVT

- Obstruction of ureter
- Seroma by imaging
- Drainage, **One**:
 - Percutaneous drainage
 - Surgical drainage (use **General Surgical subset**)
- Liver and, **Both**:
 - Finding, $\geq$ **One**:
 - Hepatic vein thrombosis by imaging
 - Portal vein thrombosis by imaging
 - Intervention, $\geq$ **One**:
 - Anticoagulant
 - Thrombectomy
- Pancreas and duodenal segment leak with percutaneous drainage
- Pancreas and thrombosis and, **One**:
 - Surgical intervention required (use **General Surgical subset**)
 - Anticoagulant
- INTERMEDIATE, $\geq$ **One**:⁽²⁰⁾
 - Cardiovascular or peripheral vascular, **One**:
 - Diagnosis, actual or suspected, **One**:
 - Atrial arrhythmia (use **Arrhythmia: Atrial subset**)
 - Heart block (use **the Arrhythmia: Blocks subset**)
 - Ventricular arrhythmia (use **the Arrhythmia: Ventricular or abnormal ECG finding subset**)
 - Chest pain and positive troponin (use **Acute Coronary Syndrome subset**)
 - Heart failure (use **Heart Failure subset**)
 - Hypertension (use **Hypertension subset**)
 - Pulmonary embolism (use **Pulmonary Embolism subset**)
 - Syncope (use **Syncope subset**)
 - Other cardiovascular diagnosis, actual or suspected, $\geq$ **One**:
 - Pacemaker or ICD replacement and, $\geq$ **One**:
 - Lead fractured or displaced
 - Malfunction
 - Pericardial effusion and high risk for cardiac tamponade $\leq$ 24h and, **Both**:^(176, 177, 178)
 - Asymptomatic
 - Effusion size $\geq$ 1 cm⁽¹⁷⁹⁾
 - Acute limb ischemia, actual or suspected $\leq$ 2d and, **Both**:
 - Finding, $\geq$ **One**:
 - Confirmed by imaging
 - Coldness, mottling, pallor, or numbness
 - Limb swelling
 - Pain at rest
 - Intervention, **Both**:
 - Anticoagulant and high risk for bleeding and, **One**:⁽¹⁸⁰⁾
 - Unfractionated heparin, continuous^(89, 90)
 - HIT, suspected or by history and anticoagulation⁽⁹¹⁾
 - Neurovascular assessment every 3-4h⁽⁹²⁾
 - Post PCI complication and continuous cardiac monitoring (excludes Holter) $\leq$ 24h^(181, 182)
 - VAD requiring readmission and, $\geq$ **One**:^(183, 184)
 - Anticoagulation, $\geq$ **One**:^(185, 186, 187)
 - Direct thrombin inhibitor
 - GP IIb or IIIa inhibitor
 - Heparin
 - Thrombolytic
 - Bleeding actual or suspected and, $\geq$ **One**:

- Hct or Hb monitoring at least 3x/24h, ≤ 3d
- Volume expander or blood product ⁽¹⁶⁴⁾
- Inadequate nutrition and, ≥ **One**:
 - Calorie count ≤ 3d
 - Requiring daily tube feeding adjustment
 - TPN initiation ≤ 24h
- Infection actual or suspected and, **Both**:
 - Finding, ≥ **One**:
 - Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁾
 - WBC ≥ 12,000/cu.mm($12 \times 10^9/L$)
 - WBC ≤ 4,000/cu.mm($4 \times 10^9/L$)
 - Bands > 10%(0.10)
 - ANC < 500/cu.mm($500 \times 10^6/L$) ⁽¹⁸⁸⁾
 - Anti-infective
- Eating disorder and, ≥ **One**:
 - Hypokalemia, **All**: ^(189, 190)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - PVCs > 6/min
 - Potassium repletion
 - Seizure and, **Both**: ⁽¹⁹¹⁾
 - Activity resolved
 - Neurological assessment every 3-4h ⁽⁸⁶⁾
- Gastrointestinal or biliary, ≥ **One**:
 - Pancreatitis (use Pancreatitis subset)
 - Hepatic encephalopathy Grade III ≤ 7d and, **Both**: ^(16, 17)
 - Finding, ≥ **One**:
 - Unusual or bizarre behavior ⁽²¹⁾
 - Confused
 - Gross disorientation ⁽²²⁾
 - Responsive to noxious stimuli only
 - Somnolence
 - Medication, ≥ **One**: ⁽¹⁹²⁾
 - Branched-chain amino acids (BCAA) (includes PO) ⁽¹³⁵⁾
 - Lactulose (includes PO or PR) ⁽¹³⁶⁾
 - L-ornithine L-aspartate (LOLA) ⁽¹³⁷⁾
 - Neomycin (includes PO) ⁽¹³⁸⁾
 - Rifaximin (includes PO) ⁽¹³⁹⁾
 - Hepatorenal syndrome ≤ 8d, **All**: ⁽¹⁹³⁾
 - Cirrhosis and ascites
 - Creatinine, ≥ **One**:
 - Creatinine increased by ≥ 50%(0.50) within last 7d
 - Creatinine increased by ≥ 0.3 mg/dL(26.5 μmol/L) within last 2d
 - Intervention, ≥ **One**:
 - Albumin
 - Octreotide and midodrine (includes PO)
 - Neurological, ≥ **One**:
 - Guillain-Barre syndrome, actual or suspected, and neurological assessment every 3-4h and, **One**: ^(86, 194)
 - Immunotherapy
 - Plasmapheresis
 - Seizure (excludes epilepsy) and neurological assessment every 3-4h and, **Both**: ⁽⁸⁶⁾
 - Activity resolved
 - Anticonvulsant, conversion from IV to PO ≤ 24h
 - Ventriculostomy awaiting V-P shunt and neurological assessment every 3-4h ⁽⁸⁶⁾

- **Respiratory, One:**
 - **Diagnosis, actual or suspected, One:**
 - COPD (**use Chronic Obstructive Pulmonary Disease subset**)
 - Cystic fibrosis exacerbation (**use Cystic Fibrosis subset**)
 - **Intervention, ≥ One:**
 - NIPPV ⁽¹⁹⁵⁾
 - Oxygen ≥ 40%(0.40) ⁽⁷⁶⁾
- **Skin, One:**
 - **Complex wound care, ≤ 5d and, ≥ One:** ⁽¹⁶³⁾
 - ≥ 3x/24h
 - ≥ 30 min in duration
- **Toxic exposure or ingestion, One:** ^(82, 87)
 - **Diagnosis, actual or suspected, One:**
 - Acetaminophen overdose (**use Acetaminophen Overdose subset**)
 - Carbon monoxide poisoning (**use Carbon Monoxide Poisoning subset**)
 - **Other toxic exposure or ingestion actual or suspected and, All:**
 - Potential for significant arrhythmia ^(196, 197)
 - Electrocardiogram (ECG) normal, unchanged, or nondiagnostic
 - Continuous cardiac monitoring (excludes Holter)
- **Transplant, ≥ One:**
 - **GVHD and, One:** ⁽¹⁷⁰⁾
 - Grade I or II and monoclonal antibody initiation ≤ 3 doses ⁽¹⁹⁸⁾
 - Grade III and immunosuppressant ^(159, 173)
 - **Post lung transplant and suspected rejection and, Both:** ⁽¹⁶⁸⁾
 - **Finding, ≥ One:**
 - Pulmonary infiltrate and abnormal vital signs
 - **Worsening pulmonary function, ≥ One:**
 - Decreased exercise tolerance
 - Decreased spirometry ⁽¹⁷¹⁾
 - Increased dyspnea
 - Increased oxygen requirement
 - **Intervention, Both:**
 - **Oxygen, ≥ One:** ⁽⁷⁶⁾
 - HFNC and respiratory monitoring every 3-4h ^(199, 200)
 - Oxygen ≥ 40%(0.40) ≤ 2d
 - **Immunosuppressant and, ≥ One:** ^(173, 201)
 - Medication adjustment (includes PO) ^(39, 174, 175)
 - New medication initiation ⁽¹⁷⁴⁾
- **CRITICAL, ≥ One:** ⁽²³⁾
 - **General, ≥ One:**
 - Exchange transfusion ⁽²⁰²⁾
 - **Hemodynamic instability, Both:**
 - **Finding, ≥ One:**
 - Heart rate > 120/min, sustained ⁽⁷⁾
 - **Systolic blood pressure, ≥ One:**
 - < 90 mmHg without chronic hypertension
 - < 110 mmHg with chronic hypertension
 - > 30 mmHg decrease from baseline ⁽⁷⁵⁾
 - Labile ⁽²⁰³⁾
 - MAP < 65 mmHg
 - **Intervention, ≥ One:**
 - Antifibrinolytic ⁽²⁰⁴⁾
 - Blood product transfusion

- IV fluid resuscitation $\leq 2d$ ⁽²⁰⁵⁾
- Hemodynamic monitoring, invasive ⁽²⁰⁶⁾
- **Hyperthermia, Both:**
 - Temperature, $\geq$ **One:**
 - $> 106.0^{\circ}F(41.1^{\circ}C)$ PR
 - $> 105.0^{\circ}F(40.6^{\circ}C)$ PR and age ≥ 65 yrs
 - Active cooling ⁽²⁰⁷⁾
- Thrombolytic ⁽²⁰⁸⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Cardiovascular or peripheral vascular, One:**
 - Diagnosis, actual or suspected, **One:**
 - Atrial arrhythmia (**use Arrhythmia: Atrial subset**)
 - Chest pain and positive troponin (**use Acute Coronary Syndrome subset**)
 - Heart block (**use Arrhythmia: Blocks subset**)
 - Heart failure (**use Heart Failure subset**)
 - Hypertension (**use Hypertension subset**)
 - Pulmonary embolism (**use Pulmonary Embolism subset**)
 - Syncope (**use Syncope subset**)
 - Ventricular arrhythmia (**use Arrhythmia: Ventricular or abnormal ECG finding subset**)
 - Other cardiovascular or peripheral vascular diagnosis actual or suspected, $\geq$ **One:** ⁽⁴¹⁾
 - Pacemaker, temporary
 - Pericardial effusion and, $\geq$ **One:** ^(209, 210)
 - Symptomatic $\leq 24h$, $\geq$ **One:** ⁽²¹¹⁾
 - Dyspnea
 - Jugular venous distention
 - Orthopnea
 - Insertion of pericardial drain performed $\leq 24h$ ⁽²¹²⁾
 - Pericardiocentesis performed $\leq 24h$ ⁽²¹²⁾
 - Acute limb ischemia, actual or suspected and, $\geq$ **One:**
 - Neurovascular assessment at least every 1-2h, $\leq 2d$ ⁽⁹²⁾
 - Catheter directed thrombolysis
 - Pulmonary arterial hypertension and, **Both:** ^(33, 34)
 - Epoprostenol, treprostinil, or iloprost (includes inhaled) initiation ⁽²¹³⁾
 - Hemodynamic monitoring, invasive ⁽²¹⁴⁾
 - VAD requiring readmission and, $\geq$ **One:** ^(183, 215)
 - Blood pressure, $\geq$ **One:**
 - $<$ baseline and inadequate perfusion ⁽²¹⁶⁾
 - MAP < 70 mmHg and continuous flow device ⁽²¹⁷⁾
 - Systolic < 90 mmHg and pulsatile flow device
 - Decreased level of consciousness
 - Decreased VAD flows and inadequate perfusion ⁽²¹⁶⁾
 - O_2 sat $< 89\%(0.89)$
 - VAD, percutaneous ^(184, 218)
 - **Eating disorder and, Both:**
 - Finding, $\geq$ **One:**
 - Magnesium < 1.0 mg/dL(0.41 mmol/L)
 - Phosphate < 1.0 mg/dL(0.32 mmol/L)
 - Potassium < 3.0 mEq/L(3.0 mmol/L) and, $\geq$ **One:** ^(189, 190, 219)
 - Multifocal PVCs
 - R on T phenomenon ⁽²²⁰⁾
 - T and U wave fusion ⁽²²¹⁾
 - U wave ⁽²²²⁾
 - Sodium > 160 mEq/L(160 mmol/L)

- Sodium < 120 mEq/L(120 mmol/L) and symptomatic ^(223, 224, 225)
- Coma, stupor, obtundation or GCS ≤ 8 ⁽²²⁶⁾
- Status epilepticus (excludes epilepsy) ⁽²²⁷⁾
- **Intervention, ≥ One:**
 - CRRT ⁽²²⁸⁾
 - Dialysis, hemodialysis or peritoneal
 - Electrolyte repletion
 - IV fluid ^(38, 229)
 - Neurological assessment every 1-2h, ≤ 2d ⁽⁸⁶⁾
 - Sodium chloride 3%(0.03) ⁽²³⁰⁾
- **Gastrointestinal or biliary, One:**
 - **Diagnosis, actual or suspected, One:**
 - Pancreatitis (**use Pancreatitis subset**)
 - Biliary sepsis (cholangitis) (**use Infection: GI/GYN subset**)
 - Ulcerative colitis or Crohn's disease (**use Inflammatory Bowel Disease subset**)
 - **Other gastrointestinal or biliary diagnosis, actual or suspected, ≥ One:** ⁽⁴¹⁾
 - **Hepatic encephalopathy Grade IV and coma ≤ 7d and, ≥ One:** ^(16, 17, 192)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽¹³⁵⁾
 - Lactulose (includes PO or PR) ⁽¹³⁶⁾
 - L-ornithine L-aspartate (LOLA) ⁽¹³⁷⁾
 - Neomycin (includes PO) ⁽¹³⁸⁾
 - Rifaximin (includes PO) ⁽¹³⁹⁾
 - **Hepatorenal syndrome ≤ 8d, All:** ⁽¹⁹³⁾
 - Cirrhosis and ascites
 - **Creatinine, ≥ One:**
 - Creatinine increased by ≥ 50%(0.50) within last 7d
 - Creatinine increased by ≥ 0.3 mg/dL(26.5 μmol/L) within last 2d
 - **Intervention, ≥ One:**
 - CRRT ⁽²²⁸⁾
 - Hemodialysis or peritoneal dialysis, ≤ 2d since initiation
 - Norepinephrine
- **Hematology or oncology, One:**
 - **Diagnosis, actual or suspected, One:**
 - Acute chest syndrome or vaso-occlusive crisis (**use Sickle Cell subset**)
 - **For bleeding, use the appropriate bleeding subset, One:**
 - Gastrointestinal (GI) Bleeding (**use Gastrointestinal (GI) Bleeding subset**)
 - Non-Traumatic Bleeding (**use Non-Traumatic Bleeding subset**)
 - **DIC, All:** ⁽²³¹⁾
 - Fibrinogen degradation product (FDP) increasing
 - Fibrinogen decreasing
 - Blood product transfusion
- **Neurological, One:**
 - **Diagnosis, actual or suspected, One:**
 - Epilepsy (**use Epilepsy subset**)
 - Stroke (**use Stroke subset**)
 - **Other neurological diagnosis and, Both:**
 - **Finding, ≥ One:**
 - Cerebral edema on imaging
 - Coma, stupor, obtundation or GCS ≤ 8 ⁽²²⁶⁾
 - Status epilepticus (excludes epilepsy) ⁽²²⁷⁾
 - **Intervention, ≥ One:**
 - **Anticonvulsant, ≥ One:**
 - > 2 medications for seizure termination within 24h

- Anticonvulsant, continuous
- Cerebral edema and, ≥ **One**:
 - ICP monitoring (includes OR placement)
 - Corticosteroid
 - Hyperosmolar therapy, ≥ **One**: ⁽²³²⁾
 - Mannitol
 - Hypertonic saline ≥ 3%(0.03) solution
 - External ventricular or lumbar drainage
 - Neurological assessment every 1-2h ≤ 2d ⁽⁸⁶⁾
- **Respiratory, One**:
 - Diagnosis, actual or suspected, **One**:
 - Asthma (**use Asthma subset**)
 - COPD (**use Chronic Obstructive Pulmonary Disease subset**)
 - Cystic fibrosis exacerbation (**use Cystic Fibrosis subset**)
 - Other respiratory diagnosis, actual or suspected, **Both**: ^(41, 59)
 - Finding, ≥ **One**:
 - Airway obstruction with impending intubation
 - Arterial or venous Pco₂ < 30 mmHg(4.0 kPa) ⁽²³³⁾
 - Arterial or venous Pco₂ > 50 mmHg(6.7 kPa) and pH < 7.25 ⁽²³³⁾
 - Arterial Po₂ < 56 mmHg(7.4 kPa)
 - Arterial or venous pH < 7.25 with mental status change ^(9, 233)
 - Arterial or venous pH > 7.50 and hemodynamic instability ⁽²³³⁾
 - O₂ sat, **One**:
 - < 89%(0.89) and < baseline ⁽⁷⁵⁾
 - Room air assessment not clinically appropriate ⁽²³⁴⁾
 - Intervention, ≥ **One**:
 - Mechanical ventilation
 - NIPPV and, ≥ **One**: ⁽¹⁹⁵⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
 - Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One**:
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
 - Nebulizer or metered dose inhaler (MDI) treatment, **One**: ⁽²³⁵⁾
 - Every 1-2h
 - Continuous
- Neuromuscular disorder, actual or suspected and, ≥ **One**: ^(236, 237)
 - Mechanical ventilation
 - NIPPV ⁽¹⁹⁵⁾
 - Respiratory interventions every 1-2h ⁽²³⁸⁾
- **Toxic exposure or ingestion, One**: ^(82, 87)
 - Diagnosis, actual or suspected, **One**:
 - Acetaminophen overdose (**use Acetaminophen Overdose subset**)
 - Carbon monoxide poisoning (**use Carbon Monoxide Poisoning subset**)
 - Other toxic exposure or ingestion diagnosis, actual or suspected, ≥ **One**: ⁽⁴¹⁾
 - General, **Both**:
 - Finding, ≥ **One**:
 - Corrosive substance ⁽²³⁹⁾
 - End organ damage, actual or potential
 - Intervention, ≥ **One**:
 - Dialysis, **One**:
 - CRRT ⁽²²⁸⁾
 - Hemodialysis
 - Peritoneal

- Artificial airway placement (excludes nasal trumpet) ⁽²⁴⁰⁾
 - Toxin-specific antidote ⁽²⁴¹⁾
- Potential for significant arrhythmia and, **Both:** ^(196, 197)
 - Abnormal electrocardiogram (ECG), ≥ **One:**
 - QRS duration ≥ 0.12 sec ^(242, 243)
 - Prolonged QT interval ^(244, 245)
 - PR interval > 0.20 sec ⁽²⁴⁶⁾
 - Continuous cardiac monitoring (excludes Holter)
- **Transplant, ≥ One:**
 - Acute humoral rejection and apheresis ^(145, 247)
- GVHD Grade IV and, ≥ **One:** ⁽¹⁷⁰⁾
 - Apheresis ⁽¹⁴⁵⁾
 - Immunosuppressant ^(159, 173)
- Post lung transplant and suspected rejection and, **Both:** ⁽¹⁶⁸⁾
 - Immunosuppressant and, ≥ **One:** ^(159, 173)
 - New medication initiation
 - Medication adjustment (includes PO) ^(39, 175)
 - Intervention, ≥ **One:**
 - Mechanical ventilation
 - NIPPV and, ≥ **One:** ⁽¹⁹⁵⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
 - Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- **Episode Day 2, One:** ⁽²⁴⁸⁾
 - **OBSERVATION, ≥ One:**
 - **General, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, ≥ **One:** ⁽²⁴⁹⁾
 - No danger to self or others in a less restrictive environment or transfer to ALOC ⁽²⁵⁰⁾
 - Pain controlled or manageable ⁽²⁵¹⁾
 - Spinal headache controlled
 - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
 - For a condition other than current condition use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **Cardiovascular, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽²⁴⁹⁾
 - Pulmonary arterial hypertension specific medication adjustment complete ⁽³⁶⁾
 - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
 - For a condition other than pulmonary arterial hypertension use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **Gastrointestinal or biliary, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **Both:** ⁽²⁴⁹⁾
 - Tolerating PO or nutritional route established ^(252, 253)
 - Clinical stability, ≥ **One:**
 - Abdominal pain controlled or manageable ⁽²⁵¹⁾
 - Laboratory values normal or within acceptable limits ⁽²⁵⁴⁾
 - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
 - For current condition use Episode Day 2 at a higher level of care

- For a condition other than current condition use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **Genitourinary, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽²⁴⁹⁾
 - Renal calculus and pain controlled or manageable ⁽²⁵¹⁾
 - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
 - For a condition other than current condition use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **Neurological, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **≥ One:** ⁽²⁴⁹⁾
 - Migraine controlled or manageable ⁽²⁵¹⁾
 - Neurological deficit resolved or resolving, **≥ One:**
 - Functional impairment and, **One:** ⁽⁴⁸⁾
 - Discharge to home and, **One:** ⁽²⁵⁵⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - Tolerating PO or nutritional route established ^(252, 253)
 - Neurological stability ⁽²⁵⁶⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **≥ One:**
 - Mental status change (excludes coma, stupor, or obtundation) unresolved and, **≥ One:** ⁽⁹⁾
 - CNS imaging scheduled or performed within 24h ⁽⁴⁶⁾
 - Culture results pending
 - Medication initiation or dose adjustment (includes PO) ⁽³⁹⁾
 - Migraine unresolved and, **≥ One:**
 - Analgesic or anti-migraine agent **≥ 2x/24h** ⁽⁵⁷⁾
 - Dihydroergotamine (DHE) and antiemetic ⁽⁵⁸⁾
 - Neurological deficit or disorder, persistent or unresolved and, **≥ One:** ^(257, 258)
 - CNS imaging scheduled or performed within 24h ⁽⁴⁶⁾
 - EEG scheduled or performed within 24h ⁽⁴⁷⁾
 - Functional impairment (new) requiring rehabilitation initiation within 24h ^(48, 49, 259)
 - Medication initiation or dose adjustment (includes PO) ⁽³⁹⁾
- **Respiratory, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **≥ One:** ⁽²⁴⁹⁾
 - Anaphylaxis resolved and no evidence of airway compromise
 - Angioedema resolved and no evidence of airway compromise
 - Dyspnea resolved, **≥ One:**
 - Arterial Po₂ **≥ 60 mmHg(7.9 kPa)** or within acceptable limits ⁽²⁵⁴⁾
 - O₂ sat **≥ 92%(0.92)** or within acceptable limits ⁽²⁵⁴⁾
 - Pneumothorax or pneumomediastinum, **Both:** ⁽⁷⁹⁾
 - Resolving on chest x-ray (CXR)
 - O₂ sat within acceptable limits or returned to baseline
 - Chest tube and, **One:**
 - Heimlich valve
 - Removed and no reaccumulation of air or fluid by chest x-ray (CXR)
 - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
 - For current condition use Episode Day 2 at a higher level of care
 - For a condition other than current condition use appropriate subset (Episode Day 1) based on clinical findings

- Refer for secondary review
- **Toxic exposure or ingestion, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽²⁴⁹⁾
 - Airway patent
 - Neurological stability ⁽²⁵⁶⁾
 - Vitals signs stable or within acceptable limits ⁽²⁵⁴⁾
 - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
 - For a condition other than current condition use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
- **ACUTE, ≥ One:**
 - **Cardiovascular or peripheral vascular, ≥ One:**
 - Pericardial effusion and, ≥ **One:**
 - Pericardial drain removed ≤ 24h
 - Post Critical care ≤ 24h
 - Acute limb ischemia and, **One:**
 - Embolectomy or thrombectomy required (**use General Surgical subset**)
 - **Treatment, Both:**
 - **Anticoagulant, One:** ⁽⁸⁹⁾
 - Unfractionated heparin, continuous ≤ 5d ⁽⁹⁰⁾
 - HIT, suspected or by history and anticoagulation ⁽⁹¹⁾
 - Neurovascular assessment at least 4x/24h ⁽⁹²⁾
 - **Eating disorder, ≥ One:**
 - **Eating disorder and, Both:**
 - **Finding, ≥ One:** ⁽⁹⁴⁾
 - Blood glucose < 60 mg/dL(3.3 mmol/L) ^(95, 96)
 - **Blood pressure, Both:**
 - Systolic < 90 mmHg ⁽⁹⁷⁾
 - Diastolic < 60 mmHg ⁽⁹⁷⁾
 - **Cardiomyopathy and, One:** ^(98, 99)
 - Cardiomegaly by imaging
 - Tricuspid or mitral valve dysfunction by physical examination
 - EF < 40%(0.40)
 - Co-occurring medical condition and unstable lab values requiring daily monitoring ⁽¹⁰⁰⁾
 - **Dehydration and, ≥ One:** ⁽²⁶⁰⁾
 - BUN 25-45 mg/dL(8.9-16.1 mmol/L)
 - Creatinine 1.5-3.0 mg/dL(132.6-265.2 μmol/L) ⁽¹⁰¹⁾
 - Heart rate sustained > 100/min ^(7, 102)
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽⁹⁾
 - Urine specific gravity > 1.030
 - **Electrolyte imbalance, ≥ One:** ^(103, 104)
 - Magnesium 1.0-1.4 mg/dL(0.41-0.58 mmol/L)
 - Phosphate 1.0-1.4 mg/dL(0.32-0.45 mmol/L)
 - Potassium < 3.0 mEq/L(3.0 mmol/L) without ECG changes
 - **Sodium, One:**
 - 151-160 mEq/L(151-160 mmol/L)
 - 120-129 mEq/L(120-129 mmol/L)
 - Heart rate < 50/min, sustained ^(7, 106)
 - **Orthostatic hypotension, Both:**
 - Sustained drop in blood pressure within 3 min of sitting or standing
 - **Blood pressure, ≥ One:**
 - Systolic drop ≥ 20 mmHg ⁽¹⁰⁷⁾

- Diastolic drop ≥ 10 mmHg ⁽¹⁰⁸⁾
- Orthostatic changes in heart rate $> 30/\text{min}$ ⁽¹⁰⁹⁾
- QTc interval > 450 ms ⁽¹¹⁰⁾
- Temperature $< 96.8^{\circ}\text{F}(36.0^{\circ}\text{C})$ ⁽¹¹¹⁾
- **Weight, $\geq$ One:**
 - Continued weight loss
 - Failure to stabilize weight
- **Intervention, Both:**
 - **Monitoring, All:**
 - Calorie count
 - Daily weight
 - Electrolytes monitored at least once daily ⁽¹¹⁴⁾
 - Observation and documentation of oral or tube feeding pattern ⁽¹¹⁵⁾
 - Vital signs at least 3x/24h
 - **Treatment, $\geq$ One:**
 - Blood glucose monitoring at least 4x/24h ⁽¹¹⁶⁾
 - Complete bed rest ⁽¹¹⁷⁾
 - Fluid, supplements, or medication to correct electrolyte abnormalities ⁽¹⁰⁴⁾
 - IV fluid ⁽³⁸⁾
 - Tube feeding or parenteral nutrition
 - **Weight-gain-specific treatment plan and, All:**
 - Accompanied or monitored bathroom visits ⁽¹¹⁸⁾
 - Nutritional assessment daily ⁽¹¹⁹⁾
 - Re-evaluation of plan effectiveness daily ⁽¹²⁰⁾
 - Supervised meals and snacks ⁽¹²¹⁾
- Post Critical care $\leq 24\text{h}$
- **Gastrointestinal or biliary, One:**
 - **Responder, discharge expected today if clinically stable last 12h, One:** ⁽²⁴⁹⁾
 - **Clinical stability, $\geq$ One:**
 - Laboratory values normal or within acceptable limits ⁽²⁵⁴⁾
 - Gastroparesis, tolerating PO or nutritional route established ^(252, 253)
 - **Partial responder, not clinically stable for discharge and requires continued stay, $\geq$ One:**
 - **Ascites $\leq 7\text{d}$ and, $\geq$ One:**
 - Albumin
 - Diuretic $\geq 2\text{x}/24\text{h}$
 - Sodium and fluid restriction $\leq 3\text{d}$
 - Therapeutic paracentesis performed within last 24h
 - Transjugular intrahepatic portosystemic shunt (TIPS) performed $\leq 24\text{h}$
 - **Bilirubin > 3.0 mg/dL(51.3 $\mu\text{mol/L}$) and, $\geq$ One:**
 - **Albumin level < 2.0 g/dL(20 g/L) and, $\geq$ One:**
 - Albumin
 - Feeding tube insertion performed within last 24h and feedings initiated
 - TPN or PPN
 - **Inadequate oral intake $\leq 4\text{d}$ or active vomiting and, $\geq$ One:** ⁽²⁶¹⁾
 - Antiemetic $\geq 3\text{x}/24\text{h}$ ⁽¹²³⁾
 - Serotonin antagonist $> 2\text{x}/24\text{h}$ ⁽¹²⁴⁾
 - IV fluid ⁽³⁸⁾
 - Liver biopsy and results pending $\leq 24\text{h}$
 - Endoscopic retrograde cholangiopancreatography (ERCP) performed within last 24h
 - Magnetic resonance cholangiopancreatography (MRCP) performed within last 24h
 - PT $\geq 1.5\text{x}$ upper limit normal (ULN) or INR ≥ 2.0 and fresh frozen plasma (FFP)
 - **Gastroparesis, actual or suspected $\leq 4\text{d}$ since initiation of treatment and, Both:** ^(125, 126)
 - Inadequate oral intake and IV fluid ^(38, 127)

- **Intervention, ≥ One:**
 - Medication discontinuation ⁽¹²⁹⁾
 - Medication ⁽¹³⁰⁾
 - Gastric scintigraphy, scheduled or performed ^(131, 132, 133)
- **Hepatic encephalopathy Grade II ≤ 7d and, ≥ One:** ^(16, 17, 134)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽¹³⁵⁾
 - Lactulose (includes PO or PR) ⁽¹³⁶⁾
 - L-ornithine L-aspartate (LOLA) (includes PO) ⁽¹³⁷⁾
 - Neomycin (includes PO) ⁽¹³⁸⁾
 - Rifaximin (includes PO) ⁽¹³⁹⁾
- **Ileus and, One:** ⁽¹⁹⁾
 - Resolved and initiating PO or tube feeding ≤ 2d
 - Unresolved and IV fluid ≤ 5d ⁽³⁸⁾
 - Post Critical care ≤ 24h
- **Genitourinary, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽²⁴⁹⁾
 - Priapism resolved
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - **Genitourinary obstruction, Both:**
 - **Condition, ≥ One:**
 - Obstruction of renal pelvis
 - Obstruction of ureter
 - Obstruction of urethra
 - **Intervention, ≥ One:**
 - Surgical intervention required (**use General Surgical subset**)
 - Nephrostomy performed within last 24h
 - Ureteral stent placed within last 24h
 - **Urinary catheter, One:**
 - Suprapubic placed within last 24h
 - Transurethral placed within last 24h
 - **Nephrotic syndrome and, ≥ One:**
 - Albumin 25%(0.25)
 - ACE inhibitor (includes PO)
 - ARB (includes PO)
 - Corticosteroid
 - Cytotoxic agent
 - **Dialysis (initial), One:** ⁽²⁶²⁾
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal ≤ 7d since initiation
 - Diuretic ≥ 2x/24h
 - **Priapism and, One:** ⁽¹⁴³⁾
 - Surgical intervention required (**use General Surgical subset**)
 - **Severe pain and alpha-adrenergic sympathomimetic and, Both:**
 - ≤ 2d since priapism onset
 - Intracavernous injection
 - Post Critical care ≤ 24h
 - **Hematology or oncology, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽²⁴⁹⁾
 - **Clinical stability, ≥ One:**
 - **Hyperviscosity syndrome resolving and, Both:**
 - Neurological stability ⁽²⁵⁶⁾
 - No evidence of bleeding
 - **Platelet count ≥ 50,000/cu.mm(50x10⁹/L) or within acceptable limits and, ≥ One:** ⁽²⁵⁴⁾

- No evidence of bleeding
- Purpura or petechiae improving or resolved
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Hyperviscosity syndrome and, ≥ **One**:⁽²⁶³⁾
 - Apheresis⁽¹⁴⁵⁾
 - IV fluid ≤ 2d⁽³⁸⁾
 - Neurological assessment at least 3x/24h, ≤ 2d⁽⁸⁶⁾
 - Thrombocytopenia and, ≥ **One**:
 - Blood product transfusion
 - Corticosteroid
 - Immunoglobulin⁽¹⁴⁷⁾
 - Post Critical care ≤ 24h
- **Neurological, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **One**:⁽²⁴⁹⁾
 - Clinical stability, **One**:
 - Unconscious and, **Both**:
 - DNR, DNI, or CMO and discharge planning⁽¹⁵⁷⁾
 - Post acute arrangements completed
 - Neurological stability⁽²⁶⁴⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - CNS evaluation performed and results pending ≤ 24h and, ≥ **One**:
 - EMG
 - Lumbar puncture (LP)
 - MRI
 - Continuous EEG ≤ 72h since initiation⁽¹⁵¹⁾
 - Functional impairment (new) requiring rehabilitation initiation within 24h^(48, 49, 259)
 - Neurological deficit or disorder, persistent or unresolved and, ≥ **One**:^(257, 258)
 - Corticosteroid ≤ 5d since initiation
 - Feeding tube, **Both**:
 - Insertion performed within last 24h
 - Feedings initiated
 - Immunotherapy
 - Medication adjustment or initiation (includes PO) within 24h⁽³⁹⁾
 - Plasmapheresis
 - Seizure evaluation (excludes epilepsy) ≤ 5d since initiation, scheduled admission^(155, 156)
 - Post Critical care ≤ 24h
 - **Respiratory, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **Both**:⁽²⁴⁹⁾
 - O₂ sat ≥ 92%(0.92) or within acceptable limits⁽²⁵⁴⁾
 - Clinical stability, ≥ **One**:
 - Dyspnea resolved
 - Chest tube and, **One**:
 - Heimlich valve
 - Removed and no reaccumulation of air or fluid by chest x-ray (CXR)
 - Pulmonary infiltrate resolving
 - Upper airway mass and airway patent
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - O₂ sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen^(75, 76)
 - Chest tube and, ≥ **One**:
 - Suction, continuous
 - Drainage ≥ 250 mL/24h
 - Repositioning within last 24h⁽²⁶⁵⁾
 - Water seal within last 24h

- Pulmonary infiltrate, non-infectious and, **Both:**
 - Oxygenation, **One:**
 - O₂ sat ≥ 92%(0.92) or baseline ⁽⁷⁵⁾
 - O₂ sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen ^(75, 76)
 - Intervention, ≥ **One:**
 - Bronchoscopy results pending ≤ 24h
 - Immunosuppressant (includes PO) ⁽¹⁵⁹⁾
 - Lung biopsy results pending ≤ 24h
- Upper airway mass and, ≥ **One:** ⁽¹⁴²⁾
 - Biopsy results pending ≤ 2d
 - Corticosteroid
 - Post Critical care ≤ 24h
- Skin, ≥ **One:**
 - Dermatitis and immunosuppressant (includes PO) ≤ 3d ⁽¹⁵⁹⁾
- Erythema multiforme major [Stevens-Johnson Syndrome or TEN] and, ≥ **One:** ⁽¹⁶²⁾
 - Complex wound care ≤ 5d and, ≥ **One:** ⁽¹⁶³⁾
 - Wound care ≥ 3x/24h and, ≥ **One:**
 - Requiring parenteral analgesic
 - ≥ 15 min in duration
 - Wound management teaching ≤ 24h ⁽²⁶⁶⁾
 - Immunoglobulin ≤ 3d ⁽¹⁴⁷⁾
 - Immunosuppressant (includes PO) ≤ 3d ⁽¹⁵⁹⁾
 - IV fluid ⁽³⁸⁾
 - Volume expander ≤ 2d ⁽¹⁶⁴⁾
- Loss or damage of skin > 10%(0.10) BSA and, ≥ **One:**
 - Analgesic ≥ 3x/24h or continuous
- Complex wound care ≤ 5d and, ≥ **One:** ⁽¹⁶³⁾
 - Wound care ≥ 3x/24h and, ≥ **One:**
 - Requiring parenteral analgesic
 - ≥ 15 min in duration
 - Wound management teaching ≤ 24h ⁽²⁶⁶⁾
 - Immunosuppressant (includes PO) ≤ 3d ⁽¹⁵⁹⁾
 - IV fluid ⁽³⁸⁾
- Psoriasis and Goeckerman regimen ≤ 5d ⁽¹⁶⁶⁾
- Wound dehiscence, ≥ **One:**
 - Complex wound care ≤ 5d and, ≥ **One:** ⁽¹⁶³⁾
 - Wound care ≥ 3x/24h and, ≥ **One:**
 - Requiring parenteral analgesic
 - ≥ 15 min in duration
 - Wound management teaching ≤ 24h ⁽²⁶⁶⁾
 - Return to OR last 24h
- Transplant, ≥ **One:**
 - Graft dysfunction and immunosuppressant and, ≥ **One:** ^(159, 168, 173)
 - Conversion from IV to PO ≤ 2d ⁽²⁰¹⁾
 - Medication adjustment (includes PO) ≤ 3d ^(39, 174, 175)
 - New medication initiation
 - Kidney and, **Both:**
 - Hematoma, seroma, or lymphocele
 - Persistent percutaneous drainage
 - Liver and, **Both:**
 - Finding, ≥ **One:**
 - Hepatic vein thrombosis
 - Portal vein thrombosis

- Intervention, ≥ **One**:
 - Anticoagulant
 - Thrombectomy performed within last 24h
- Pancreas and duodenal segment leak with percutaneous drainage
- Pancreas and thrombosis and, **One**:
 - Surgical intervention required (**use General Surgical subset**)
 - Anticoagulant
- **INTERMEDIATE, ≥ One**:
 - **Cardiovascular or peripheral vascular, ≥ One**:
 - Epoprostenol, treprostinil, or iloprost (includes continuous infusion or inhaled) initiation ≤ 2d ⁽²⁶⁷⁾
 - Pacemaker or ICD placement performed within last 24h
 - Pericardial drain and post Critical care ≤ 24h
 - Pericardial effusion and high risk for cardiac tamponade ≤ 24h and, **Both**: ^(176, 177, 178)
 - Asymptomatic
 - Effusion size ≥ 1 cm ⁽¹⁷⁹⁾
 - Acute limb ischemia ≤ 2d and, **Both**:
 - Anticoagulant and high risk for bleeding and, **One**: ⁽¹⁸⁰⁾
 - Unfractionated heparin, continuous ^(89, 90)
 - HIT, suspected or by history and anticoagulation ⁽⁹¹⁾
 - Neurovascular assessment every 3-4h ⁽⁹²⁾
 - VAD requiring readmission and, ≥ **One**: ^(183, 184)
 - Anticoagulation, ≥ **One**: ^(185, 186, 187)
 - Direct thrombin inhibitor
 - GP IIb or IIIa inhibitor
 - Heparin
 - Thrombolytic
 - Bleeding actual or suspected and, ≥ **One**:
 - Hct or Hb monitoring at least 3x/24h, ≤ 3d
 - Volume expander or blood product ⁽¹⁶⁴⁾
 - Inadequate nutrition and, ≥ **One**:
 - Calorie count ≤ 3d
 - Requiring daily tube feeding adjustment
 - TPN initiation ≤ 24h
 - Infection actual or suspected and, **Both**:
 - Finding, ≥ **One**:
 - Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁾
 - WBC ≥ 12,000/cu.mm($12 \times 10^9/L$)
 - WBC ≤ 4,000/cu.mm($4 \times 10^9/L$)
 - Bands > 10%(0.10)
 - ANC < 500/cu.mm($500 \times 10^6/L$) ⁽¹⁸⁸⁾
 - Anti-infective
 - Post acute facility unavailable and, ≥ **One**:
 - Functional impairment requiring therapy ⁽⁴⁸⁾
 - Inadequate social or family support
 - Patient or caregiver unable to demonstrate functional ability to maintain and operate VAD
 - Post Critical care ≤ 3d
 - **Eating disorder and, ≥ One**:
 - Hypokalemia, **All**: ^(189, 190)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - PVCs > 6/min
 - Potassium repletion
 - Seizure and, **Both**: ⁽¹⁹¹⁾
 - Activity resolved

- Neurological assessment every 3-4h ⁽⁸⁶⁾
- **Gastrointestinal or biliary, ≥ One:**
 - Hepatic encephalopathy Grade III ≤ 7d and, ≥ **One:** ^(16, 17, 192)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽¹³⁵⁾
 - Lactulose (includes PO or PR) ⁽¹³⁶⁾
 - L-ornithine L-aspartate (LOLA) ⁽¹³⁷⁾
 - Neomycin (includes PO) ⁽¹³⁸⁾
 - Rifaximin (includes PO) ⁽¹³⁹⁾
 - Hepatorenal syndrome ≤ 8d and, ≥ **One:** ⁽¹⁹³⁾
 - Albumin
 - Octreotide and midodrine (includes PO)
- **Neurological, ≥ One:**
 - External ventricular or lumbar drainage and neurological assessment every 3-4h ⁽⁸⁶⁾
 - Guillain-Barre syndrome, actual or suspected, and neurological assessment every 3-4h and, **One:** ^(86, 194)
 - Immunotherapy
 - Plasmapheresis
 - Neurological deficit or disorder, persistent or unresolved, and neurological assessment every 3-4h ≤ 2d ^(86, 257, 258)
 - Seizure (excludes epilepsy) and neurological assessment every 3-4h and, **Both:** ⁽⁸⁶⁾
 - Activity resolved
 - Anticonvulsant, conversion from IV to PO ≤ 24h
 - Ventriculostomy awaiting V-P shunt and neurological assessment every 3-4h ⁽⁸⁶⁾
- **Respiratory, ≥ One:**
 - NIPPV ⁽¹⁹⁵⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽⁷⁶⁾
- **Skin, One:**
 - Complex wound care, ≤ 5d and, **Both:** ⁽¹⁶³⁾
 - ≥ 3x/24h
 - ≥ 30 min in duration
- **Toxic exposure or ingestion and, All:** ^(82, 87)
 - Potential for significant arrhythmia ^(196, 197)
 - Electrocardiogram (ECG) normal, unchanged, or nondiagnostic
 - Continuous cardiac monitoring (excludes Holter)
- **Transplant, ≥ One:**
 - Acute humoral rejection and post Critical care ≤ 24h ⁽²⁴⁷⁾
 - GVHD and, **One:** ⁽¹⁷⁰⁾
 - Grade I or II and monoclonal antibody initiation ≤ 3 doses ⁽¹⁹⁸⁾
 - Grade III and immunosuppressant ^(159, 173)
 - Post lung transplant and suspected rejection and, **Both:** ⁽¹⁶⁸⁾
 - Oxygen, ≥ **One:** ⁽⁷⁶⁾
 - HFNC and respiratory monitoring every 3-4h ^(199, 200)
 - Oxygen ≥ 40%(0.40) ≤ 2d
 - Immunosuppressant and, ≥ **One:** ⁽¹⁷³⁾
 - Conversion from IV to PO ≤ 2d ⁽²⁰¹⁾
 - Medication adjustment (includes PO) ^(39, 175)
 - New medication initiation
- **CRITICAL, ≥ One:**
 - **General, ≥ One:**
 - Exchange transfusion ⁽²⁰²⁾
 - Heart failure and, **Both:**
 - Finding, ≥ **One:**
 - Arterial Po₂ < 56 mmHg(7.4 kPa)
 - O₂ sat < 89%(0.89) and < baseline ⁽⁷⁵⁾

- Diuretic every 1-2h or continuous
- Hemodynamic instability, **Both:**
 - Finding, $\geq$ **One:**
 - Heart rate $> 120/\text{min}$, sustained ⁽⁷⁾
 - Systolic blood pressure, $\geq$ **One:**
 - < 90 mmHg without chronic hypertension
 - < 110 mmHg with chronic hypertension
 - > 30 mmHg decrease from baseline ⁽⁷⁵⁾
 - Labile ⁽²⁰³⁾
 - MAP < 65 mmHg
 - Intervention, $\geq$ **One:**
 - Antifibrinolytic ⁽²⁰⁴⁾
 - Blood product transfusion
 - IV fluid resuscitation $\leq 2\text{d}$ ⁽²⁰⁵⁾
- Hemodynamic monitoring, invasive ⁽²⁰⁶⁾
- Hyperthermia and active cooling ⁽²⁰⁷⁾
- Thrombolytic ⁽²⁰⁸⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- Cardiovascular or peripheral vascular, $\geq$ **One:**
 - Pericardial effusion and, $\geq$ **One:** ^(209, 210)
 - Pericardial window (**use General Surgical subset**)
 - Symptomatic $\leq 24\text{h}$, $\geq$ **One:** ⁽²¹¹⁾
 - Dyspnea
 - Jugular venous distention
 - Orthopnea
 - Insertion of pericardial drain performed $\leq 24\text{h}$ ⁽²¹²⁾
 - Pericardiocentesis performed $\leq 24\text{h}$ ⁽²¹²⁾
- Acute limb ischemia and, $\geq$ **One:**
 - Neurovascular assessment at least every 1-2h, $\leq 2\text{d}$ ⁽⁹²⁾
 - Catheter directed thrombolysis performed within last 24h
- Pulmonary arterial hypertension and, **Both:** ⁽³⁴⁾
 - Epoprostenol, treprostinil, or iloprost (includes continuous infusion or inhaled) initiation ⁽²¹³⁾
 - Hemodynamic monitoring, invasive ⁽²¹⁴⁾
- Temporary pacemaker insertion performed within last 2d
- VAD requiring readmission and, $\geq$ **One:** ^(183, 215)
 - Blood pressure, $\geq$ **One:**
 - $<$ baseline and inadequate perfusion ⁽²¹⁶⁾
 - MAP < 70 mmHg and continuous flow device ⁽²¹⁷⁾
 - Systolic < 90 mmHg and pulsatile flow device
 - Decreased level of consciousness
 - Decreased VAD flows and inadequate perfusion ⁽²¹⁶⁾
 - O_2 sat $< 89\%$ ^(0.89)
- VAD, percutaneous ^(184, 218)
- Eating disorder and, **Both:**
 - Finding, $\geq$ **One:**
 - Magnesium < 1.0 mg/dL (0.41 mmol/L)
 - Phosphate < 1.0 mg/dL (0.32 mmol/L)
 - Potassium < 3.0 mEq/L (3.0 mmol/L) and, $\geq$ **One:** ^(190, 219)
 - Multifocal PVCs
 - R on T phenomenon ⁽²²⁰⁾
 - T and U wave fusion ⁽²²¹⁾
 - U wave ⁽²²²⁾
 - Sodium > 160 mEq/L (160 mmol/L)

- Sodium < 120 mEq/L(120 mmol/L) and symptomatic ^(223, 224, 225)
- Coma, stupor, obtundation or GCS ≤ 8 ⁽²²⁶⁾
- **Intervention, ≥ One:**
 - CRRT ⁽²²⁸⁾
 - Dialysis, hemodialysis or peritoneal
 - Electrolyte repletion
 - IV fluid ^(38, 229)
 - Neurological assessment every 1-2h, ≤ 2d ⁽⁸⁶⁾
 - Sodium chloride 3%(0.03) ⁽²³⁰⁾
- **Gastrointestinal or biliary, ≥ One:**
 - **Hepatic encephalopathy Grade IV ≤ 7d and, ≥ One:** ^(16, 17, 192)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽¹³⁵⁾
 - Lactulose (includes PO or PR) ⁽¹³⁶⁾
 - L-ornithine L-aspartate (LOLA) ⁽¹³⁷⁾
 - Neomycin (includes PO) ⁽¹³⁸⁾
 - Rifaximin (includes PO) ⁽¹³⁹⁾
 - **Hepatorenal syndrome ≤ 8d, ≥ One:** ⁽¹⁹³⁾
 - CRRT ⁽²²⁸⁾
 - Hemodialysis or peritoneal dialysis ≤ 2d since initiation
 - Norepinephrine
- **Hematology or oncology, One:**
 - DIC and blood product transfusion ⁽²³¹⁾
- **Neurological, ≥ One:**
 - **Anticonvulsant, ≥ One:**
 - > 2 medications for seizure termination within 24h
 - Anticonvulsant continuous
 - **Cerebral edema and, ≥ One:**
 - ICP monitoring (includes OR placement)
 - Corticosteroid
 - **Hyperosmolar therapy, ≥ One:** ⁽²³²⁾
 - Mannitol
 - Hypertonic saline ≥ 3%(0.03) solution
 - External ventricular or lumbar drainage and neurological assessment every 1-2h ⁽⁸⁶⁾
 - Neurological deficit or disorder, persistent or unresolved, and neurological assessment every 1-2h ≤ 2d ^(86, 257, 258)
- **Respiratory, ≥ One:**
 - Mechanical ventilation
 - **NIPPV and, ≥ One:** ^(195, 238)
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽²⁶⁸⁾
 - **Nebulizer or metered dose inhaler (MDI) treatment, One:** ⁽²³⁵⁾
 - Every 1-2h
 - Continuous
- **Toxic exposure or ingestion, ≥ One:** ^(82, 87)
 - **General, Both:**
 - **Finding, ≥ One:**
 - Corrosive substance ⁽²³⁹⁾
 - End organ damage, actual or potential
 - **Intervention, ≥ One:**
 - **Dialysis, One:**
 - CRRT ⁽²²⁸⁾
 - Hemodialysis
 - Peritoneal

- Artificial airway placement (excludes nasal trumpet) ⁽²⁴⁰⁾
 - Toxin-specific antidote ⁽²⁴¹⁾
- Potential for significant arrhythmia and, **Both:** ^(196, 197)
 - Abnormal electrocardiogram (ECG), ≥ **One:**
 - QRS duration ≥ 0.12 sec ^(242, 243)
 - Prolonged QT interval ^(244, 245)
 - PR interval > 0.20 sec ⁽²⁴⁶⁾
 - Continuous cardiac monitoring (excludes Holter)
- **Transplant, ≥ One:**
 - Acute humoral rejection and apheresis ^(145, 247)
- GVHD, Grade IV and, ≥ **One:** ⁽¹⁷⁰⁾
 - Apheresis ⁽¹⁴⁵⁾
 - Immunosuppressant ^(159, 173)
- Post lung transplant and suspected rejection and, **Both:** ⁽¹⁶⁸⁾
 - Immunosuppressant and, ≥ **One:** ^(159, 173)
 - New medication initiation
 - Medication adjustment (includes PO) ^(39, 175)
 - Intervention, ≥ **One:**
 - Mechanical ventilation
 - NIPPV and, **One:** ⁽¹⁹⁵⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽²⁶⁸⁾
- **Episode Day 3-X, One:** ⁽²⁶⁹⁾
 - **OBSERVATION, One:**
 - **Neurological, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, ≥ **One:** ⁽²⁴⁹⁾
 - Migraine controlled or manageable ⁽²⁵¹⁾
 - Neurological deficit resolved or resolving, ≥ **One:**
 - Functional impairment and, **One:** ⁽⁴⁸⁾
 - Discharge to home and, **One:** ⁽²⁵⁵⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - Tolerating PO or nutritional route established ^(252, 253)
 - Neurological stability ⁽²⁵⁶⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
 - For current condition use Episode Day 3 at a higher level of care
 - For a condition other than current condition use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, ≥ One:**
 - **Cardiovascular or peripheral vascular, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽²⁴⁹⁾
 - Clinical stability, ≥ **One:**
 - Pericardial effusion resolving and pain controlled or manageable ^(251, 270)
 - Acute limb ischemia resolving, **All:**
 - Limb swelling decreased
 - Antithrombotic therapy, **One:** ⁽²⁷¹⁾
 - Antiplatelet regimen established
 - Anticoagulation regimen established
 - Warfarin with no evidence of bleeding and, **One:** ⁽²⁷²⁾
 - INR ≥ 2.0

- INR ≥ 2.5 and mechanical valve (excludes bioprosthetic valves) ^(273, 274)
 - Pain controlled or manageable ⁽²⁵¹⁾
 - Peripheral pulses adequate
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One**:
 - Pericardial drain removed ≤ 24 h
 - Acute limb ischemia ≤ 7 d since onset and, $\geq$ **One**:
 - Embolectomy or thrombectomy required (**use General Surgical subset**)
 - Anticoagulation, **One**:
 - Unfractionated heparin, continuous ≤ 5 d ^(90, 275)
 - Oral anticoagulation and, **Both**:
 - Patient or caregiver unable to administer
 - Outpatient services unavailable and, **One**: ⁽²⁷⁶⁾
 - Warfarin (includes PO) and INR non-therapeutic and, **Both**: ^(89, 272)
 - Finding, **One**:
 - INR < 2.0
 - INR < 2.5 and mechanical valve (excludes bioprosthetic valves) and, **One**: ^(273, 274)
 - Mitral
 - Mitral and aortic
 - Unfractionated heparin, continuous ⁽²⁷⁵⁾
 - HIT and anticoagulation ^(91, 277)
 - Bleeding actual or suspected and, **Both**: ⁽²⁷⁸⁾
 - Finding, $\geq$ **One**:
 - Heart rate 100-120/min, sustained ⁽⁷⁾
 - Hct or Hb decreased within last 24h
 - Orthostatic hypotension, **Both**: ⁽²⁷⁹⁾
 - Sustained drop in blood pressure within 3 min of sitting or standing ⁽⁷⁾
 - Blood pressure, $\geq$ **One**:
 - Systolic drop ≥ 20 mmHg
 - Diastolic drop ≥ 10 mmHg
 - Intervention, $\geq$ **One**:
 - Blood product transfusion and anticoagulation reversed ⁽²⁸⁰⁾
 - Requiring Hct or Hb monitoring at least 2x/24h, ≤ 2 d
 - Neurovascular assessment at least 4x/24h, ≤ 24 h ⁽⁹²⁾
 - Post Critical care ≤ 24 h
 - **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria, **One**:
 - **Use Extended Stay subset**
 - Refer for secondary review
- **Eating disorder**, **One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽²⁴⁹⁾
 - Vital signs within acceptable limits ⁽²⁵⁴⁾
 - Lab values within acceptable limits ⁽²⁵⁴⁾
 - Treatment regimen or nutritional route established ⁽²⁵³⁾
 - Weight increasing or stabilized
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One**:
 - Eating disorder and, **Both**:
 - Finding, $\geq$ **One**:
 - Blood glucose < 60 mg/dL (3.3 mmol/L) ^(95, 96)
 - Blood pressure, **Both**:
 - Systolic < 90 mmHg ⁽⁹⁷⁾
 - Diastolic < 60 mmHg ⁽⁹⁷⁾
 - Cardiomyopathy and, $\geq$ **One**: ^(98, 99)
 - Cardiomegaly by imaging
 - Tricuspid or mitral valve dysfunction by physical examination

- EF < 40%(0.40)
- Co-occurring medical condition and unstable lab values requiring daily monitoring ⁽¹⁰⁰⁾
- Dehydration and, ≥ **One**:
 - BUN 25-45 mg/dL(8.9-16.1 mmol/L)
 - Creatinine 1.5-3.0 mg/dL(132.6-265.2 µmol/L) ⁽¹⁰¹⁾
 - Heart rate sustained > 100/min ^(7, 102)
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽⁹⁾
 - Urine specific gravity > 1.030
- Edema or rapid weight gain ⁽²⁸¹⁾
- Electrolyte imbalance, ≥ **One**: ^(103, 104)
 - Magnesium 1.0-1.4 mg/dL(0.41-0.58 mmol/L)
 - Phosphate 1.0-1.4 mg/dL(0.32-0.45 mmol/L)
 - Potassium < 3.0 mEq/L(3.0 mmol/L) without ECG changes
- Sodium, **One**:
 - 151-160 mEq/L(151-160 mmol/L)
 - 120-129 mEq/L(120-129 mmol/L)
- Heart rate < 50/min, sustained ^(7, 106)
- Orthostatic hypotension, **Both**:
 - Sustained drop in blood pressure within 3 min of sitting or standing
 - Blood pressure, ≥ **One**:
 - Systolic drop ≥ 20 mmHg ⁽¹⁰⁷⁾
 - Diastolic drop ≥ 10 mmHg ⁽¹⁰⁸⁾
- Orthostatic changes in heart rate > 30/min ⁽¹⁰⁹⁾
- QTc interval > 450 ms ⁽¹¹⁰⁾
- Temperature < 96.8°F(36.0°C) ⁽¹¹¹⁾
- Weight, ≥ **One**:
 - Continued weight loss
 - Failure to stabilize weight
- Intervention, **Both**:
 - Monitoring, **All**:
 - Calorie count
 - Daily weight
 - Electrolytes monitored at least once daily ⁽¹¹⁴⁾
 - Observation and documentation of oral or tube feeding pattern ⁽¹¹⁵⁾
 - Vital signs at least 3x/24h
 - Treatment, ≥ **One**:
 - Blood glucose monitoring at least 4x/24h ⁽¹¹⁶⁾
 - Complete bed rest ⁽¹¹⁷⁾
 - Fluid, supplements, or medication to correct electrolyte abnormalities ⁽¹⁰⁴⁾
 - IV fluid ⁽³⁸⁾
 - Tube feeding or parenteral nutrition
 - Weight-gain-specific treatment plan and, **All**:
 - Accompanied or monitored bathroom visits ⁽¹¹⁸⁾
 - Nutritional assessment daily ⁽¹¹⁹⁾
 - Re-evaluation of plan effectiveness daily ⁽¹²⁰⁾
 - Supervised meals and snacks ⁽¹²¹⁾
- Post Critical care ≤ 24h
- **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria, **One**:
 - **Use Extended Stay subset**
 - Refer for secondary review
- **Gastrointestinal or biliary, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **One**: ⁽²⁴⁹⁾
 - Clinical stability, ≥ **One**:

- Ascites controlled or improving and tolerating PO or nutritional route established ^(252, 253)
- **Bilirubin, All:**
 - Laboratory values normal or within acceptable limits ⁽²⁵⁴⁾
 - Neurological stability ⁽²⁵⁶⁾
 - Tolerating PO or nutritional route established ^(252, 253)
- **Gastroparesis, Both:** ^(125, 126)
 - Laboratory values normal or within acceptable limits ⁽²⁵⁴⁾
 - Tolerating PO or nutritional route established ^(252, 253)
- **Hepatic encephalopathy, Both:** ⁽¹⁶⁾
 - Grade I or within acceptable limits ⁽²⁵⁴⁾
 - Neurological stability last 24h ⁽²⁵⁶⁾
 - Ileus resolved and tolerating PO or nutritional route established ^(252, 253)
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Ascites ≤ 7d and, **Both:**
 - Finding, ≥ **One:**
 - Abdominal girth or weight, ≥ **One:**
 - Increasing
 - Subtherapeutic response to treatment ⁽²⁸²⁾
 - Unchanged from admission
 - Dyspnea > baseline ⁽⁷⁵⁾
 - Intervention, ≥ **One:**
 - Albumin
 - Diuretic ≥ 2x/24h
 - Sodium and fluid restriction ≤ 3d
 - Therapeutic paracentesis performed within last 24h
 - Transjugular intrahepatic portosystemic shunt (TIPS) performed ≤ 24h
 - Bilirubin > 3.0 mg/dL(51.3 μmol/L) and, ≥ **One:**
 - Albumin level < 2.0 g/dL(20 g/L) and, ≥ **One:**
 - Albumin
 - Feeding tube insertion performed within last 24h and feedings initiated
 - TPN or PPN
 - Inadequate oral intake ≤ 4d or active vomiting and, ≥ **One:** ⁽²⁶¹⁾
 - Antiemetic ≥ 3x/24h ⁽¹²³⁾
 - Serotonin antagonist > 2x/24h ⁽¹²⁴⁾
 - IV fluid ⁽³⁸⁾
 - Liver biopsy results pending ≤ 24h
 - PT ≥ 1.5x upper limit normal (ULN) or INR ≥ 2.0 and fresh frozen plasma (FFP)
 - Gastroparesis, actual or suspected ≤ 4d since initiation of treatment and, **Both:** ^(125, 126)
 - Inadequate oral intake and IV fluid ^(38, 127, 128)
 - Intervention, ≥ **One:**
 - Medication discontinuation ⁽¹²⁹⁾
 - Medication ⁽¹³⁰⁾
 - Gastric scintigraphy, scheduled or performed ^(131, 132, 133)
 - Hepatic encephalopathy Grade II ≤ 7d and, ≥ **One:** ^(16, 17, 134)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽¹³⁵⁾
 - Lactulose (includes PO or PR) ⁽¹³⁶⁾
 - L-ornithine L-aspartate (LOLA) (includes PO) ⁽¹³⁷⁾
 - Neomycin (includes PO) ⁽¹³⁸⁾
 - Rifaximin (includes PO) ⁽¹³⁹⁾
 - Ileus and, **One:** ⁽¹⁹⁾
 - Resolved and initiating PO or tube feeding ≤ 2d
 - Unresolved and ≤ 5d and, ≥ **One:**
 - IV fluid ⁽³⁸⁾

- TPN or PPN
- Post Critical care ≤ 24h
- **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria, **One**:
 - **Use Extended Stay subset**
 - Refer for secondary review
- **Genitourinary, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **One**:⁽²⁴⁹⁾
 - Clinical stability, ≥ **One**:
 - Obstruction of ureter, renal pelvis, urethra and, **Both**:
 - Creatinine < 1.5 mg/dL(132.6 μmol/L)
 - Finding, **One**:
 - Obstruction resolved
 - Patient or caregiver able to manage catheter at home or post acute care arranged
 - Nephrotic syndrome controlled and, **Both**:
 - Edema resolving
 - Kidney function improving or within acceptable limits⁽²⁵⁴⁾
 - Priapism resolved
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Genitourinary obstruction, **Both**:
 - Condition, ≥ **One**:
 - Obstruction of renal pelvis
 - Obstruction of ureter
 - Obstruction of urethra
 - Intervention, ≥ **One**:
 - Surgical intervention (**use General Surgical subset**)
 - Creatinine ≥ 1.5 mg/dL(132.6 μmol/L) and > baseline requiring urinary catheter and, **One**:
 - Suprapubic placed within last 24h
 - Transurethral placed within last 24h
 - Nephrostomy performed within last 24h
 - Ureteral stent placed within last 24h
 - Nephrotic syndrome ≤ 7d since onset and, **Both**:
 - Finding, ≥ **One**:
 - Blood urea nitrogen (BUN) > 45 mg/dL(16.1 mmol/L)
 - Generalized edema unchanged or worsening
 - Intervention, ≥ **One**:
 - Albumin 25%(0.25)
 - ACE inhibitor (includes PO)
 - ARB (includes PO)
 - Corticosteroid
 - Cytotoxic agent
 - Dialysis (initial), **One**:⁽²⁶²⁾
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal ≤ 7d since initiation
 - Diuretic ≥ 2x/24h
 - Priapism and surgical intervention required (**use General Surgical subset**)⁽¹⁴³⁾
 - Post Critical care ≤ 24h
 - **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria, **One**:
 - **Use Extended Stay subset**
 - Refer for secondary review
- **Hematology or oncology, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **One**:⁽²⁴⁹⁾
 - Clinical stability, ≥ **One**:
 - Hyperviscosity syndrome resolving and, **Both**:

- Neurological stability ⁽²⁵⁶⁾
- No evidence of bleeding
- Platelet count $\geq 50,000/\text{cu.mm}(50 \times 10^9/\text{L})$ or within acceptable limits and, $\geq$ **One:** ⁽²⁵⁴⁾
 - No evidence of bleeding
 - Purpura or petechiae improving or resolved
- **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - Hyperviscosity syndrome $\leq 6\text{d}$ since onset and, **Both:** ⁽²⁶³⁾
 - Finding, $\geq$ **One:**
 - Mucosal bleeding
 - Persistent neurological symptoms
 - Visual changes
 - Intervention, $\geq$ **One:**
 - Apheresis ⁽¹⁴⁵⁾
 - IV fluid $\leq 2\text{d}$ ⁽³⁸⁾
 - Neurological assessment at least $3\text{x}/24\text{h}$, $\leq 2\text{d}$ ⁽⁸⁶⁾
 - Thrombocytopenia $\leq 7\text{d}$ since onset and, **Both:**
 - Finding, $\geq$ **One:**
 - Bleeding actual or suspected and, $\geq$ **One:** ⁽²⁷⁸⁾
 - Hct $< 25\%(0.25)$ or Hb $< 8.3\text{ g/dL}(83\text{ g/L})$
 - Platelet count $< 60,000/\text{cu.mm}(60 \times 10^9/\text{L})$
 - Heart rate $100\text{-}120/\text{min}$, sustained ⁽⁷⁾
 - Hct or Hb decreased last 24h
 - Orthostatic hypotension, **Both:** ⁽²⁷⁹⁾
 - Sustained drop in blood pressure within 3 min of sitting or standing ⁽⁷⁾
 - Blood pressure, $\geq$ **One:**
 - Systolic drop $\geq 20\text{ mmHg}$
 - Diastolic drop $\geq 10\text{ mmHg}$
 - PT $\geq 1.5\text{x}$ upper limit normal (ULN) or INR ≥ 2.0
 - PTT $\geq 1.5\text{x}$ upper limit normal (ULN)
 - Platelet count $< 50,000/\text{cu.mm}(50 \times 10^9/\text{L})$
 - Purpura or petechiae, progressive or unknown etiology
 - Intervention, $\geq$ **One:**
 - Blood product transfusion
 - Corticosteroid
 - Immunoglobulin ⁽¹⁴⁷⁾
 - Post Critical care $\leq 24\text{h}$
 - **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
 - **Neurological, One:**
 - **Responder**, discharge expected today if clinically stable last 12h , **Both:** ⁽²⁴⁹⁾
 - Neurological stability ⁽²⁵⁶⁾
 - Clinical stability, $\geq$ **One:**
 - Functional impairment and, **One:** ⁽⁴⁸⁾
 - Discharge to home and, **One:** ⁽²⁵⁵⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - Neurological deficit, **Both:**
 - Resolved or resolving
 - Tolerating PO or nutritional route established ^(252, 253)
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - Functional impairment (new) requiring rehabilitation initiation within 24h ^(48, 49, 259)

- Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 and, **One:** ^(9, 226)
 - Observation work-up non-diagnostic ⁽²⁸³⁾
- Neurological deficit or disorder, persistent or unresolved and, **≥ One:**
 - CNS evaluation performed and results pending $\leq 24\text{h}$ and, **≥ One:**
 - EMG
 - Lumbar puncture (LP)
 - MRI
 - Continuous EEG $\leq 72\text{h}$ since initiation ⁽¹⁵¹⁾
 - Corticosteroid $\leq 5\text{d}$ since initiation
 - Dihydroergotamine (DHE) and antiemetic $\leq 5\text{d}$ since initiation ⁽⁵⁸⁾
 - Feeding tube, **Both:**
 - Insertion performed within last 24h
 - Feedings initiated
 - Immunotherapy
 - Inadequate oral intake and IV fluid $\leq 4\text{d}$ ^(38, 261)
 - Medication adjustment or initiation (includes PO) $\leq 2\text{d}$ ⁽³⁹⁾
 - Plasmapheresis
 - Seizure evaluation (excludes epilepsy) $\leq 5\text{d}$ since initiation, scheduled admission ^(155, 156)
 - Post Critical care $\leq 24\text{h}$
- **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria, **One:**
 - Use **Extended Stay subset**
 - Refer for secondary review
- **Respiratory, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **Both:** ⁽²⁴⁹⁾
 - O_2 sat $\geq 92\%(0.92)$ or within acceptable limits ⁽²⁵⁴⁾
 - Clinical stability, **≥ One:**
 - Dyspnea resolved
 - Chest tube and, **One:**
 - Heimlich valve
 - Removed and no reaccumulation of air or fluid by chest x-ray (CXR)
 - Pulmonary infiltrate resolving
 - Upper airway mass and airway patent
- **Partial responder**, not clinically stable for discharge and requires continued stay, **≥ One:**
 - O_2 sat $\leq 91\%(0.91)$ and $<$ baseline requiring supplemental oxygen ^(75, 76)
 - Chest tube and, **≥ One:**
 - Suction, continuous
 - Drainage $\geq 250\text{ mL}/24\text{h}$
 - Repositioning within last 24h ⁽²⁶⁵⁾
 - Water seal within last 24h
 - Pulmonary infiltrate (non-infectious) $\leq 6\text{d}$ since onset and, **Both:**
 - Oxygenation, **One:**
 - O_2 sat $\geq 92\%(0.92)$ or baseline ⁽⁷⁵⁾
 - O_2 sat $\leq 91\%(0.91)$ and $<$ baseline requiring supplemental oxygen ^(75, 76)
 - Intervention, **≥ One:**
 - Bronchoscopy results pending $\leq 24\text{h}$
 - Immunosuppressant and, **≥ One:** ⁽¹⁵⁹⁾
 - New medication initiation (includes PO)
 - Medication adjustment (includes PO) ⁽³⁹⁾
 - Lung biopsy results pending $\leq 24\text{h}$
 - Upper airway mass $\leq 7\text{d}$ since onset and, **Both:** ⁽¹⁴²⁾
 - Risk of airway compromise
 - Intervention, **≥ One:**
 - Biopsy results pending $\leq 2\text{d}$

- Corticosteroid
- Post Critical care $\leq 24\text{h}$
- **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria, **One**:
 - Use **Extended Stay subset**
 - Refer for secondary review
- **Skin, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽²⁴⁹⁾
 - Pain controlled or manageable ⁽²⁵¹⁾
 - Skin healing or skin care regimen established and tolerated
 - Tolerating PO ⁽²⁵²⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One**:
 - Dermatitis and immunosuppressant (includes PO) $\leq 3\text{d}$ ⁽¹⁵⁹⁾
 - Erythema multiforme major [Stevens-Johnson Syndrome or TEN] and, **Both**: ⁽¹⁶²⁾
 - Inadequate skin healing or fluid imbalance
 - Intervention, $\geq$ **One**:
 - Complex wound care $\leq 5\text{d}$ and, $\geq$ **One**: ⁽¹⁶³⁾
 - Wound care $\geq 3\text{x}/24\text{h}$ and, $\geq$ **One**:
 - Requiring parenteral analgesic
 - $\geq 15\text{ min}$ in duration
 - Wound management teaching $\leq 24\text{h}$ ⁽²⁶⁶⁾
 - Immunoglobulin $\leq 3\text{d}$ ⁽¹⁴⁷⁾
 - Immunosuppressant (includes PO) $\leq 3\text{d}$ ⁽¹⁵⁹⁾
 - IV fluid ⁽³⁸⁾
 - Volume expander $\leq 2\text{d}$ ⁽¹⁶⁴⁾
 - Loss or damage of skin $> 10\%(0.10)$ BSA and, $\geq$ **One**:
 - Pain uncontrolled and, **Both**:
 - Pain assessment ⁽³²⁾
 - Requiring change in medication, dose, or frequency ^(39, 40)
 - Complex wound care $\leq 5\text{d}$ and, $\geq$ **One**: ⁽¹⁶³⁾
 - Wound care $\geq 3\text{x}/24\text{h}$ and, $\geq$ **One**:
 - Requiring parenteral analgesic
 - $\geq 15\text{ min}$ in duration
 - Wound management teaching $\leq 24\text{h}$ ⁽²⁶⁶⁾
 - Immunosuppressant (includes PO) $\leq 3\text{d}$ ⁽¹⁵⁹⁾
 - IV fluid ⁽³⁸⁾
 - Psoriasis and Goeckerman regimen $\leq 5\text{d}$ ⁽¹⁶⁶⁾
 - Wound dehiscence, $\geq$ **One**:
 - Complex wound care $\leq 5\text{d}$ and, $\geq$ **One**: ⁽¹⁶³⁾
 - Wound care $\geq 3\text{x}/24\text{h}$ and, $\geq$ **One**:
 - Requiring parenteral analgesic
 - $\geq 15\text{ min}$ in duration
 - Wound management teaching $\leq 24\text{h}$ ⁽²⁶⁶⁾
 - Return to OR last 24h
 - Post Critical care $\leq 24\text{h}$
- **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria, **One**:
 - Use **Extended Stay subset**
 - Refer for secondary review
- **Transplant, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **One**: ⁽²⁴⁹⁾
 - Clinical stability, $\geq$ **One**:
 - Graft function stable and, **Both**:
 - Rejection controlled
 - Immunosuppressant drug tolerated

- Kidney and fluid collection resolving
- Liver and thrombosis resolving and, **One:**
 - Anticoagulation regimen established or not indicated ⁽²⁸⁴⁾
 - Warfarin with no evidence of bleeding and, **One:** ⁽²⁷²⁾
 - INR ≥ 2.0
 - INR ≥ 2.5 and mechanical valve (excludes bioprosthetic valves) ^(273, 274)
- Pancreas and, $\geq$ **One:**
 - Duodenal segment leak resolved
 - Thrombosis resolving, **One:**
 - Anticoagulation regimen established or not indicated ⁽²⁸⁴⁾
 - Warfarin with no evidence of bleeding and, **One:** ⁽²⁷²⁾
 - INR ≥ 2.0
 - INR ≥ 2.5 and mechanical valve (excludes bioprosthetic valves) ^(273, 274)
- **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - Graft dysfunction and immunosuppressant and, $\geq$ **One:** ^(159, 168, 173)
 - Conversion from IV to PO $\leq 2d$ ⁽²⁰¹⁾
 - Medication adjustment (includes PO) $\leq 3d$ ^(39, 174, 175)
 - New medication initiation
 - Kidney and, **Both:**
 - Hematoma, seroma, or lymphocele
 - Persistent percutaneous drainage
 - Liver and, **Both:**
 - Finding, $\geq$ **One:**
 - Hepatic vein thrombosis
 - Portal vein thrombosis
 - Intervention, $\geq$ **One:**
 - Anticoagulation, **One:**
 - LMWH $\leq 3d$ ⁽²⁸⁵⁾
 - Unfractionated heparin $\leq 5d$ ⁽²⁷⁵⁾
 - Oral anticoagulation and, **Both:**
 - Patient or caregiver unable to administer
 - Outpatient services unavailable and, **One:** ⁽²⁷⁶⁾
 - Warfarin (includes PO) and INR non-therapeutic and, **Both:** ⁽²⁷²⁾
 - Finding, **One:**
 - INR < 2.0
 - INR < 2.5 and mechanical valve (excludes bioprosthetic valves) and, **One:** ^(273, 274)
 - Mitral
 - Mitral and aortic
 - Intervention, **One:** ⁽⁸⁹⁾
 - Fondaparinux ⁽²⁸⁵⁾
 - LMWH ⁽²⁸⁵⁾
 - Unfractionated heparin ⁽²⁷⁵⁾
 - Non-vitamin K oral anticoagulant (includes PO) $\leq 5d$ and, **One:** ^(286, 287)
 - LMWH ⁽²⁸⁵⁾
 - Fondaparinux ⁽²⁸⁵⁾
 - Unfractionated heparin ⁽²⁷⁵⁾
 - HIT and anticoagulation ^(91, 277)
 - Bleeding actual or suspected and, **Both:** ⁽²⁷⁸⁾
 - Finding, $\geq$ **One:**
 - Heart rate 100-120/min, sustained ⁽⁷⁾
 - Hct or Hb decreased within last 24h
 - Orthostatic hypotension, **Both:** ⁽²⁷⁹⁾
 - Sustained drop in blood pressure within 3 min of sitting or standing ⁽⁷⁾

- Blood pressure, $\geq$ **One**:
 - Systolic drop $\geq$ 20 mmHg
 - Diastolic drop $\geq$ 10 mmHg
- Intervention, $\geq$ **One**:
 - Blood product transfusion and anticoagulation reversed
 - Requiring Hct or Hb monitoring at least $2x/24h \leq 2d$
- Thrombectomy performed within last 24h
- Pancreas and duodenal segment leak with percutaneous drainage
- Pancreas and thrombosis and, $\geq$ **One**:
 - Surgical intervention required (**use General Surgical subset**)
- Anticoagulation, **One**:
 - LMWH $\leq 3d$ ⁽²⁸⁵⁾
 - Unfractionated heparin $\leq 5d$ ⁽²⁷⁵⁾
- Oral anticoagulation and, **Both**:
 - Patient or caregiver unable to administer
- Outpatient services unavailable and, **One**: ⁽²⁷⁶⁾
 - Warfarin (includes PO) and INR non-therapeutic and, **Both**: ⁽²⁷²⁾
 - Finding, **One**:
 - INR < 2.0
 - INR < 2.5 and mechanical valve (excludes bioprosthetic valves) and, **One**: ^(273, 274)
 - Mitral
 - Mitral and aortic
 - Intervention, **One**: ⁽⁸⁹⁾
 - Fondaparinux ⁽²⁸⁵⁾
 - LMWH ⁽²⁸⁵⁾
 - Unfractionated heparin ⁽²⁷⁵⁾
 - Non-vitamin K oral anticoagulant (includes PO) $\leq 5d$ and, **One**: ^(286, 287)
 - LMWH ⁽²⁸⁵⁾
 - Fondaparinux ⁽²⁸⁵⁾
 - Unfractionated heparin ⁽²⁷⁵⁾
- HIT and anticoagulation ^(91, 277)
- Bleeding, actual or suspected and, **Both**: ⁽²⁷⁸⁾
 - Finding, $\geq$ **One**:
 - Hct $< 25\%(0.25)$ or Hb < 8.3 g/dL(83 g/L)
 - Platelet count $< 60,000/cu.mm(60 \times 10^9/L)$ or $> 1,000,000/cu.mm(1000 \times 10^9/L)$
 - Heart rate 100-120/min, sustained ⁽⁷⁾
 - Hct or Hb decreased last 24h
 - Orthostatic hypotension, **Both**: ⁽²⁷⁹⁾
 - Sustained drop in blood pressure within 3 min of sitting or standing ⁽⁷⁾
 - Blood pressure, $\geq$ **One**:
 - Systolic drop ≥ 20 mmHg
 - Diastolic drop ≥ 10 mmHg
 - PT $\geq 1.5x$ upper limit normal (ULN) or INR ≥ 2.0
 - PTT $\geq 1.5x$ upper limit normal (ULN)
 - Intervention, $\geq$ **One**:
 - Blood product transfusion and anticoagulation reversed
 - Requiring Hct or Hb monitoring at least $2x/24h, \leq 2d$
- Post Critical care $\leq 24h$
- **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria, **One**:
 - **Use Extended Stay subset**
 - Refer for secondary review
- **INTERMEDIATE, One**:
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One**:

- **Cardiovascular or peripheral vascular, ≥ One:**
 - Epoprostenol, treprostinil, or iloprost (includes continuous infusion or inhaled) initiation ≤ 2d ⁽²⁶⁷⁾
 - Pacemaker or ICD placement performed within last 24h
 - Pericardial drain and post Critical care ≤ 24h
- **Pericardial effusion and high risk for cardiac tamponade ≤ 24h and, Both:** ^(176, 177, 178)
 - Asymptomatic
 - Effusion size ≥ 1 cm ⁽¹⁷⁹⁾
- **VAD requiring readmission and, ≥ One:** ^(183, 184)
 - **Anticoagulation, ≥ One:** ^(185, 186, 187)
 - Direct thrombin inhibitor
 - GP IIb or IIIa inhibitor
 - Heparin
 - Thrombolytic
 - **Bleeding actual or suspected and, ≥ One:**
 - Hct or Hb monitoring at least 3x/24h, ≤ 3d
 - Volume expander or blood product ⁽¹⁶⁴⁾
 - **Inadequate nutrition and, ≥ One:**
 - Calorie count ≤ 3d
 - Requiring daily tube feeding adjustment
 - TPN initiation ≤ 24h
 - **Infection actual or suspected and, Both:**
 - **Finding, ≥ One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁾
 - WBC ≥ 12,000/cu.mm($12 \times 10^9/L$)
 - WBC ≤ 4,000/cu.mm($4 \times 10^9/L$)
 - Bands > 10%(0.10)
 - ANC < 500/cu.mm($500 \times 10^6/L$) ⁽¹⁸⁸⁾
 - Anti-infective
 - **Post acute facility unavailable and, ≥ One:**
 - Functional impairment requiring therapy ⁽⁴⁸⁾
 - Inadequate social or family support
 - Patient or caregiver unable to demonstrate functional ability to maintain and operate VAD
 - Post Critical care ≤ 3d
- **Eating disorder and, ≥ One:**
 - **Hypokalemia, All:** ^(189, 190)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - PVCs > 6/min
 - Potassium repletion
 - **Seizure and, Both:** ⁽¹⁹¹⁾
 - Activity resolved
 - Neurological assessment every 3-4h ⁽⁸⁶⁾
- **Gastrointestinal or biliary, ≥ One:**
 - **Hepatic encephalopathy Grade III ≤ 7d and, ≥ One:** ^(16, 17, 192)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽¹³⁵⁾
 - Lactulose (includes PO or PR) ⁽¹³⁶⁾
 - L-ornithine L-aspartate (LOLA) ⁽¹³⁷⁾
 - Neomycin (includes PO) ⁽¹³⁸⁾
 - Rifaximin (includes PO) ⁽¹³⁹⁾
 - **Hepatorenal syndrome ≤ 8d and, ≥ One:** ⁽¹⁹³⁾
 - Albumin
 - Octreotide and midodrine (includes PO)
- **Neurological, ≥ One:**
 - External ventricular or lumbar drainage and neurological assessment every 3-4h ⁽⁸⁶⁾

- Guillain-Barre syndrome, actual or suspected, and neurological assessment every 3-4h and, **One:** ^(86, 194)
 - Immunotherapy
 - Plasmapheresis
- Neurological deficit or disorder, persistent or unresolved, and neurological assessment every 3-4h ≤ 2d ^(86, 257, 258)
- Seizure (excludes epilepsy) and neurological assessment every 3-4h and, **Both:** ⁽⁸⁶⁾
 - Activity resolved
 - Anticonvulsant, conversion from IV to PO ≤ 24h
 - Ventriculostomy awaiting V-P shunt and neurological assessment every 3-4h, ≤ 6d since onset ⁽⁸⁶⁾
- **Respiratory, ≥ One:**
 - NIPPV ⁽¹⁹⁵⁾
 - Oxygen ≥ 40%(0.40) ≤ 2d ⁽⁷⁶⁾
- **Skin, One:**
 - Complex wound care ≤ 5d and, **Both:** ⁽¹⁶³⁾
 - ≥ 3x/24h
 - ≥ 30 min in duration
- **Transplant, ≥ One:**
 - Acute humoral rejection and post Critical care ≤ 24h ⁽²⁴⁷⁾
 - GVHD ≤ 7d since onset and, **One:** ⁽¹⁷⁰⁾
 - Grade I or II and monoclonal antibody initiation ≤ 3 doses ⁽¹⁹⁸⁾
 - Grade III and immunosuppressant ^(159, 173)
 - Post lung transplant and suspected rejection ≤ 8d since onset and, **Both:** ⁽¹⁶⁸⁾
 - Oxygen, ≥ **One:** ⁽⁷⁶⁾
 - HFNC and respiratory monitoring every 3-4h ^(199, 200)
 - Oxygen ≥ 40%(0.40)
 - Immunosuppressant and, ≥ **One:** ^(159, 173)
 - Conversion from IV to PO ≤ 2d ⁽²⁰¹⁾
 - Medication adjustment (includes PO) ^(39, 175)
 - New medication initiation
- **Non-responder, not clinically stable for discharge, One:**
 - Does not meet partial responder criteria or at the Acute level of care, **One:**
 - Use **Extended Stay subset**
 - Refer for secondary review
- **CRITICAL, One:**
 - **Partial Responder, not clinically stable for discharge and requires continued stay, ≥ One:**
 - **General, ≥ One:**
 - Exchange transfusion ⁽²⁰²⁾
 - Heart failure and, **Both:**
 - Finding, ≥ **One:**
 - Arterial Po₂ < 56 mmHg(7.4 kPa)
 - O₂ sat < 89%(0.89) and < baseline ⁽⁷⁵⁾
 - Diuretic every 1-2h or continuous
 - Hemodynamic instability, **Both:**
 - Finding, ≥ **One:**
 - Heart rate > 120/min, sustained ⁽⁷⁾
 - Systolic blood pressure, ≥ **One:**
 - < 90 mmHg without chronic hypertension
 - < 110 mmHg with chronic hypertension
 - > 30 mmHg decrease from baseline ⁽⁷⁵⁾
 - Labile ⁽²⁰³⁾
 - MAP < 65 mmHg
 - Intervention, ≥ **One:**

- Antifibrinolytic ⁽²⁰⁴⁾
- Blood product transfusion
- IV fluid resuscitation $\leq 2d$ ⁽²⁰⁵⁾
- Hemodynamic monitoring, invasive ⁽²⁰⁶⁾
- Thrombolytic ⁽²⁰⁸⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Cardiovascular or peripheral vascular, $\geq$ One:**
 - Pericardial effusion and, $\geq$ **One:** ^(209, 210)
 - Pericardial window (**use General Surgical subset**)
 - Symptomatic $\leq 24h$, $\geq$ **One:** ⁽²¹¹⁾
 - Dyspnea
 - Jugular venous distention
 - Orthopnea
 - Insertion of pericardial drain performed $\leq 24h$ ⁽²¹²⁾
 - Pericardiocentesis performed $\leq 24h$ ⁽²¹²⁾
 - Acute limb ischemia and, $\geq$ **One:**
 - Neurovascular assessment at least every 1-2h, $\leq 2d$ ⁽⁹²⁾
 - Catheter directed thrombolysis performed within last 24h
 - Pulmonary arterial hypertension and, **Both:** ⁽³⁴⁾
 - Epoprostenol, treprostinil, or iloprost (includes continuous infusion or inhaled) initiation ⁽²¹³⁾
 - Hemodynamic monitoring, invasive ⁽²¹⁴⁾
 - Temporary pacemaker insertion performed within last 2d
 - VAD requiring readmission and, $\geq$ **One:** ^(183, 215)
 - Blood pressure, $\geq$ **One:**
 - $<$ baseline and inadequate perfusion ⁽²¹⁶⁾
 - MAP < 70 mmHg and continuous flow device ⁽²¹⁷⁾
 - Systolic < 90 mmHg and pulsatile flow device
 - Decreased level of consciousness
 - Decreased VAD flows and inadequate perfusion ⁽²¹⁶⁾
 - O₂ sat $< 89\%$ (0.89)
 - VAD, percutaneous ^(184, 218)
- **Eating disorder and, Both:**
 - Finding, $\geq$ **One:**
 - Magnesium < 1.0 mg/dL(0.41 mmol/L)
 - Phosphate < 1.0 mg/dL(0.32 mmol/L)
 - Potassium < 3.0 mEq/L(3.0 mmol/L) and, $\geq$ **One:** ^(190, 219)
 - Multifocal PVCs
 - R on T phenomenon ⁽²²⁰⁾
 - T and U wave fusion ⁽²²¹⁾
 - U wave ⁽²²²⁾
 - Sodium > 160 mEq/L(160 mmol/L)
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ^(223, 224, 225)
 - Coma, stupor, obtundation or GCS ≤ 8 ⁽²²⁶⁾
 - Intervention, $\geq$ **One:**
 - CRRT ⁽²²⁸⁾
 - Dialysis, hemodialysis or peritoneal
 - Electrolyte repletion
 - IV fluid ^(38, 229)
 - Neurological assessment every 1-2h, $\leq 2d$ ⁽⁸⁶⁾
 - Sodium chloride 3%(0.03) ⁽²³⁰⁾
- **Gastrointestinal or biliary, $\geq$ One:**
 - Hepatic encephalopathy Grade IV $\leq 7d$ and, $\geq$ **One:** ^(16, 17, 192)

- Branched-chain amino acids (BCAA) (includes PO) ⁽¹³⁵⁾
- Lactulose (includes PO or PR) ⁽¹³⁶⁾
- L-ornithine L-aspartate (LOLA) ⁽¹³⁷⁾
- Neomycin (includes PO) ⁽¹³⁸⁾
- Rifaximin (includes PO) ⁽¹³⁹⁾
- Hepatorenal syndrome ≤ 8d, ≥ **One**: ⁽¹⁹³⁾
 - CRRT ⁽²²⁸⁾
 - Hemodialysis or peritoneal dialysis ≤ 2d since initiation
 - Norepinephrine
- **Hematology or oncology, One**:
 - DIC and blood product transfusion ≤ 7d since onset ⁽²³¹⁾
- **Neurological, ≥ One**:
 - Anticonvulsant, ≥ **One**:
 - > 2 medications for seizure termination within 24h
 - Anticonvulsant, continuous
 - Cerebral edema and, ≥ **One**:
 - ICP monitoring (includes OR placement)
 - Corticosteroid
 - Hyperosmolar therapy, ≥ **One**: ⁽²³²⁾
 - Mannitol
 - Hypertonic saline ≥ 3%(0.03) solution
 - External ventricular or lumbar drainage and neurological assessment every 1-2h ⁽⁸⁶⁾
 - Neurological deficit or disorder, persistent or unresolved, and neurological assessment every 1-2h ≤ 2d ^(86, 257, 258)
- **Respiratory, ≥ One**:
 - Mechanical ventilation
 - NIPPV and, ≥ **One**: ⁽¹⁹⁵⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ^(238, 268)
 - Nebulizer or metered dose inhaler (MDI) treatment, **One**: ⁽²³⁵⁾
 - Every 1-2h
 - Continuous
- **Toxic exposure or ingestion, ≥ One**: ^(82, 87)
 - General, **Both**:
 - Finding, ≥ **One**:
 - Corrosive substance ⁽²³⁹⁾
 - End organ damage, actual or potential
 - Intervention, ≥ **One**:
 - Dialysis, **One**:
 - CRRT ⁽²²⁸⁾
 - Hemodialysis
 - Peritoneal
 - Artificial airway placement (excludes nasal trumpet) ⁽²⁴⁰⁾
 - Toxin-specific antidote ⁽²⁴¹⁾
 - Potential for significant arrhythmia and, **Both**: ^(196, 197)
 - Abnormal electrocardiogram (ECG), ≥ **One**:
 - QRS duration ≥ 0.12 sec ^(242, 243)
 - Prolonged QT interval ^(244, 245)
 - PR interval > 0.20 sec ⁽²⁴⁶⁾
 - Continuous cardiac monitoring (excludes Holter)
- **Transplant, ≥ One**:
 - Acute humoral rejection and apheresis ^(145, 247)
 - GVHD, Grade IV, ≤ 7d since onset and, ≥ **One**: ⁽¹⁷⁰⁾

- Apheresis ⁽¹⁴⁵⁾
- Immunosuppressant ^(159, 173)
- Post lung transplant and suspected rejection $\leq$ 8d since onset and, **Both:** ⁽¹⁶⁸⁾
 - Immunosuppressant and, $\geq$ **One:** ^(159, 173)
 - New medication initiation
 - Medication adjustment (includes PO) ^(39, 175)
 - Intervention, $\geq$ **One:**
 - Mechanical ventilation
 - NIPPV and, **One:** ⁽¹⁹⁵⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽²⁶⁸⁾
- **Non-responder**, not clinically stable for discharge, **One:**
 - Does not meet partial responder criteria or at the Acute or Intermediate level of care, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, Discharge Screens, **One:** ⁽²⁸⁸⁾

- **HOME, Both:**
 - Level of care appropriateness, **Both:**
 - Home environment safe and accessible ⁽²⁸⁹⁾
 - Patient or caregiver demonstrates ability to manage care
 - Discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽²⁹⁰⁾
 - Medication reconciliation ⁽²⁹¹⁾
 - Follow-up care planned ⁽²⁹²⁾
 - Identify and address transportation needs
- **HOME CARE, Both:**
 - Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽²⁸⁹⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(276, 293)
 - Skilled services, $\geq$ **One:** ⁽²⁹⁴⁾
 - Skilled nursing ⁽²⁹⁵⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽²⁹⁶⁾
 - Discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽²⁹⁰⁾
 - Medication reconciliation ⁽²⁹¹⁾
 - Follow-up care planned ⁽²⁹²⁾
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽²⁸⁾
 - Skilled treatment, $\geq$ **One:**
 - Clinical assessment ⁽²⁹⁷⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽²⁹⁸⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq$ 1x/wk
 - Treatment precluded in a lower level
 - Services, $\geq$ **One:**
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽²⁹⁵⁾

- **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(299, 300)
 - Functional impairments requiring at least supervision ⁽³⁰¹⁾
 - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽²⁹¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
 - **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽²⁹⁵⁾
 - **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(299, 300)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽³⁰¹⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽²⁹¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Treatment precluded in a lower level
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(299, 300)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽³⁰¹⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽²⁹¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - **Level of care appropriateness, Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(299, 300)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽³⁰¹⁾
 - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽³⁰²⁾
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽²⁹¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - **Level of care appropriateness, Both:**
 - Treatment precluded in a lower level
 - **In lieu of acute or continued hospitalization or failed lower level of care, ≥ One:**

- Primary condition or illness and treatment of active comorbid condition(s) ⁽³⁰³⁾
- Ventilator dependent ≥ 6h/d and weaning planned ^(304, 305)
- Acute and comorbid conditions requiring prolonged hospitalization ⁽³⁰⁶⁾
- Complete prior to facility transfer, All:
 - Medication reconciliation ⁽²⁹¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
052 SPINAL DISORDERS AND INJURIES WITH CC/MCC	n/a	4.6	CMS GMLOS
053 SPINAL DISORDERS AND INJURIES WITHOUT CC/MCC	n/a	2.8	CMS GMLOS
056 DEGENERATIVE NERVOUS SYSTEM DISORDERS WITH MCC	n/a	6.1	CMS GMLOS
057 DEGENERATIVE NERVOUS SYSTEM DISORDERS WITHOUT MCC	n/a	4	CMS GMLOS
058 MULTIPLE SCLEROSIS AND CEREBELLAR ATAXIA WITH MCC	n/a	4.7	CMS GMLOS
059 MULTIPLE SCLEROSIS AND CEREBELLAR ATAXIA WITH CC	n/a	3.7	CMS GMLOS
060 MULTIPLE SCLEROSIS AND CEREBELLAR ATAXIA WITHOUT CC/MCC	n/a	2.9	CMS GMLOS
080 NONTRAUMATIC STUPOR AND COMA WITH MCC	n/a	5.4	CMS GMLOS
081 NONTRAUMATIC STUPOR AND COMA WITHOUT MCC	n/a	2.7	CMS GMLOS
091 OTHER DISORDERS OF NERVOUS SYSTEM WITH MCC	n/a	4.5	CMS GMLOS
092 OTHER DISORDERS OF NERVOUS SYSTEM WITH CC	n/a	3.1	CMS GMLOS
093 OTHER DISORDERS OF NERVOUS SYSTEM WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
094 BACTERIAL AND TUBERCULOUS INFECTIONS OF NERVOUS SYSTEM WITH MCC	n/a	8.6	CMS GMLOS
095 BACTERIAL AND TUBERCULOUS INFECTIONS OF NERVOUS SYSTEM WITH CC	n/a	5.9	CMS GMLOS
096 BACTERIAL AND TUBERCULOUS INFECTIONS OF NERVOUS SYSTEM WITHOUT CC/MCC	n/a	4.5	CMS GMLOS
100 SEIZURES WITH MCC	n/a	4.5	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
101 SEIZURES WITHOUT MCC	n/a	2.7	CMS GMLOS
102 HEADACHES WITH MCC	n/a	3.1	CMS GMLOS
103 HEADACHES WITHOUT MCC	n/a	2.3	CMS GMLOS
149 DYSEQUILIBRIUM	n/a	1.9	CMS GMLOS
154 OTHER EAR, NOSE, MOUTH AND THROAT DIAGNOSES WITH MCC	n/a	4.2	CMS GMLOS
155 OTHER EAR, NOSE, MOUTH AND THROAT DIAGNOSES WITH CC	n/a	2.9	CMS GMLOS
156 OTHER EAR, NOSE, MOUTH AND THROAT DIAGNOSES WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
199 PNEUMOTHORAX WITH MCC	n/a	5.1	CMS GMLOS
200 PNEUMOTHORAX WITH CC	n/a	3.2	CMS GMLOS
201 PNEUMOTHORAX WITHOUT CC/MCC	n/a	2.4	CMS GMLOS
204 RESPIRATORY SIGNS AND SYMPTOMS	n/a	2.1	CMS GMLOS
205 OTHER RESPIRATORY SYSTEM DIAGNOSES WITH MCC	n/a	4.5	CMS GMLOS
206 OTHER RESPIRATORY SYSTEM DIAGNOSES WITHOUT MCC	n/a	2.5	CMS GMLOS
296 CARDIAC ARREST, UNEXPLAINED WITH MCC	n/a	2	CMS GMLOS
297 CARDIAC ARREST, UNEXPLAINED WITH CC	n/a	1.3	CMS GMLOS
298 CARDIAC ARREST, UNEXPLAINED WITHOUT CC/MCC	n/a	1.1	CMS GMLOS
299 PERIPHERAL VASCULAR DISORDERS WITH MCC	n/a	4	CMS GMLOS
300 PERIPHERAL VASCULAR DISORDERS WITH CC	n/a	3.1	CMS GMLOS
301 PERIPHERAL VASCULAR DISORDERS WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
314 OTHER CIRCULATORY SYSTEM DIAGNOSES WITH MCC	n/a	5.1	CMS GMLOS
315 OTHER CIRCULATORY SYSTEM DIAGNOSES WITH CC	n/a	2.7	CMS GMLOS
316 OTHER CIRCULATORY SYSTEM DIAGNOSES WITHOUT CC/MCC	n/a	1.8	CMS GMLOS
388 GASTROINTESTINAL OBSTRUCTION WITH MCC	n/a	4.6	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
389 GASTROINTESTINAL OBSTRUCTION WITH CC	n/a	3.1	CMS GMLOS
390 GASTROINTESTINAL OBSTRUCTION WITHOUT CC/MCC	n/a	2.3	CMS GMLOS
391 ESOPHAGITIS, GASTROENTERITIS AND MISCELLANEOUS DIGESTIVE DISORDERS WITH MCC	n/a	3.9	CMS GMLOS
392 ESOPHAGITIS, GASTROENTERITIS AND MISCELLANEOUS DIGESTIVE DISORDERS WITHOUT MCC	n/a	2.6	CMS GMLOS
432 CIRRHOSIS AND ALCOHOLIC HEPATITIS WITH MCC	n/a	4.9	CMS GMLOS
433 CIRRHOSIS AND ALCOHOLIC HEPATITIS WITH CC	n/a	3.3	CMS GMLOS
434 CIRRHOSIS AND ALCOHOLIC HEPATITIS WITHOUT CC/MCC	n/a	2.3	CMS GMLOS
438 DISORDERS OF PANCREAS EXCEPT MALIGNANCY WITH MCC	n/a	4.7	CMS GMLOS
439 DISORDERS OF PANCREAS EXCEPT MALIGNANCY WITH CC	n/a	3.1	CMS GMLOS
440 DISORDERS OF PANCREAS EXCEPT MALIGNANCY WITHOUT CC/MCC	n/a	2.4	CMS GMLOS
441 DISORDERS OF LIVER EXCEPT MALIGNANCY, CIRRHOSIS OR ALCOHOLIC HEPATITIS WITH MCC	n/a	4.8	CMS GMLOS
442 DISORDERS OF LIVER EXCEPT MALIGNANCY, CIRRHOSIS OR ALCOHOLIC HEPATITIS WITH CC	n/a	3.2	CMS GMLOS
443 DISORDERS OF LIVER EXCEPT MALIGNANCY, CIRRHOSIS OR ALCOHOLIC HEPATITIS WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
595 MAJOR SKIN DISORDERS WITH MCC	n/a	5.7	CMS GMLOS
596 MAJOR SKIN DISORDERS WITHOUT MCC	n/a	3.5	CMS GMLOS
606 MINOR SKIN DISORDERS WITH MCC	n/a	4.7	CMS GMLOS
607 MINOR SKIN DISORDERS WITHOUT MCC	n/a	3	CMS GMLOS
640 MISCELLANEOUS DISORDERS OF NUTRITION, METABOLISM, FLUIDS AND ELECTROLYTES WITH MCC	n/a	3.6	CMS GMLOS
641 MISCELLANEOUS DISORDERS OF NUTRITION, METABOLISM, FLUIDS AND ELECTROLYTES WITHOUT MCC	n/a	2.6	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
689 KIDNEY AND URINARY TRACT INFECTIONS WITH MCC	n/a	3.9	CMS GMLOS
690 KIDNEY AND URINARY TRACT INFECTIONS WITHOUT MCC	n/a	2.9	CMS GMLOS
693 URINARY STONES WITH MCC	n/a	3.8	CMS GMLOS
694 URINARY STONES WITHOUT MCC	n/a	2.1	CMS GMLOS
695 KIDNEY AND URINARY TRACT SIGNS AND SYMPTOMS WITH MCC	n/a	3.7	CMS GMLOS
696 KIDNEY AND URINARY TRACT SIGNS AND SYMPTOMS WITHOUT MCC	n/a	2.3	CMS GMLOS
698 OTHER KIDNEY AND URINARY TRACT DIAGNOSES WITH MCC	n/a	4.9	CMS GMLOS
699 OTHER KIDNEY AND URINARY TRACT DIAGNOSES WITH CC	n/a	3.3	CMS GMLOS
700 OTHER KIDNEY AND URINARY TRACT DIAGNOSES WITHOUT CC/MCC	n/a	2.3	CMS GMLOS
729 OTHER MALE REPRODUCTIVE SYSTEM DIAGNOSES WITH CC/MCC	n/a	3	CMS GMLOS
730 OTHER MALE REPRODUCTIVE SYSTEM DIAGNOSES WITHOUT CC/MCC	n/a	1.9	CMS GMLOS
808 MAJOR HEMATOLOGICAL AND IMMUNOLOGICAL DIAGNOSES EXCEPT SICKLE CELL CRISIS AND COAGULATION DISORDERS WITH MCC	n/a	5.3	CMS GMLOS
809 MAJOR HEMATOLOGICAL AND IMMUNOLOGICAL DIAGNOSES EXCEPT SICKLE CELL CRISIS AND COAGULATION DISORDERS WITH CC	n/a	3.4	CMS GMLOS
810 MAJOR HEMATOLOGICAL AND IMMUNOLOGICAL DIAGNOSES EXCEPT SICKLE CELL CRISIS AND COAGULATION DISORDERS WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
813 COAGULATION DISORDERS	n/a	3.7	CMS GMLOS
814 RETICULOENDOTHELIAL AND IMMUNITY DISORDERS WITH MCC	n/a	4.8	CMS GMLOS
815 RETICULOENDOTHELIAL AND IMMUNITY DISORDERS WITH CC	n/a	3	CMS GMLOS
816 RETICULOENDOTHELIAL AND IMMUNITY DISORDERS WITHOUT CC/MCC	n/a	2.3	CMS GMLOS
864 FEVER AND INFLAMMATORY CONDITIONS	n/a	2.7	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
885 PSYCHOSES	n/a	6.4	CMS GMLOS
887 OTHER MENTAL DISORDER DIAGNOSES	n/a	3.3	CMS GMLOS
896 ALCOHOL, DRUG ABUSE OR DEPENDENCE WITHOUT REHABILITATION THERAPY WITH MCC	n/a	5	CMS GMLOS
897 ALCOHOL, DRUG ABUSE OR DEPENDENCE WITHOUT REHABILITATION THERAPY WITHOUT MCC	n/a	3.4	CMS GMLOS
915 ALLERGIC REACTIONS WITH MCC	n/a	3.8	CMS GMLOS
916 ALLERGIC REACTIONS WITHOUT MCC	n/a	1.8	CMS GMLOS
917 POISONING AND TOXIC EFFECTS OF DRUGS WITH MCC	n/a	3.9	CMS GMLOS
918 POISONING AND TOXIC EFFECTS OF DRUGS WITHOUT MCC	n/a	2.5	CMS GMLOS
919 COMPLICATIONS OF TREATMENT WITH MCC	n/a	4.4	CMS GMLOS
920 COMPLICATIONS OF TREATMENT WITH CC	n/a	2.9	CMS GMLOS
921 COMPLICATIONS OF TREATMENT WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
922 OTHER INJURY, POISONING AND TOXIC EFFECT DIAGNOSES WITH MCC	n/a	4.5	CMS GMLOS
923 OTHER INJURY, POISONING AND TOXIC EFFECT DIAGNOSES WITHOUT MCC	n/a	2.9	CMS GMLOS
947 SIGNS AND SYMPTOMS WITH MCC	n/a	3.7	CMS GMLOS
948 SIGNS AND SYMPTOMS WITHOUT MCC	n/a	2.6	CMS GMLOS
ABDOMINAL PAIN	81%	2.5	InterQual
ALCOHOL INTOXICATION	47%	3.1	InterQual
ANGIOEDEMA	57%	2.8	InterQual
ANOREXIA	18%	11.3	InterQual
ASCITES	72%	3.1	InterQual
CARDIAC ARREST	12%	6.2	InterQual
DRUG INTOXICATION	47%	3.2	InterQual
GRAFT DYSFUNCTION, KIDNEY	7%	5.5	InterQual

Condition or Procedure	% Pd Obs	LOS (days)	Type
GRAFT DYSFUNCTION, LIVER	8%	5.9	InterQual
HEPATIC ENCEPHALOPATHY	0%	5	InterQual
HEPATORENAL SYNDROME	3%	7.2	InterQual
HYDRONEPHROSIS	50%	2	InterQual
ILEUS	24%	3.8	InterQual
INTRACTABLE MIGRAINE	45%	3.6	InterQual
JAUNDICE OR ELEVATED BILIRUBIN	42%	3.1	InterQual
MIGRAINE	57%	2.4	InterQual
NEPHROTIC SYNDROME	24%	6.2	InterQual
PARALYSIS OF EXTREMITIES	7%	13.8	InterQual
PARALYSIS OF FACE	60%	2	InterQual
PERICARDIAL EFFUSION	n/a	5.4	InterQual
PERICARDITIS	28%	2.9	InterQual
PERIPHERAL ARTERY OCCLUSION	27%	3.3	InterQual
PNEUMOMEDIASTINUM	38%	3.2	InterQual
PNEUMOTHORAX, SPONTANEOUS	11%	5.3	InterQual
POST LUNG TRANSPLANT AND SUSPECTED REJECTION	5%	6.5	InterQual
POST TRANSPLANT COMPLICATION, KIDNEY	7%	4.8	InterQual
POST TRANSPLANT COMPLICATION, LIVER	8%	5.6	InterQual
PRIAPISM	53%	2	InterQual
PULMONARY ARTERIAL HYPERTENSION	18%	5.7	InterQual
PULMONARY INFILTRATE	41%	2.9	InterQual
RENAL CALCULUS	61%	2.3	InterQual
THROMBOCYTOPENIA (EXCLUDES KNOWN VON WILLEBRAND DISEASE)	21%	3.5	InterQual
TOXIC INGESTION	30%	3.5	InterQual
WOUND DEHISCENCE	27%	5.4	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

These criteria apply to patients with abdominal pain of unknown etiology who are not clinically stable for discharge and in need of a diagnostic workup. If a patient is diagnosed with a new condition necessitating continued stay, the reviewer should refer to Episode Day 1 or the Operative Day within the most appropriate subset. For example:

- If a patient has pancreatitis, refer to the Pancreatitis subset.
- If a patient requires surgical intervention, refer to the General Surgical subset.

5:

Abdominal pain may be caused by a variety of conditions from a self-limiting diagnosis to a surgical emergency. A diagnostic workup is necessary for patients presenting with acute abdominal pain; to determine the cause of their symptoms, establish a diagnosis and initiate appropriate treatment (Broder et al., Acad Emerg Med 2022, 29: 526-60). For older patients, an abdominal pain differential diagnosis may be more difficult to identify because physical findings in this age group are often less distinct (e.g., high body temperatures may be a late finding for infection) (Henden Cam et al., Current gerontology and geriatrics research 2018, 2018:).

6:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

7:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

8:

Abdominal pain may account for up to 10% of emergency department (ED) visits. A history of abdominal surgery raises concerns for abdominal adhesions, ileus, or obstruction (Bruze et al., Surg Obes Relat Dis 2017, 13: 962-9; Strik et al., Surgery 2016, 159: 1351-9; Natesan et al., Emerg Med Clin North Am 2016, 34: 165-90). In a study of over 700 patients who underwent elective open or laparoscopic general abdominal surgeries, 6% developed adhesive bowel obstruction within 4 years of surgery (Strik et al., Surgery 2016, 159: 1351-9). For gastric bypass patients, abdominal pain was the most frequent ED presentation during a year-long study (Mora-Pinzon et al., Surgery 2017, 162: 1155-62). A national cohort study found bariatric bypass patients were at risk for readmission greater than 6 years post-surgery, with approximately 24% admitted with an ileus or obstruction (Bruze et al., Surg Obes Relat Dis 2017, 13:

962-9).

9:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

10:

Patients with mild to moderate active vomiting should be treated with intravenous hydration and antiemetic therapy. A dehydrated patient who cannot tolerate oral hydration should not be discharged (Joshua Guttman, In: Rosen's Emergency Medicine, 9 edn. 2018:230-41.e1).

11:

Abdominal imaging includes ultrasound (US), computed tomography (CT), and magnetic resonance imaging (MRI) studies. A first line imaging study may be an US. If there is a further need to clarify results, for adults with abdominal pain of unknown etiology, abdominal CT may be recommended or, if contraindicated, abdominal MRI. In pediatric patients, an MRI may be performed to avoid radiation exposure associated with CT imaging.

12:

Imaging includes:

- Ultrasound (US)
- Computed tomography (CT)
- Magnetic resonance imaging (MRI)

13:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

14:

For adults with ascites, abdominal and pelvis computer tomography (CT) may be needed if an ultrasound is nondiagnostic. For pediatric patients, if an abdominal ultrasound (US) is nondiagnostic, magnetic resonance imaging (MRI) is recommended.

15:

Refractory ascites is ascites that cannot be managed by maximum doses of diuretics due to a lack of response or the development of side effects. Large volume therapeutic paracentesis, a first-line treatment for refractory ascites, usually is accompanied by the removal of 5 or more liters in combination with albumin infusion (Zhao et al., J Int Med Res 2018, 46: 1138-45). Large volume ascites, also called massive or tense ascites, is defined as an amount of intraperitoneal fluid that significantly limits the activities of daily living.

16:

Hepatic encephalopathy (HE) is a complication of acute or chronic liver disease that is primarily characterized by changes in personality, consciousness, cognition, and motor function. West Haven criteria for grading HE includes (Montrief et al., Am J Emerg Med 2019, 37: 329-37; Weissenborn, Drugs 2019, 79: 5-9; Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47):

- Grade I - Lack of awareness, anxiety, short attention span, euphoria, inability to add/subtract
- Grade II - Disorientation for time, inappropriate behavior, lethargy/apathy, obvious personality change, dyspraxia, or asterixis
- Grade III - Confusion, bizarre behavior, gross disorientation, responsive to verbal stimuli, somnolence to semi-stupor

- Grade IV - Coma

17:

While an elevated ammonia level is the classic laboratory finding used to diagnose hepatic encephalopathy, serial ammonia measurements are inferior to clinical assessment in gauging improvement or deterioration in a patient receiving treatment (Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47).

18:

With Grade II hepatic encephalopathy, patients may experience disorientation for time to 3 of 5 of the following: day of the month, day of the week, month, season, and/or year.

19:

An ileus is a blockage, partial or complete, or paralysis of the bowel that prevents the passage of intestinal contents from moving forward. It is more common in the small bowel, as opposed to large bowel. It is often the result of surgical adhesions or hernia. Small bowel ileus symptoms typically present as acute and severe, with complaints of nausea and vomiting, cramps, bloating, and the retention of stool and flatus. It can often be treated medically. Large bowel ileus usually has a milder presentation and often is due to cancer. It may also be the result of adhesions or stenoses from recurrent diverticulitis or, less commonly, is associated with a sigmoid volvulus or hernia (Vilz et al., Dtsch Arztebl Int 2017, 114: 508-18).

20:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

21:

West Haven criteria for hepatic encephalopathy (HE) includes the criterion "bizarre behavior" for Grade III HE. This behavior may be viewed as "out of sync" from societal norms; it varies from patient to patient and is often communicated to a healthcare provider by a patient's family member or caregiver (Ferenci et al., Hepatology 2002, 35: 716-21).

22:

Gross disorientation is defined as being disoriented to time and space (Biggins et al., Hepatology 2021, 74: 1014-48; Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47).

Disorientation to time includes at least 3 of the following:

- Disoriented to the date of the month
- Disoriented to the day of the week
- Unable to discern the month or season
- Disoriented to the year

Disorientation to space includes all of the following:

- Unable to identify the country
- Unable to identify the city, state, or region
- Disoriented to place

23:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

24:

A patient presenting to an acute care facility with a suspected psychiatric problem should undergo an evaluation by a qualified behavioral health professional within a specified time frame, which may be dictated by contractual, organizational, or regulatory mandate.

Instruction: This criterion is appropriate to use when a patient presents to an acute hospital and requires further monitoring and/or treatment prior to transferring to an appropriate behavioral health facility.

25:

A neurocognitive disorder is defined by a decline in cognitive function, and may result from a multitude of conditions including traumatic brain injury, Alzheimer's disease, Parkinson's disease, Huntington's disease, substance use, infection, and cerebrovascular disease (Sachdev et al., Nat Rev Neurol 2014, 10: 634-42).

26:

Agitation refers to excessive motor or verbal activity that requires psychiatric intervention to control.

27:

Unavailable or inadequate support system means the patient's support system is unable to take responsibility to provide support, supervision, and safety between outpatient appointments. A strong therapeutic relationship where the patient trusts the therapist may substitute for lack of supportive family or friends.

28:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

29:

Suicidal ideation includes not only active ideation that entails serious thoughts or plans to die by suicide but also passive ideation without an active plan, intent, or means. Passive ideation can present as a recurrent wish to die, or thoughts of dying in one's sleep (American Psychiatric Association, Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition Text Revision, DSM-5-TR, 2022; American Psychiatric Association. Practice guideline for the assessment and treatment of patients with suicidal behaviors. 2006, p. 1315-496).

30:

Antipsychotic medication administration in the setting of acute agitation or self-destructive behavior should be reserved for cases that fail to respond to non-pharmacological interventions, such as de-escalation techniques (Wilson et al., West J Emerg Med 2012, 13: 26-34).

31:

Psychiatric crisis intervention and stabilization include:

- Establishing a safety plan including removing weapons or access to drugs from the home or work setting
- Obtaining information from collateral sources (e.g., prior ED visit records, current or former treaters, primary care provider (PCP), family, significant others, law enforcement)
- Formulating and implementing both treatment and discharge plans
- Identifying and involving the patient and patient's support system in the treatment and discharge plan

32:

Pain assessment includes quality, intensity, location, frequency, duration, and exacerbating or alleviating factors. Uncontrolled pain should be monitored, assessed, and documented. Appropriate medication adjustment and other interventions should be taken to reduce pain.

33:

Pulmonary arterial hypertension (PAH) is a syndrome caused by restricted blood flow through the pulmonary arteries which leads to right ventricular hypertrophy and dilatation, and eventually to the development of right heart failure. The prognosis for PAH is poor and depending on the severity of the disease, mortality within one year of diagnosis is high (Vazquez and Klinger, Lung 2020, 198: 581-96; Hoeper et al., European Respiratory Journal 2017, 50: 1602425). PAH includes the following subgroups (Galie et al., Eur Respir J 2019, 53;; Klinger et al., Chest 2019,

155: 565-86):

- Idiopathic
- Heritable
- Drugs and toxins induced
- Associated with:

-Connective tissue disease

-Human immunodeficiency virus (HIV) infection

-Portal hypertension

-Congenital heart disease

-Schistosomiasis

Common symptoms of PAH include dyspnea, fatigue, syncope, palpitations, chest pain, and lower extremity edema. Clinical evaluation of patients suspected of having PAH includes physical examination, chest x-ray, electrocardiogram, echocardiogram, and, in some cases, cardiac magnetic resonance imaging (MRI), computed tomography (CT), or pulmonary angiography. Right heart catheterization is required to confirm a PAH diagnosis (Vazquez and Klinger, Lung 2020, 198: 581-96; Galie et al., European heart journal 2015, 37: 67-119; Klinger et al., Chest 2019, 155: 565-86).

34:

The World Health Organization classifies pulmonary arterial hypertension (PAH) into 4 functional classes (Vazquez and Klinger, Lung 2020, 198: 581-96; Klinger et al., Chest 2019, 155: 565-86):

- Class I: Patients without physical limitations
- Class II: Patients with some physical limitations, comfortable at rest, may experience dyspnea, fatigue, and chest pain with ordinary activities
- Class III: Patients with marked physical limitations, comfortable at rest, may experience dyspnea, fatigue, and chest pain with minimal activity
- Class IV: Patients with severe limitations, unable to perform any physical activity without symptoms, have signs of right heart failure at rest

35:

Therapy specific to pulmonary arterial hypertension (PAH) includes the following (Klinger et al., Chest 2019, 155: 565-86; Galie et al., Eur Respir J 2019, 53:):

- Endothelin receptor antagonists (ERA)- ambrisentan, bosentan, macitentan
- Phosphodiesterase-5 inhibitors (PDE-5I)- tadalafil, sildenafil citrate
- Prostacyclin analogues- epoprostenol, treprostinil, iloprost
- Guanylate cyclase activators- riociguat
- Prostacyclin receptor agonists- selexipag

36:

Recent guidelines define an adequate response to pulmonary arterial hypertension treatment as the ability to achieve a low-risk profile classified as being World Health Organization functional class I or II and near normalization of hemodynamic parameters. Regular assessments should be performed every 3 to 6 months in stable patients (Vazquez and Klinger, Lung 2020, 198: 581-96; Galie et al., European heart journal 2015, 37: 67-119; Galie et al., Eur Respir J 2019, 53:).

37:

There are many medications used in the treatment of pulmonary arterial hypertension (PAH). Although most are safe with only mild side effects, some, primarily the intravenous or inhalation administration of prostacyclins (e.g., epoprostenol, treprostinil, iloprost), can cause hypotension, palpitations, bradycardia, tachycardia, chest pain, dyspnea, and marked thrombocytopenia. Hemodynamic monitoring may be required due to the potential for severe adverse effects during medication initiation or dosage adjustments (Vazquez and Klinger, Lung 2020, 198: 581-96; Barnes et al., Cochrane Database Syst Rev 2019, 1: Cd012621; Galie et al., European heart journal 2015, 37: 67-119; McLaughlin et al., J Am Coll Cardiol 2015, 65: 1976-97).

38:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and

excludes the patient who has IV fluid ordered to keep vein open (KVO).

39:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

40:

For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

41:

Instruction: Selection of this criteria is appropriate for patients who have a symptom or finding not associated with a condition specific subset. Refer to the review process or subset level note for more information.

42:

Actual or suspected neurological disorders may include:

- Amyotrophic lateral sclerosis (ALS)
- Dementia (including Alzheimer's disease)
- Guillain-Barre syndrome (GBS)
- Multiple sclerosis (MS)
- Myasthenia gravis (MG)
- Parkinson's disease (PD)
- Transverse myelitis

43:

Instruction: The Observation level of care is appropriate for patients presenting with a mild deterioration of a pre-existing neurologic disorder. A more intensive level of care is appropriate for patients presenting with new or severe neurological deficit(s).

44:

Patients with multiple sclerosis (MS) may experience exacerbation of current symptoms and/or the onset of a new symptom. MS relapses may present as numbness and tingling (paresthesia), visual signs and/or symptoms, vertigo, weakness, and bladder/bowel dysfunction (Wang et al., J Immunol Clin Res 2018, 5:). The severity of the relapse

determines which level of care is appropriate (National Institute for Health and Care Excellence (NICE). Multiple sclerosis in adults: management. NG220. London: NICE; 2022).

45:

Patients presenting with myasthenia gravis exacerbation without signs and/or symptoms of respiratory distress, impending myasthenic crisis, or bulbar weakness may be managed at the Observation level of care. Patients who require respiratory monitoring or interventions, plasma exchange, or intravenous immunoglobulin (IVIG) require a more intensive level of care.

46:

Appropriate CNS imaging for this criterion include:

- Computed tomography (CT)
- Magnetic resonance imaging (MRI)

47:

Instruction: This criterion refers to a 20-30 minute routine (standard) electroencephalogram (EEG) or a prolonged EEG lasting 1-2 hours which may or may not include video.

48:

Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). In pediatric patients, functional impairment includes limitations that impact the patient's developmental progress, such as deficits in mental status, sensory functioning, communication, motor functioning, feeding, and respiratory status. The patient will require assistance with one or more limitations.

49:

Rehabilitation disciplines include:

- Occupational therapy
- Physical therapy
- Speech therapy (includes cognitive and swallowing therapy)

50:

Acute symptomatic seizures or provoked seizures are common in patients with an acute central nervous system insult. Seizure within the first 7 days post-stroke or traumatic brain injury are considered an acute symptomatic seizure. However, these patients are at a higher risk of later developing epilepsy. Seizure due to multiple sclerosis (MS) is considered an acute symptomatic seizure if it occurs as the first presenting symptom or within seven days of a relapse. If seizure recurs in patients with MS that is not associated with a relapse, as evidenced by brain imaging and/or the absence of other neurologic symptoms, epilepsy may be diagnosed. Seizure due to progressive neurodegenerative disease such as dementia is considered progressive symptomatic seizure. Epilepsy may be diagnosed after first time progressive symptomatic seizure if there is a risk of seizure recurrence greater than 60% (0.60). Patients may be admitted for seizure greater than or equal to 1 and no diagnosis of epilepsy (Mauritz et al., Epileptic Disord 2022, 24: 26-49; Johnson, Med Clin North Am 2019, 103: 309-24). These types of seizure(s) may include:

- Neurodegenerative disease progressive symptomatic seizure, new-onset (e.g., Alzheimer's disease, dementia)
- Acute symptomatic seizure or provoked seizure (e.g., multiple sclerosis, post-stroke, post-traumatic brain injury)

51:

Instruction: This criteria point may be selected for seizures witnessed in the emergency department (ED), by providers in the hospital setting, or those described pre-hospital by clinicians or non-clinicians that result in the provider ordering the required interventions. The decision to provide these interventions highlights a clinical determination that the events described were seizures and not another condition that mimics a seizure (e.g., hypoglycemia, stroke, transient ischemic attack, syncope).

52:

Following a first time seizure, patients with a normal neurological examination and negative emergent workup, may be discharged from the emergency department (ED) and complete further evaluation on an outpatient basis. However, if the patient does not regain full recovery and return to baseline, or if there are abnormalities on the seizure workup, the patient may require admission for further intervention (Wheatley and Ross, Emerg Med Clin North Am 2017, 35: 603-23).

53:

Instruction: The Observation level of care is appropriate for patients with vertigo likely caused by a peripheral process (e.g., benign paroxysmal peripheral vertigo (BPPV), Meniere's disease). A more intensive level of care may be appropriate if central vertigo is suspected based on clinical examination. Patients with multiple sclerosis (MS) may present with acute vestibular syndrome (persistent, acute onset of dizziness/vertigo) as the initial presenting symptom or as an acute exacerbation (Gurley and Edlow, Emerg Med Clin North Am 2021, 39: 181-201; Wheatley and Ross, Emerg Med Clin North Am 2017, 35: 603-23).

54:

These criteria refer to the lack of response to treatment received through the medical practitioner's office, outpatient clinic (e.g., urgent care), teleconsultation, trial of home care services, emergency department (prior to decision to admit), or hospital Observation setting (if appropriate for this condition). These criteria can be applied when the patient fails to respond to more intensive treatment beyond the routine treatments or medications delivered for a chronic condition.

55:

Migraine associated focal neurological deficits may include the following: visual deficit (e.g., diplopia, blurred vision, visual field loss), sensory deficit (e.g., numbness, tingling), speech and/or language changes (e.g., aphasia, dysarthria), and/or motor deficits (e.g., ataxia, paresis).

56:

Intractable migraine is defined as a migraine headache in which symptoms persist for more than 72 hours and is refractory to treatment.

57:

Anti-migraine agents include ergotamine and triptans.

58:

Placebo-controlled studies of intravenous dihydroergotamine (DHE) did not clearly establish its efficacy in the acute treatment of migraines (Snow et al., Ann Intern Med 2002; 137(10): 840-849). However, when combined with an antiemetic, it is an effective initial therapy (Marmura et al., Headache: The Journal of Head and Face Pain 2015, 55: 3-20; Colman et al., Ann Emerg Med 2005; 45(4): 393-401).

59:

Electronic- (E-) cigarette vaping associated lung injury (EVALI) was identified as a new pulmonary disease in early 2019 by the CDC. Criteria selection for a patient with EVALI should be based on the presenting symptoms or necessary treatments (e.g., mechanical ventilation, dehydration requiring IV fluids). Applicable criteria for review may include respiratory criteria in this subset. Criteria in the Dehydration or Gastroenteritis subset may also be appropriate. Common symptoms include cough, dyspnea, and chest pain as well as gastrointestinal symptoms including abdominal pain, nausea, vomiting, and diarrhea. Hypoxemia is seen in more than half of patients, but lung auscultation is often unremarkable.

Most patients with EVALI have required hospital admission (98%) while some remain appropriate for outpatient treatment. Nearly half have been admitted to ICU, and more than 20% have needed mechanical ventilation. Complications such as SIRS and ARDS have been reported. Work-up includes lab testing for respiratory viruses and community-acquired pneumonia pathogens, CXR, and Urine toxicology testing (including THC). Antimicrobial/antiviral agents should be considered for all patients (due to the overlap in clinical presentation of EVALI and pneumonia/influenza). Corticosteroids may benefit patients with severe disease. Pulmonologist

consultation may be useful in guiding medical treatment and possible procedures (e.g., bronchoalveolar lavage). The long-term outcome of EVALI is unknown. The importance of abstaining from E-cigarette/vape usage should be emphasized, and ambulatory care follow-up should be arranged within 1-2 weeks of discharge. (Layden et al., N Engl J Med 2019;; Siegel et al., MMWR Morb Mortal Wkly Rep 2019, 68: 919-27)

60:

Anaphylaxis is a systemic hypersensitivity reaction that ranges in severity and can lead to airway compromise and cardiac arrest. Management is based primarily on expert opinion and experience due to a limited evidence base. Prompt administration of epinephrine remains the first-line treatment in the management of anaphylaxis (Dhami et al., Allergy 2014, 69: 168-75; Sheikh et al., Allergy 2009, 64: 204-12). Repeated doses in 5-minute increments may be required to control symptoms. Secondary interventions include intravenous fluid, supplemental oxygen administration, antihistamines, corticosteroids, and inhaled short-acting bronchodilators or epinephrine. Subsequent observation and management are dependent on the severity of the reaction, response to treatment, and risk of recurrence. Guidelines recommend that observation periods following the initial anaphylactic episode be individualized. Prolonged monitoring is recommended for patients with severe reactions upon presentation or risk of recurrence (Boyce et al., J Allergy Clin Immunol 2010, 126: S1-58; Lieberman et al., J Allergy Clin Immunol 2010, 126: 477-80 e1-42; Muraro et al., Allergy 2014, 69: 1026-45).

61:

Hemodynamic stability is evidenced by adequate tissue and organ perfusion. Blood pressure and heart rate are parameters commonly used to assess hemodynamic stability. Clinical findings associated with hemodynamic stability include normal peripheral pulses, warm extremities, adequate urine output, and neurological stability. The Critical level of care is appropriate for patients who develop hemodynamic instability.

62:

Instruction: This criterion applies to patients requiring observation after receiving treatment for an acute reaction with demonstrated improvement. For patients that are hypoxic or hemodynamically unstable, consider a more intensive level of care.

63:

Anaphylaxis that recurs after apparent resolution is termed a biphasic reaction. Most biphasic reactions occur within 8 to 10 hours after the initial onset of symptoms (Lieberman, Am J Med 2014, 127: S6-11; Lieberman et al., J Allergy Clin Immunol 2010, 126: 477-80 e1-42).

64:

Continuous oximetry should be used to guide the need for supplemental oxygen therapy in patients presenting with anaphylaxis (Lieberman et al., J Allergy Clin Immunol 2010, 126: 477-80 e1-42).

65:

H₁ and H₂ antihistamines are administered after epinephrine to control cutaneous and cardiovascular symptoms associated with anaphylaxis (Boyce et al., J Allergy Clin Immunol 2010, 126: S1-58; Lieberman et al., J Allergy Clin Immunol 2010, 126: 477-80 e1-42; Muraro et al., Allergy 2014, 69: 1026-45).

66:

Corticosteroids are administered to minimize the risk of biphasic reaction or protracted anaphylaxis (Boyce et al., J Allergy Clin Immunol 2010, 126: S1-58; Lieberman et al., J Allergy Clin Immunol 2010, 126: 477-80 e1-42; Muraro et al., Allergy 2014, 69: 1026-45).

67:

Angioedema is characterized by acute localized edema of the upper respiratory or gastrointestinal tract. In the hospital setting, maintaining the airway is the primary goal. Subsequent pharmacological treatment depends on the type of angioedema (Jaiganesh et al., Eur J Emerg Med 2013, 20: 10-7; Wilkerson, Emerg Med Pract 2012, 14: 1-21).

68:

The treatment of acute angioedema is dependent on the cause. Antihistamines are effective in the treatment of allergen-induced angioedema, which involves mast cell degranulation. For bradykinin-mediated, hereditary, or acquired angioedema, immunomodulator therapy is recommended and includes C1 inhibitor replacement, as well as bradykinin inhibitors or receptor antagonists (Lang et al., Ann Allergy Asthma Immunol 2012, 109: 395-402; Jaiganesh et al., Eur J Emerg Med 2013, 20: 10-7; Wilkerson, Emerg Med Pract 2012, 14: 1-21).

69:

Fresh frozen plasma may be used in the treatment of acute hereditary angioedema when immunomodulators are not available. Evidence of efficacy is limited and caution should be used due to the possibility of worsening symptoms (Lang et al., Ann Allergy Asthma Immunol 2012, 109: 395-402; Wilkerson, Emerg Med Pract 2012, 14: 1-21).

70:

Corticosteroids are administered to prevent a recurrence of the angioedema episode. Although limited evidence exists, corticosteroids are considered a reasonable secondary treatment due to the low risk of harm (Wilkerson, Emerg Med Pract 2012, 14: 1-21).

71:

Following an acute episode of angioedema, indications for prolonged observation include severe presenting symptoms and lack of response to treatment. Severe presenting symptoms may include anaphylaxis, voice change, hoarseness, stridor, and dyspnea (Wilkerson, Emerg Med Pract 2012, 14: 1-21).

72:

Instruction: This criterion applies to those patients that demonstrate severe symptoms prior to treatment. At the time the decision is made to prolong monitoring at the Observation level of care, symptoms should be resolving post-treatment.

73:

Continuous oximetry is indicated in the setting of acute angioedema of the upper respiratory tract to monitor oxygenation status and guide the need for supplemental oxygen therapy (Wilkerson, Emerg Med Pract 2012, 14: 1-21).

74:

B-type natriuretic peptide (BNP) level is an adjunctive test for the evaluation of acute dyspnea and is useful in distinguishing dyspnea due to heart failure (HF) from that caused by intrinsic pulmonary disease. To date, however, no clear guidelines exist to aid in the interpretation of BNP levels and additional clinical studies are needed. BNP results should not be used to determine whether a hospital admission is warranted or to determine if a specific level of care is indicated. Preliminary evidence suggests that levels less than 100 pg/mL exclude HF as a cause of dyspnea. BNP levels between 100 and 500 pg/mL are indeterminate, and BNP levels greater than 500 pg/mL are likely to be associated with HF. However, elevated BNP levels have been identified in patients without HF. Associated conditions include pulmonary embolism, cor pulmonale, and advanced renal insufficiency (de Denus et al., Chest 2004; 125(2): 652-668; Mueller et al., N Engl J Med 2004; 350(7): 647-654; Rodeheffer, J Am Coll Cardiol 2004; 44(4): 740-749).

75:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

76:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is

required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO_2 . The following are estimates of O_2 delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula

Room air 21%(0.21)

1L 24%(0.24) 4L 36%(0.36)

2L 28%(0.28) 5L 40%(0.40)

3L 32%(0.32) 6L 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO_2 (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO_2 in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO_2 (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO_2 (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO_2 (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS

Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO_2

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO_2

77:

A pneumothorax is the accumulation of air within the pleural space. Types of pneumothorax include:

- Spontaneous - occurs without a preceding event
- Traumatic - occurs as a result of a physical chest injury either penetrating (open) or nonpenetrating (closed) or following a surgical procedure
- Tension - occurs when air moves into the pleural space causing a mediastinal shift to the uninvolved side and may lead to cardiovascular collapse

78:

Instruction: Spontaneous pneumothorax is reviewed in the General Medical subset. Traumatic pneumothorax is to be reviewed in the General Trauma subset.

79:

Clinically stable patients who have experienced a small/stable pneumothorax can be discharged home based on clinical judgment and repeated chest x-ray excludes progression of the pneumothorax and symptoms are resolving. Patients requiring the placement of a small bore ($\leq 14\text{F}$) chest tube whose symptoms have resolved may be discharged after stable repeat chest x-ray. Frequently, these patients are otherwise young and healthy with no other comorbidities or underlying lung disease. If discharged with a small bore chest tube, these patients require thorough discharge instructions, chest tube or Heimlich valve care instruction and a follow up plan of care (Sale et al., Respir Med 2020, 166: 105931; Tschopp et al., Eur Respir J 2015, 46: 321-35).

80:

- Small bore chest tubes are $\leq 14\text{F}$
- Large bore chest tubes are $> 14\text{F}$

81:

Small bore chest tubes are less than or equal to 14F and large bore chest tubes are greater than 14F (Mummadi et

al., Ann Emerg Med 2020, 76: 88-102; Chang et al., Chest 2018, 153: 1201-12).

82:

Criteria located in the "Toxic exposure or ingestion" category is not inclusive of all potential toxic effects. When criteria within this category do not reflect the patient's clinical signs or symptoms, refer to an alternative category within the General Medical subset according to the findings or interventions driving the hospitalization. For example, if a patient hospitalized with neurotoxicity does not meet the "Toxic exposure or ingestion" criteria, refer to the "Neurological" category within the General Medical subset.

83:

Agitation refers to combative or resistive behaviors that sometimes require medical or psychiatric intervention in a safe environment.

84:

Confusion is characterized by fluctuating levels of consciousness and attention manifested by the inability to maintain a coherent stream of thought or action.

85:

Disorientation refers to confusion about who the person is, place, and time (American Psychiatric Association, Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition Text Revision, DSM-5-TR, 2022).

86:

A neurological assessment establishes a baseline so that subtle changes can be monitored. A comprehensive neurological assessment often includes an evaluation of:

- Mental status (e.g., level of consciousness, orientation, insight, calculation ability)
- Cranial nerves (e.g., olfactory, optic, vestibulocochlear)
- Motor system (e.g., muscle tone, strength, reflexes)
- Sensory system (e.g., light touch, pain, temperature)
- Coordination (e.g., orchestration and fluidity of movement)
- Gait (e.g., heel-to-toe straight-line walking)

87:

Toxic substance ingestion refers to the accidental or intentional ingestion of a:

- Substance that is toxic in any amount (e.g., lye, bleach, anti-freeze, or other chemicals or household cleaners)
- Medication (prescribed or over the counter) taken over a prolonged period of time or in a dosage that reaches a toxic level
- Substance of abuse; including, alcohol, cannabis, hallucinogens, phencyclidine (PCP), inhalants, steroids, or stimulants that are ingested to a toxic level for that individual

88:

Initial management of a patient who has ingested a potentially toxic substance incorporates the following ABCD₃EF assessment (Calello and Henretig, Emerg Med Clin North Am 2014, 32: 29-52; Thim et al., Int J Gen Med 2012, 5: 117-21):

- **Airway** (protection and monitor for possible compromise)
- **Breathing** (ensure adequate oxygenation and ventilation)
- **Circulation** (close monitoring of vital signs or capillary perfusion and assess the need for intravenous access)
- **Disability** (determine level of consciousness, pupillary size and reactivity)
- **Drugs** (as needed administration of dextrose, oxygen, naloxone, or other emergency resuscitation drugs)
- **Decontamination** (removal of toxic substances from skin, eyes, or gastrointestinal tract)
- **Electrocardiogram** (establish baseline of rate, rhythm, QRS width, QTc length)
- **Fever** (monitor for hypothermia or hyperthermia)

89:

Guidelines indicate that warfarin should be initiated on day 1 or 2 of intravenous (IV) anticoagulation therapy in patients with thromboembolic disease, unless contraindicated (Holbrook et al., Chest 2012, 141: e152S-84S).

90:

Continuous unfractionated heparin is the first-line treatment for acute limb ischemia (ALI) or peripheral artery occlusion unless contraindicated. Goals of care include timely assessment for viability by vascular surgeon, immediate anticoagulation after imaging confirmation, and evaluation for urgent revascularization, especially when neurological deficit is present (Bjorck et al., Eur J Vasc Endovasc Surg 2020, 59: 173-218; Aboyans et al., Eur J Vasc Endovasc Surg 2018, 55: 305-68; Gerhard-Herman et al., Circulation 2017, 135: e686-e725).

91:

Heparin-induced thrombocytopenia (HIT) results from sensitization to the heparin molecule. Usually, it occurs between 5 to 10 days after the current heparin exposure but may occur as early as minutes to hours after exposure. HIT is a greater than 50%(0.50) serial reduction in a patient's baseline platelet count. A direct thrombin inhibitor (DTI) (e.g., argatroban, bivalirudin) is the initial treatment for patients with active HIT, a history of HIT, or those with a strong clinical suspicion of HIT who require anticoagulation. Although the American Society of Hematology (ASH) guidelines suggest the use of fondaparinux or direct oral anticoagulants (DOACs), the recommendation is conditional and based on very low certainty of evidence. In addition, these agents are considered off-label use in the treatment of HIT since they have yet to be approved by the U.S. Food and Drug Administration (FDA) for this condition. For patients who require long-term anticoagulation, warfarin should not be initiated until platelet count recovers (e.g., platelet count greater than $150 \times 10^9/L$). Additionally, a 5-day overlap with warfarin and DTI therapy is recommended (Cuker et al., Blood Adv 2018, 2: 3360-92; Guyatt et al., Chest 2012, 141: 48S-52S).

92:

A neurovascular assessment is done to identify any signs of neurovascular compromise and should include the following:

- Evaluation of the quality of the peripheral pulses and capillary refill rate
- Pain assessment
- Assessment for warmth in the extremity
- Gross motor and sensory function of fingertips and toes

Presence of any of the following signs suggests neurovascular compromise: painful passive motion, pallor of the extremity, paresthesias, paralysis, pulselessness or diminished pulses, or poikilothermia (fluctuating temperature) (Gerhard-Herman et al., Circulation 2017, 135: e686-e725).

93:

Instruction: Inpatient medical stabilization is required in an acute care facility when the level of acuity necessary for the medical complications of an eating disorder is greater than the capability of the inpatient psychiatric facility (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023).

94:

Although no standardized set of indications for hospitalization of patients with eating disorders exists, recommendations may be found within society guidelines (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023). Indications generally include significant weight loss, vital sign instability, dehydration, hypothermia, electrolyte imbalances, and cardiac rhythm disturbances and their potential complications (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023; Mehler et al., J Eat Disord 2023, 11: 16; Ralph et al., J Eat Disord 2022, 10: 121).

95:

According to the American Psychiatric Association, glucose of less than 60 mg/dL is a factor supporting hospitalization for adults on a specialized eating disorder unit (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023).

Hypoglycemia is a common complication of anorexia nervosa (AN) and is linked to severity of disease. It is considered to be a negative prognostic indicator for patients with serious eating disorders and may be a precursor

of serious conditions including, but not limited, to hepatic failure which may result in sudden death (Puckett et al., J Clin Med 2021, 10;; Jeana et al., Cleveland Clinic Journal of Medicine 2020, 87: 361; Chidiac, Curr Opin Pediatr 2019, 31: 448-53; Rosen et al., Dig Dis Sci 2017, 62: 2977-81; Mehler and Brown, J Eat Disord 2015, 3: 11).

96:

Frequent point-of-care blood glucose testing is necessary for patients during refeeding or for patients with eating disorders experiencing hypoglycemia (i.e., low blood sugar) (Mehler et al., J Eat Disord 2023, 11: 16; Royal College of Psychiatrists UK, Medical emergencies in eating disorders (MEED). 2022).

97:

Inpatient treatment of eating disorders is indicated for a blood pressure less than 90/60 in adults (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023; Pompili et al., Int J Environ Res Public Health 2021, 18;; Hay et al., Aust N Z J Psychiatry 2014, 48: 977-1008; Pritts and Susman, American Academy of Family Physicians 2003, 67: 297-304).

98:

Chronic ingestion of ipecac, which is used by patients with eating disorders to induce vomiting, is associated with serious cardiac toxicity that can cause damage to the myocardium. Emetine is an alkaloid found in ipecac which is cardiotoxic. Much of the damage this causes is the result of cumulative exposure from repeated use (Mehler et al., J Eat Disord 2023, 11: 16).

99:

Cardiomyopathy can be a complication of anorexia nervosa and severe malnutrition. Although not as frequent, this complication requires medical attention and monitoring (Mehler et al., J Eat Disord 2023, 11: 16).

100:

There are medical conditions that may be sequelae of an eating disorder. These include, but are not limited to, gastroesophageal reflux disease, gastroparesis, irritable bowel syndrome, or other gastrointestinal (GI) motility disorders (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023).

Additionally, there are other co-occurring health conditions including, but not limited to, celiac disease or inflammatory bowel disease, which may place restrictions on an individual's eating behaviors and available dietary options. These other health conditions may exacerbate an eating disorder or increase the likelihood an individual developing an eating disorder (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023).

101:

Patients with anorexia nervosa may have renal insufficiency identified by an abnormal creatinine level possibly secondary to moderate to severe dehydration requiring inpatient admission. Medical refeeding and IV fluid support could alleviate and improve renal function (Downey et al., J Adolesc Health 2022;).

102:

Although society guidelines may vary, it is recommended that patients with an eating disorder who present with dehydration and an elevated heart rate of greater than 100 beats per minute be admitted for monitoring and hydration (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023; Royal College of Psychiatrists UK, Medical emergencies in eating disorders (MEED). 2022).

103:

The American Psychiatric Association recommends that laboratory assessment and monitoring of a patient with an eating disorder include a comprehensive metabolic panel with electrolytes such as magnesium and phosphorus (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023). Electrolyte imbalances may be the result of many factors related to eating disorders including, but not limited to,

diuretic use, laxative abuse, restricted dietary intake, or vomiting. Once treatment has been initiated in these patients, risk for refeeding syndrome must be carefully monitored which includes daily labs with particular attention for hypophosphatemia, hypokalemia and/or hypomagnesemia. Additionally, patients with confirmed diagnosis of malnutrition who have started treatment are at risk for refeeding syndrome (Stoody et al., J Pediatr 2023; 113482; Peebles and Sieke, Child Adolesc Psychiatr Clin N Am 2019, 28: 593-615).

104:

Fluid, supplements, or medication to correct electrolyte abnormalities may include, but are not limited to, potassium supplements given to correct hypokalemia, phosphate replacement to correct refeeding hypophosphatemia, fluid given to correct hypernatremia, and certain medications used to treat specific electrolyte abnormalities (Peebles and Sieke, Child Adolesc Psychiatr Clin N Am 2019, 28: 593-615; Society for Adolescent Health and Medicine, The Journal of adolescent health: official publication of the Society for Adolescent Medicine 2014, 55: 455-7).

105:

The Academy for Eating Disorders (AED) suggests that patients who have attempted lower levels of care and continue to struggle with eating disorder symptoms may require inpatient care (Academy for Eating Disorders, Eating Disorders: A Guide to Medical Care, 3rd edition. 2016).

106:

Society guidelines recommend that patients with a known eating disorder presenting with a sustained heart rate of less than 50 beats per minute be admitted for inpatient treatment (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023; Royal College of Psychiatrists UK, Medical emergencies in eating disorders (MEED). 2022).

107:

Inpatient treatment is indicated for a patient with an eating disorder whose systolic blood pressure drops 20 mmHg or more upon sitting or standing (Hay et al., Aust N Z J Psychiatry 2014, 48: 977-1008; Pritts and Susman, American Academy of Family Physicians 2003, 67: 297-304).

108:

Inpatient treatment is indicated for a patient with an eating disorder whose diastolic blood pressure drops 10 mmHg or more upon sitting or standing (Hay et al., Aust N Z J Psychiatry 2014, 48: 977-1008; Pritts and Susman, American Academy of Family Physicians 2003, 67: 297-304).

109:

Inpatient treatment is indicated for an adult patient with an eating disorder who has orthostatic changes in heart rate greater than 30 beats per minute (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023).

110:

Guidelines recommend that patients with an eating disorder who present with a QTc of greater than 450 milliseconds be admitted for inpatient treatment (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023).

111:

Patients who are very thin and malnourished are prone to develop cold intolerance due to the loss of body fat and the prevalence of peripheral vascular abnormalities (Friars et al., J Eat Disord 2023, 11: 13). Inpatient treatment is indicated for adults with eating disorders with an oral temperature of less than 96.8°F or 36° C (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023).

112:

Although body mass index (BMI) alone is not an indicator of overall health and well-being, patients with a BMI of less than 15 require close monitoring and initiation of a specialized nutritional plan by a dietitian (American

Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023).

113:

Guidelines recommend that patients who have rapidly lost weight (i.e., greater than 10%(0.10) weight loss in the last 6 months or greater than 20%(0.20) weight loss in the last year) be admitted for close monitoring and nutritional management (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023).

114:

A complete panel of all electrolytes is not required. This criterion is met if the specific electrolytes that were abnormal are checked.

115:

Observation of the patient may be required to ensure tube feeds and/or oral feeds are not interrupted. Nasogastric feeding (tube feeding) is recommended in individuals who refuse to eat and require life-preserving nutrition (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023).

116:

Literature is inconsistent in support of a frequency of hypoglycemia monitoring for patients with eating disorders. However, clinicians suggest that blood glucose levels are monitored before the three meals of the day and at bedtime.

117:

Complete bedrest is an activity restriction placed on those patients who are experiencing unstable medical complications as a result of their eating disorder.

Medically compromised patients with eating disorders admitted to the inpatient level of care may be placed on strict bedrest until vital signs improve (Derenne, Child Adolesc Psychiatr Clin N Am 2019, 28: 573-82). Once vital signs have normalized and sustained, patient's activity level may be upgraded to standard activity (Derenne, Child Adolesc Psychiatr Clin N Am 2019, 28: 573-82).

118:

For patients who have been purging, accompanied or monitored bathroom visits are necessary to prevent use of the bathroom for purging, which can impair weight gain. For patients who have been restricting, accompanied or monitored bathroom visits are necessary to prevent feigning of weight gain by deliberate water loading or by hiding of weight-bearing objects in the mouth or other body orifices.

119:

Nutritional counseling and assessment are important components of eating disorder treatment (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023). The initial nutritional assessment should be done by a dietician. Subsequent daily nutritional assessments may be done by any clinician, including nurses or other clinical staff, as long as a dietician does a nutritional assessment at least once a week (Gbareen et al., BMC Nurs 2021, 20: 157; Reber et al., J Clin Med 2019, 8:).

120:

Re-evaluation of the effectiveness of the individualized, recovery-oriented treatment plan should be done by the physician or the clinical staff, and includes (Coffey et al., BMC Psychiatry 2019, 19: 115; Mee-Lee et al., The ASAM criteria: treatment criteria for addictive, substance-related, and co-occurring conditions, 3rd Ed. ed. 2013):

- Involvement of the patient and their formal or informal support system (if appropriate)
- Modification of the individualized, recovery-oriented treatment based upon the patient response to the plan

121:

Treatment-specific meal plans teach patients healthy eating patterns. A treatment-specific meal plan is tailored to caloric needs and includes several nutritionally balanced meals and snacks daily. During eating disorder treatment, staff may discourage the use of eating disordered behaviors (i.e., restriction or compensatory behaviors) by providing supervision before (i.e., food portioning), after (i.e. monitoring bathroom use), and during meal and snack consumption (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023; Levinson et al., Behav Res Ther 2018, 105: 36-42; Lock et al., J Am Acad Child Adolesc Psychiatry 2015, 54: 412-25; Long et al., Int J Eat Disord 2012, 45: 241-6; Pritts and Susman, American Academy of Family Physicians 2003, 67: 297-304).

122:

Abdominal ultrasound is the first line imaging study to evaluate suspected biliary obstruction, cirrhosis, or vessel patency. Computed tomography (CT) or magnetic resonance imaging (MRI) is usually performed if the ultrasound is non-diagnostic. More advanced imaging includes magnetic resonance cholangiopancreatography (MRCP) and endoscopic retrograde cholangiopancreatography (ERCP), both allowing for the visualization of the intra- and extrahepatic biliary tree. The advantage of the ERCP is that a stent may be placed if an obstruction is encountered (Fargo et al., Am Fam Physician 2017, 95: 164-8).

123:

The following list of generic medications may be prescribed for nausea and/or vomiting: ondansetron, promethazine, metoclopramide, prochlorperazine, meclizine, lorazepam, hydroxyzine, scopolamine, dimenhydrinate or diphenhydramine, haloperidol, trimethobenzamide, chlorpromazine, droperidol, perphenazine, pyridoxine, or doxylamine/pyridoxine.

124:

Serotonin 5-HT₃ receptor antagonists are typically given to help prevent nausea and vomiting. Examples of these drugs include: ondansetron, dolasetron, granisetron, and palonosetron.

125:

Gastroparesis is a chronic disorder of gastric motility characterized by delayed gastric emptying in the absence of mechanical obstruction.

126:

Gastroparesis is a common complication of long-standing (10 years or longer) diabetes mellitus (DM) and is believed to be a form of neuropathy affecting the vagus nerve (Krishnasamy and Abell, Diabetes Therapy 2018, 9: 1-42). Although more common in Type 1 DM, the increasing prevalence of Type 2 DM has resulted in larger numbers of patients with Type 2 MD who have gastroparesis. Hyperglycemia is thought to be an aggravating factor, while optimal glucose control can reduce gastrointestinal symptoms (Camilleri et al., Am J Gastroenterol 2022, 117: 1197-220).

127:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirement. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit.

128:

Recommended dietary modifications for patients with gastroparesis include eating frequent, smaller sized meals, replacing solid foods with liquid nutritional supplements, and adhering to a low fat, low fiber diet. In addition to dietary modifications, cessation of alcohol and tobacco use is recommended, as their use is associated with slowed gastric emptying (American Diabetes Association Professional Practice, Diabetes Care 2024, 47: S1-S321; Camilleri et al., Am J Gastroenterol 2022, 117: 1197-220).

129:

Medications such as opiates, certain diabetes medications, muscle relaxants, cyclosporine, and anticholinergic agents can cause delayed gastric emptying (Zhao and Mashimo, Curr Treat Options Gastroenterol 2015, 13: 452-72;

Camilleri et al., The American journal of gastroenterology 2013, 108: 18). Treatment will likely not be effective if tried while the patient is on any of these medications.

130:

Prokinetic agents (e.g., metoclopramide, erythromycin) and antiemetics (e.g., promethazine, ondansetron) are currently the mainstay of medical treatment for symptomatic gastroparesis (Asha and Khalil, Sultan Qaboos Univ Med J 2019, 19: e291-e304; Navas et al., Dig Dis Sci 2017:).

Domperidone (Motilium), which is both a prokinetic and antiemetic medication, can be used for the treatment of gastroparesis but can only be prescribed under a special FDA-approved program, as this drug has been shown to dangerously prolong QT intervals (Camilleri et al., Am J Gastroenterol 2022, 117: 1197-220; U.S. Food and Drug Administration (FDA), How to request Domperidone for expanded access use. 2021; Zhao and Mashimo, Curr Treat Options Gastroenterol 2015, 13: 452-72).

131:

The most reliable method for diagnosis of gastroparesis is gastric scintigraphy (Stein et al., J Clin Gastroenterol 2015, 49: 550-8). An alternative to scintigraphy is the use of a wireless motility capsule to quantify luminal pH and pressure, or breath test for gastric emptying (Pasricha and Parkman, Gastroenterol Clin North Am 2015, 44: 1-7). If a diagnosis of gastroparesis is questionable, additional imaging studies or upper endoscopy may be required to exclude a mechanical obstruction.

132:

Radionuclide gastric emptying scintigraphy is the standard imaging study used to establish the diagnosis of gastroparesis. Patients consume a radiolabeled meal and following ingestion, gastric emptying is measured at various intervals. A 4-hour assessment period is considered the minimal timeframe for measuring gastric emptying, although there is a lack of standardization of procedure protocols and outcome measurements (Stein et al., J Clin Gastroenterol 2015, 49: 550-8; Camilleri et al., The American journal of gastroenterology 2013, 108: 18).

133:

Guidelines recommend deferring gastric emptying testing until relative euglycemia (blood glucose < 275 mg/dL (15.3 mmol/L)) is achieved, as hyperglycemia (glucose levels > 200 mg/dL (11.1 mmol/L)) will delay gastric emptying (Camilleri et al., The American journal of gastroenterology 2013, 108: 18).

134:

Instruction: Patients with end-stage liver disease are generally on chronic outpatient lactulose, rifaximin, or neomycin at home to control symptoms of hepatic encephalopathy (HE). Admission to a skilled nursing facility (SNF) may be appropriate to initiate or adjust the dosage of these medications due to worsening hepatic encephalopathy.

135:

Oral branched-chain amino acid (BCAA) enriched formulations may improve the manifestations of occult or minimal episodic hepatic encephalopathy (HE). Intravenous administration is ineffective (Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47; J Hepatol 2014, 61: 642-59). Although BCAA has a beneficial effect on HE, there is no evidence regarding an effect on mortality, quality of life, or nutritional parameters (Gluud et al., Cochrane Database Syst Rev 2017, 5: CD001939).

136:

Lactulose is recommended by the American Association for the Study of Liver Diseases/European Association for the Study of the Liver guidelines as the first-line treatment for episodes of hepatic encephalopathy. It is a nonabsorbable substance used to treat an elevation in blood ammonia (NH₃) and reduces plasma ammonia concentration by converting soluble ammonia into insoluble ammonia in the colon. The laxative effect of lactulose results in the expulsion of trapped ammonium in the form of diarrhea. Lactulose can be administered orally, via a nasogastric tube, or as a retention enema (Swaminathan et al., Hepat Med 2018, 10: 1-11; J Hepatol 2014, 61: 642-59).

137:

Evidence has shown that L-ornithine L-aspartate (LOLA) is effective in lowering ammonia and improving mental status in patients with cirrhosis and hepatic encephalopathy (HE), with the oral form of LOLA being especially beneficial (Butterworth et al., J Clin Exp Hepatol 2018, 8: 301-13). In a 2017 randomized, controlled trial, patients with persistent HE showed improvement from intravenous LOLA administration, whereas oral supplementation was determined to be ineffective (Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47). A recent systematic review reassessed both oral and intravenous (IV) LOLA for the treatment of low-grade or minimal hepatic encephalopathy (MHE) and high grade or overt hepatic encephalopathy (OHE). The relative efficacy of LOLA depended on the type of HE as well as the route of administration. Patients with MHE, defined as patients with cirrhosis and alterations in mental status diagnosed by psychometric testing, appear to benefit more from the oral administration of LOLA than patients with high grade overt HE (Weissenborn, Drugs 2019, 79: 5-9; Butterworth et al., J Clin Exp Hepatol 2018, 8: 301-13).

138:

The renal and ototoxicity of neomycin have rendered this medication nearly obsolete in the treatment of hepatic encephalopathy (Tabbers et al., J Pediatr Gastroenterol Nutr 2014, 58: 258-74).

139:

Rifaximin is an anti-infective recommended by evidence-based clinical guidelines as an effective add-on therapy to lactulose for the treatment of overt hepatic encephalopathy (OHE). It has been shown to reduce healthcare resource utilization over the long term by lessening OHE recurrences and rehospitalization (Hudson and Schuchmann, European journal of gastroenterology & hepatology 2019, 31: 434). In a systematic review comparing the effectiveness and safety of interventions in patients with OHE, Rifaximin showed the greatest reduction in blood ammonia concentration (Zhu et al., Aliment Pharmacol Ther 2015, 41: 624-35).

140:

Imaging studies used in the diagnosis of ileus include abdominal computed tomography (CT) scan and abdominal x-ray.

141:

Hydronephrosis is the dilation of the pelvis of one or both kidneys resulting from the obstruction of the flow of urine.

142:

Instruction: For patients with this condition who undergo surgical intervention, refer to Operative Day of the General Surgical subset and conduct a review.

143:

The inpatient treatment of priapism should begin with alpha-adrenergic sympathomimetic agent (e.g., phenylephrine, epinephrine, norepinephrine, ephedrine, and metaraminol) administration via irrigation or intracavernous injection. Phenylephrine is preferred in order to reduce the occurrence of cardiovascular side effects. Intracavernous injection should be initiated if priapism does not resolve following irrigation. Surgical intervention is reserved for cases that do not respond to a trial of intracavernous injection. The decision to proceed with surgery depends on the duration of the priapism. After 48 to 72 hours of priapism onset, successful resolution with intracavernous injection is unlikely and surgical intervention is often required (American Urologic Association, Guideline on the management of priapism, 2003 (cited Feb 2020)).

144:

Hyperviscosity syndrome is caused by elevated immunoglobulin levels which coat red and white blood cells, causing increased blood viscosity, sludging of blood and hypoperfusion. It can be seen in diseases such as multiple myeloma, leukemia, polycythemia, and other myeloproliferative disorders. Symptoms of hyperviscosity syndrome include headache, tinnitus, stupor, coma, and uncontrolled or recurrent epistaxis. In the presence of symptoms, therapeutic plasma exchange is indicated (Halaseh et al., QJM 2022, 115: 30-1; Emerson and Kaide, Emerg Med Clin North Am 2018, 36: 603-8; Gertz, Blood 2018, 132: 1379-85).

145:

Apheresis is a procedure that separates and removes elements in the blood (e.g., white blood cells (WBCs), platelets, stem cells, plasma bound organic compounds, or antibodies). Subsequent replacement with blood components, solvent detergent plasma (SDP), albumin, and/or normal saline may follow apheresis.

146:

Instruction: For known Von Willebrand disease, use the Anemia/Bleeding subset.

147:

Intravenous immunoglobulin (IVIG) is a blood product which contains immunoglobulins extracted, then pooled, from the plasma of thousands of blood donors. Its administration carries the same risks as those of any blood product transfusion, including headache, backache, chills, nausea, muscle pain, and, although rare, anaphylaxis. IVIG is used for a wide range of disorders, and its therapeutic mechanism varies based on the dose given and the patient's disease. IVIG is used for the treatment of diseases in neurology, hematology, immunology, dermatology, and nephrology.

148:

Dysphagia is an impairment in swallowing which may lead to aspiration and malnourishment. Neurological conditions that may cause dysphagia include (McCarty and Chao, Med Clin North Am 2021, 105: 939-54; Panebianco et al., Neurol Sci 2020, 41: 3067-73):

- Amyotrophic lateral sclerosis (ALS)
- Multiple sclerosis
- Multiple system atrophy
- Parkinson's disease

A more intensive level of care is appropriate for patients with myasthenia gravis presenting with an acute exacerbation of dysphagia due to bulbar weakness as this may be an indication of potential respiratory failure (i.e., myasthenic crisis) (Roper, The Journal of Emergency Medicine 2017, 53: 843-53).

Instruction: Patients experiencing dysphagia from a non-neurologic cause are excluded from these criteria.

149:

Paresis refers to a reduction in voluntary movement while paralysis refers to a complete loss of movement. These criteria include paresis or paralysis of extremities and the face. Myasthenia gravis typically presents as bulbar weakness, ptosis, and muscle weakness that worsens with repetition (fatigable). A more intensive level of care is appropriate for patients with myasthenia gravis presenting with an acute exacerbation of bulbar weakness or any indication of potential respiratory failure (i.e., myasthenic crisis) (Khamess, Emergency Medicine Clinics of North America 2020, 39: 173-80; Roper, The Journal of Emergency Medicine 2017, 53: 843-53).

Instruction: Patients experiencing general weakness or inability to move limbs from a non-neurologic cause are excluded from these criteria.

150:

InterQual® external peer reviewers agree that patients who experience more than two seizures within 24 hours require further intervention in a hospital setting.

151:

Instruction: This criterion refers to a long-term, continuous electroencephalogram (EEG) which may last less than or equal to 72 hours and may or may not include video.

152:

Guillain-Barre (GBS) syndrome is a rapidly progressive inflammatory disease of the peripheral nervous system. Patients with GBS syndrome often present with hyporeflexia (decreased) or areflexia (absent). There are several variants of GBS in which normal deep tendon reflexes or hyperreflexia (overresponsive) may be present (Shahrizaila et al., Lancet 2021, 397: 1214-28; Leonhard et al., Nat Rev Neurol 2019, 15: 671-83). Transverse myelitis is an inflammatory condition of the spinal cord causing rapid (hours to days) onset of symptoms. Acutely,

hyporeflexia may be present (Smith and Siket, Emerg Med Clin North Am 2020, 38: 523-37).

153:

New onset diplopia, or double vision, may be the presenting symptom of neurological disorders such as (Norse, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:158-66; Margolin and Lam, J Emerg Med 2018, 54: 799-806):

- Miller Fisher syndrome (a variant of Guillain-Barre syndrome)
- Multiple sclerosis
- Myasthenia gravis

154:

Postictal period refers to a variable time interval following a seizure, usually characterized by impaired consciousness (excludes lethargy), and is sometimes also marked by self-limited focal paralysis (Subota et al., Epilepsy Behav 2019, 94: 243-51). Following a seizure, most neurological deficits should resolve within 60 minutes. If deficits persists longer than 60 minutes, nonconvulsive status epilepticus should be suspected (Pottkmper et al., Epilepsia 2020, 61: 1045-61).

155:

Patients with seizure(s) may require a planned, elective admission to the epilepsy monitoring unit (EMU) for continuous electroencephalography (EEG) and video monitoring. EMU length of stay varies based on the reason for admission and seizure frequency. Admission to the EMU may be indicated for (Shih et al., Epilepsia 2018, 59: 27-36):

- Classifying seizure type
- Determining seizure frequency
- Differential diagnosis of epileptic seizures versus other etiologies
- Differentiating between seizures versus medication side effects
- Medication adjustment

156:

Seizure evaluation admission may be indicated for:

- Classifying seizure type
- Determining seizure frequency
- Differential diagnosis of epileptic seizures versus other etiologies
- Differentiating between seizures versus medication side effects
- Medication adjustment

157:

Instruction: Application of this criteria is appropriate for patients who are receiving care in the hospital or at a lower level of care, including the home setting, and experience a deterioration of their condition which leaves the caregiver unable to manage care. This criteria may be appropriate while transfer to a lower level of care is arranged.

158:

Instruction: These criteria are meant to include patients who have had a chest tube placed during their current admission, not for patients who had a chest tube placed during a prior episode of care.

159:

Immunosuppressant medications are used to treat conditions such as graft-versus-host disease (GVHD), hemolytic anemia, organ transplant, graft failure, rejection, inflammatory bowel disease, acute glomerulonephritis, and inflammatory cellulitis. Medications used for immunosuppression include prednisone, prednisolone, cyclosporine, azathioprine, mycophenolate mofetil, tacrolimus, sirolimus, everolimus, belatacept, antithymocyte globulin (ATG), and basiliximab.

160:

Inflammatory dermatitis is a cutaneous hypersensitivity reaction that may be triggered by a wide variety of stimuli

including infections, rheumatologic diseases, inflammatory bowel disease, medications, autoimmune disorders, pregnancy, or malignancy. Inpatient treatment of inflammatory dermatitis with corticosteroids or immunosuppression may be required in severe cases or when outpatient treatment is ineffective (Requena and Yus, *Dermatol Clin* 2008; 26(4): 425-438, v; Lamoreux et al., *Am Fam Physician* 2006; 74(11): 1883-1888).

161:

Outpatient treatment may include anti-infectives, anti-inflammatory medications (e.g., steroids), or warm moist dressings.

162:

Stevens-Johnson Syndrome (SJS) and Toxic Epidermal Necrolysis (TEN) are characterized by acute necrosis of the skin and mucous membranes. Compared to SJS, TEN is more severe and involves necrosis of greater than 30% of the body surface area. These conditions result most often from a medication reaction, and less commonly from infection. Management is aimed at identifying and eliminating the cause (e.g., discontinuing the precipitating medication or treating the infection). Supportive treatment involves meticulous skin and oral care, pain control, and adequate nutrition. Use of immunosuppressant therapy and intravenous immunoglobulin may be indicated in the treatment of SJS and TEN; however, further evidence is necessary to support the safety and efficacy of these treatment options.

163:

Although the vast majority of wound care is readily handled in the post-acute setting, there are situations when a patient requires care at the acute level, such as high-frequency wound care required, prolonged length of time required performing the procedure, and the type or amount of medication required to keep the patient comfortable during the procedure. For children less than 12 years of age (or less than 30 kg), the wound care regimen may need to be adjusted to a lesser frequency and/or duration (e.g., twice per day, 10 minutes), though still requiring the current level of care, due to the patient's size, pain tolerance, fluid shifts, and vital sign lability.

To determine the appropriate discharge destination, an assessment is required of the patient and/or caregiver's ability to learn the necessary wound care regimen, complexity of the treatment regimen, and a plan for managing contributory factors. For discharge to the home setting, the patient and/or caregiver must be able and willing to participate in the care plan, or there is sufficient agency assistance available to adequately meet the patient's needs. Patient and/or caregiver instruction should include all the following (Bryant and Nix, *Acute & chronic wounds: current management concepts*, 5th ed. 2015, xiv, 655 p.):

- Wound care regimen that includes a plan to monitor patient-specific contributory factors
- Adjunctive therapies (medications, nutritional plan, and supplements)
- Re-ordering medications and wound management supplies
- When to seek medical care (e.g., signs, symptoms, and complications to report to the health care provider)
- Continued follow-up and support with health care providers

164:

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., *Crit Care Med* 2021, 49: e1063-e143; Rhodes et al., *Crit Care Med* 2017, 45: 486-552; Winters et al., *J Emerg Med* 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, *J Crit Care* 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.

Instruction: In order to apply criteria for volume expanders, there should be documentation of a volume deficit supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

165:

Outpatient treatment of psoriasis includes prescription topical medications, phototherapy, and systemic therapy such as cyclosporine or methotrexate (Strowd et al., *J Am Acad Dermatol* 2009; 60(3): 478-481). Inpatient admission for treatment with biologic agents, tumor necrosis factor (TNF) inhibitors, or IV systemic therapy may be required

for patients who do not respond to oral therapy or phototherapy 3 times a week for 7 weeks. (Woods et al., Br J Dermatol 2008; 158(2): 266-272 ; Menter et al., J Am Acad Dermatol 2008; 58(5): 826-850).

166:

The Goeckerman treatment regimen is reserved for treating moderate to severe psoriasis with an application of prescription coal tar and ultraviolet B (UVB) light.

167:

This criterion is intended to be used for the patient who returns to the operating room (OR) in the event of a complication (e.g., bleeding, fistula repair, intra-abdominal infection, wound or ostomy revision) or staged procedure (e.g., open fracture definitive soft tissue coverage and internal fixation).

168:

Graft dysfunction is usually treated empirically with anti-rejection therapy while awaiting the results of the biopsy. Immunosuppressive medications are initiated based on the signs and symptoms of organ dysfunction or graft versus host disease (GVHD).

169:

Transplant graft dysfunction can be categorized as graft failure or graft rejection. In solid organ transplants, graft failure refers to the improper function or failure of the newly transplanted organ. The mechanism is due to an intrinsic problem with the grafted organ, not an immune reaction. Transplant rejection is a process in which a transplant recipient's immune system attacks the transplanted organ or tissue which results in graft dysfunction. In bone marrow transplant (BMT) and stem cell transplant (SCT) graft failure refers to the lack of engraftment, usually caused by something attacking the graft or initial graft strength being insufficient for implantation. Graft versus host disease (GVHD) is a complication of BMT/SCT. In this immune related complication, the newly transplanted donor cells attack the transplant recipient's body.

170:

Graft versus Host Disease (GVHD) is a transplant complication that occurs when immunocompetent cells from a non-identical donor are grafted into an immunosuppressed recipient. The immunocompetent donor/graft cells recognize alloantigens in the host cells and the host develops a Graft vs Host (GVH) reaction. Classic acute GVHD typically presents during the first 100 days after a hematopoietic stem cell transplantation (HSCT). Classic chronic GVHD presents beyond the first 100 days post HSCT and may include overlap syndrome, with acute GVHD features. Classic acute GVHD affects the skin, gut, and liver. Diagnosis is strongly supported if symptoms include exanthema (an eruptive skin rash oftentimes associated with a fever or systemic symptoms), diarrhea, and elevated bilirubin levels. Chronic GVHD, in addition to skin, liver, or gut involvement, may also affect the mouth, eyes, respiratory tract, musculoskeletal system, and peripheral nerves. It presents with symptomatology resembling autoimmune diseases (e.g., systemic lupus, erythematosus, or rheumatoid arthritis).

Chronic GVHD grade can be categorized as mild, moderate, or severe. In patients with GVHD, hyperbilirubinemia and small intestinal/colonic involvement are known risk factors for increased mortality (Hill et al., Annu Rev Immunol 2021, 39: 19-49; Penack et al., Lancet Haematol 2020, 7: e157-e67; Rodrigues et al., American journal of clinical dermatology 2018, 19: 33-50).

In determining GVHD grade, stages for skin, liver and gut are scored.

Acute GVHD Staging and Grading (Rodrigues et al., American journal of clinical dermatology 2018, 19: 33-50):

Grade I: Stage 1-2 skin organ involvement, only

Grade II: Stage 3 skin; Stage 1 liver and gut

Grade III: Stage 1-3 skin; Stage 2-3 liver and Stage 2-4 gut

Grade IV: Stage 4 skin and liver

Stage 1

- Skin: Rash < 25%(0.25) of body surface area
- Liver: Bilirubin 2-3 mg/dL (35-40 µmol/L)
- Gut: Diarrhea > 500-1000 ml/day

Stage 2

- Skin: Rash 25%(0.25) to 50%(0.50) of body surface area with associated symptoms
- Liver: Bilirubin 3-6 mg/dL (53-102 µmol/L)

- Gut: Diarrhea > 1000 ml/day

Stage 3

- Skin: Generalized rash covering > 50%(0.50) of body surface area
- Liver: Bilirubin 6-15 mg/dL (104-225 µmol/L)
- Gut: Diarrhea > 1500 ml/day

Stage 4

- Skin: Generalized erythroderma (reddening of skin often associated with exfoliation) or bullous eruption
- Liver: Bilirubin > 15 mg/dL (225 µmol/L)
- Gut: Severe abdominal pain with or without ileus

Chronic GVHD grading (scores are based on a 4-point scale of 0-3):

- Mild: 1 or 2 organs, with score no more than 1 with lung score 0
- Moderate: 3 or more organs with no more than score 1 or > 1 organ (not lung) with score 2 or lung score 1
- Severe: At least 1 organ with score of 3 or lung score of 2 or 3

171:

In an effort to address inequities in medicine and improve appropriate access to care, multiple medical societies have been reevaluating the use of adjustments made for a patient's race and updating their calculators and reference equations when reporting diagnostic results.

In 2023, the American Thoracic Society issued updated guidance on reporting of pulmonary function tests (PFTs), advocating for a race-neutral average reference equation to determine the baseline (i.e., predicted normal) for a given patient (Bhakta et al., Am J Respir Crit Care Med 2023, 207: 978-995). The reference equation previously included lower predicted normal values for patients of certain races and ethnicities (in the US, for Black and Asian patients); a practice which has not been found to have biological validity and may lead to misclassification of diagnoses and disease severity in racial or ethnic minorities (Vyas et al., N Engl J Med 2020, 383: 874-882). Since not all spirometer manufacturers and PFT laboratories have switched over to the new race-neutral average reference equation, clinical judgement and consideration of patient-specific factors should be used when interpreting PFT results and deciding whether patients with borderline reported PFT values in the presence of concerning symptoms or signs should have further evaluation or treatment.

172:

Instruction: This criteria point includes primary graft non-function related to all types of organ transplant.

173:

There have been significant improvements in immunosuppressant therapy for transplant recipients. Early protocols included azathioprine and corticosteroids, but with the addition of calcineurin inhibitors in solid organ transplantation, long-term survival and recovery has been improved. A multi-drug approach to therapy allows lower dosing of potentially very toxic individual agents. The 3 phases of immunosuppressive therapy include (Enderby and Keller, The American journal of managed care 2015, 21: s12-23):

- *Induction* - Immediately after transplant, when the risk for rejection is highest, intense antibody therapy or higher doses of maintenance therapy immunosuppressive agents are given depending on the organ being transplanted and the transplant center. Antibody therapies are T-cell depleting polyclonal (e.g., antithymocyte globulin (ATG)) and monoclonal (e.g., alemtuzumab or anti- CD52 antibody) or nondepleting agents (e.g., basiliximab).
- *Maintenance* - Transplant recipients remain on one or a combination of maintenance immunosuppressive therapy for the rest of their lives. Calcineurin inhibitors (e.g., cyclosporine, tacrolimus) are the mainstay of maintenance therapy with additional agents including antiproliferative/antimetabolites (e.g., mycophenolate mofetil (MMF), mycophenolic acid (MPA)), corticosteroids, mammalian target of rapamycin (mTOR) inhibitors (e.g., sirolimus), or T-cell costimulation blockers. The most common combination of these post transplant (kidney, liver, lung, heart, or pancreas) include tacrolimus, MMF/MPA, and prednisone. Immunosuppressant therapy may cause opportunistic infections, cancers and other adverse side effects. To limit these effects and increase survival and quality of life, research is underway that includes the infusion of marrow-derived

autologous mesenchymal stem cells (MSCs), xenotransplantation, and advancements in regenerative medicine.

- *Rejection* - Anti-rejection therapy is given to reverse an episode of rejection and is based on the severity and type of rejection, as well as the hemodynamic consequences of rejection. Treatment includes immunosuppressant medications given for treating an acute rejection during the initial post-transplant and may consist of a medication adjustment, steroid administration, or the addition of T-cell depleting or nondepleting agents.

174:

All medications commonly used to prevent rejection or opportunistic infections in the post-transplant period cause some degree of side effects. One of the important tasks of the transplant team is to titrate the dosage to minimize toxicity while maximizing benefit. Often this fine-tuning is empiric. If a patient has an intolerable side effect from a particular drug, an alternative medication or different combination or dosage of medications is attempted.

Tacrolimus and cyclosporine (CSA) are immunosuppressive agents that inhibit calcineurin and are given to solid organ transplant recipients to prevent and treat allograft rejection. Tacrolimus is less nephrotoxic and may be more cost-effective than CSA. However, new-onset diabetes should be closely monitored during the medication period. The dosing of CSA is more difficult to adjust appropriately in children, especially in younger/smaller children (less than 2

years). This is thought to be due to their shorter intestine, leading to less absorption of the drug, and also their faster metabolism. More time may be needed to adjust the dose and determine the appropriate medication management in this population than in older children or adults. The goal is to preserve the graft without seriously harming the patient.

175:

Medication adjustments tend to be driven by patient response. Patients being treated for symptoms of rejection usually have a biopsy and their medications adjusted based on the biopsy results to control the rejection.

176:

Pericardial diseases that are prone to developing cardiac tamponade include (Mekontso Dessap and Chew, Intensive Care Med 2018, 44: 936-9):

- Human immunodeficiency virus infection
- Tuberculosis
- Malignancies
- Renal failure
- Trauma/iatrogenic
- Hemopericardium in aortic dissection
- Rupture of the heart after acute myocardial infarction
- Bacterial infections
- Autoimmune disorders
- Radiation-induced
- Pneumopericardium
- Large (> 2 cm) idiopathic chronic pericardial effusions (which occur mostly in individuals whose sex at birth is female and who are at least 50 years of age)

177:

Patients that are at high risk for cardiac tamponade and not exhibiting symptoms can be monitored at the Intermediate level of care. These patients are hemodynamically stable; however, the pericardial effusion size is moderate (1cm to less than 2cm) to large (greater than 2 cm) requiring more intensive monitoring and potentially the need for serial echocardiograms (Mekontso Dessap and Chew, Intensive Care Med 2018, 44: 936-9; Adler et al., Eur Heart J 2015, 36: 2921-64).

178:

The medical therapy used to treat pericarditis/pericardial effusions depends on the etiology of the effusion. Medical therapy for these patients often involves treatment with NSAIDs, corticosteroids, and colchicine (Adler et al., Eur Heart J 2015, 36: 2921-64).

179:

Transthoracic echocardiogram is the initial modality used in the evaluation of suspected pericardial disease. It is inexpensive, rapidly performed, and available in virtually all hospital settings (Doherty et al., J Am Coll Cardiol 2019, 73: 488-516; Expert Panel on Cardiac et al., J Am Coll Radiol 2017, 14: S127-S37; Adler et al., Eur Heart J 2015, 36: 2921-64; Cosyns et al., Eur Heart J Cardiovasc Imaging 2015, 16: 12-31; Klein et al., J Am Soc Echocardiogr 2013, 26: 965-1012 e15).

180:

Patients at high risk for bleeding include those with thrombocytopenia, those taking medications which affect coagulation (i.e., aspirin, warfarin, antiplatelet agents) or those with a history of coagulopathies, active malignancies, or a creatinine greater than 3 mg/dL (265 µmol/L).

181:

Complications requiring an Intermediate level of care admission for cardiac monitoring may include transient occlusion of a coronary vessel, arrhythmia, chest pain, or dyspnea.

182:

Percutaneous coronary intervention (PCI) is the opening of a stenosed vessel by means of balloon angioplasty, stent insertion, atherectomy, or combination thereof (Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).

183:

Cardiac transplantation provides the greatest survival benefit and remains the gold standard treatment in patients with heart failure (HF), but limited availability of donor hearts make ventricular assist devices (VAD) a viable alternative. A VAD is a mechanical pump that assumes the function of the damaged ventricle and helps restore normal hemodynamics and end-organ blood flow. Pumps provide either pulsatile or continuous flow and may be implanted in one or both ventricles:

- Bi-ventricular assist device (BiVAD)-replaces the function of both ventricles
- Left ventricular assist device (LVAD)-replaces the function of the left ventricle
- Right ventricular assist device (RVAD)-replaces the function of the right ventricle

VAD insertion is currently approved as:

- Bridge to recovery: in patients with reversible ventricular dysfunction after cardiac surgery or myocardial infarction (MI)
- Bridge to transplantation: as temporary support for patients with end-stage HF awaiting transplant
- Destination therapy: as a permanent cardiac replacement device for patients who are not candidates for heart transplant

There are some devices that are not portable and require the patient to remain hospitalized while the device is in place. Some VADs can be inserted percutaneously and therefore, may be used more quickly than traditional, more invasive VADs (Miller and Rogers, JAMA Cardiol 2018;; Mancini and Colombo, Journal of the American College of Cardiology 2015, 65: 2542-55; Maini et al., Catheter Cardiovasc Interv 2014, 83: E183-92).

184:

In some facilities a ventricular assist device (VAD) will be used as a bridge to heart or heart-lung transplant. This may stabilize a patient long enough to survive until transplant, but the patient is still fragile and relatively unstable.

Instruction: In this clinical setting, the patient may require Critical care until a donor organ is available. Due to the restrictions associated with these devices, the Critical level of care is required for percutaneous VADs.

185:

There are 2 types of thrombi which can develop in a ventricular assist device (VAD) (Bartoli et al., J Card Surg 2014, 29: 83-94):

- A red thrombus occurs when stagnating blood coagulates due to low pressure and is formed when fibrin deposits trap erythrocytes. This type of thrombus is best managed with anticoagulation medications.
- A white thrombus occurs in areas of turbulence and is composed of activated platelets and is treated with antiplatelet agents.

186:

Absolute contraindications to thrombolytic therapy include prior or suspected intracranial bleed, prior stroke, intracranial aneurysm, arteriovenous malformation, neoplasm, closed head or facial trauma, active internal bleeding, suspected aortic dissection, known bleeding diathesis, or uncontrolled hypertension (greater than 180 systolic, greater than 100 diastolic) (Bartoli et al., J Card Surg 2014, 29: 83-94).

187:

Reference: (Rutledge et al., J Heart Lung Transplant 2013, 32: 569-73)

188:

The absolute neutrophil count (ANC) is the percentage of available polymorphonuclear cells (also referred to as polys, PMNs, or segs) and bands multiplied by the white blood cells (WBC). The resulting count predicts the body's ability to respond to an infection. Neutropenia is the condition of a low ANC. Patients with an ANC less than 500/cu. mm($500 \times 10^6/L$) are severely neutropenic and have a very limited immunologic response to infection. To determine the ANC, a differential WBC count must be performed. Segs and bands may be reported as a percentage of the WBC or can be reported in total number.

The formula for calculating an ANC is neutrophils (segs + bands) x WBC.

Examples:

ANC calculation using a percentage of total WBC count:

WBC count: 1,500/cu.mm($1.5 \times 10^9/L$)

Segs: 20%(0.20) of the WBCs

Bands: 2%(0.02) of the WBCs

$(20\% + 2\%(0.20 + 0.02)) \times 1,500/\text{cu.mm} (1.5 \times 10^9/L)$

$= 330/\text{cu.mm} (330 \times 10^6/L)$

Normal range: 1,500-8,000/cu.mm($1.5-8.0 \times 10^9/L$)

An ANC of less than 500/cu.mm($500 \times 10^6/L$) is considered severely neutropenic.

ANC Calculation using the above numbers as whole numbers:

WBC count: 1.5

Segs: 0.2

Bands: 0.02

$(0.2 + 0.02) \times 1.5 = 0.33$

Normal range: 1.5 to 8.0

An ANC of less than 0.5 is considered severely neutropenic.

189:

Hypokalemia rarely occurs in isolation and is usually secondary to a disease process or overuse of diuretics.

Criteria should be reviewed for the underlying cause of the hypokalemia (e.g., diabetic ketoacidosis, gastroenteritis) to ensure the patient is managed appropriately.

190:

When a patient with hypokalemia is not responding to potassium repletion, the magnesium level should be checked since hypomagnesemia could be contributing to the refractory hypokalemia.

191:

Seizures are a serious complication of severe low blood glucose levels and/or electrolyte imbalances (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023; Mehler et al., J Eat Disord 2023, 11: 16).

192:

Instruction: Patients with end-stage liver disease are generally on chronic outpatient lactulose, rifaximin, or neomycin at home to control symptoms of hepatic encephalopathy (HE). Hospitalization may be appropriate to initiate or adjust the dosage of these medications due to worsening HE.

193:

Hepatorenal syndrome (HRS) has been defined as acute renal failure in the presence of cirrhosis that manifests in profound circulatory dysfunction. A recent update in the diagnosis of HRS focuses on changes in serum creatinine levels instead of a fixed high serum creatinine value. Standard treatment for HRS includes drugs that increase blood pressure along with the albumin, a protein that is a volume expander. Per HRS guidelines, terlipressin with albumin is the recommended treatment, although in the U.S., because the use of terlipressin has not yet been approved, octreotide and midodrine are used in combination with albumin. A Cochrane Database Systematic Review analyzing the benefit or harmful effects of terlipressin given in combination with albumin found insufficient evidence in several small randomized controlled trials to support or refute terlipressin's superiority. Terlipressin was also associated with an increased risk of abdominal pain or diarrhea (Israelsen et al., Cochrane Database Syst Rev 2017, 9: Cd011532). Norepinephrine (also named noradrenaline) with albumin has been studied in relation to terlipressin and albumin and found to have a comparable efficacy. The use of norepinephrine in adult patients was validated by InterQual® consultants. Both regimens showed about a 40-50% response rate and up to a 50% rate of relapsing post-treatment. There is a high morbidity and mortality associated with HRS. Liver transplant screening is recommended for patients with this syndrome (Piano et al., Hepatol Int 2018, 12: 122-34; Acevedo and Cramp, World journal of hepatology 2017, 9: 293).

194:

Guillain-Barre syndrome (GBS) is an inflammatory peripheral nervous system disease. GBS typically presents as progressive, ascending limb weakness and diminished or absent reflexes, although variations of clinical presentations exist. The autonomic nervous system may be affected causing changes in blood pressure and cardiac arrhythmias. Over 20%(0.20) of patients with GBS will require mechanical ventilation (Willison et al., Lancet 2016, 388: 717-27). Patients with GBS require supportive care and close monitoring for deterioration.

195:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

196:

Significant cardiac arrhythmias may include any of the following: Atrioventricular nodal reentry tachycardia (AVNRT), sinus pauses of 3 seconds or more, bradycardia (heart rate less than 60), multifocal atrial tachycardia, 2nd degree atrioventricular (AV) block (Mobitz II), 3rd degree (complete) AV block, AV junctional tachycardia, ventricular tachycardia, ventricular fibrillation, torsade de pointes, and uncontrolled ventricular response with either atrial fibrillation or flutter or Wolff-Parkinson-White (WPW) syndrome (Jang et al., Emerg Med Clin North Am 2014, 32: 79-102).

197:

Cardiotoxic effects account for a large percentage of the overdose emergencies reported to the American Association of Poison Control Centers National Data Systems (NPDS). These can result from the following overdoses (Jang et al., Emerg Med Clin North Am 2014, 32: 79-102):

- Tricyclic antidepressants
- Sympathomimetics
- Methylxanthines

- Aconite or agents that can open cardiac sodium channels
- Alpha₂-adrenergic agonists
- β- adrenergic antagonists
- Calcium channel blockers
- Digoxin
- Cholinergic agents
- Opioids
- Plants such as lily of the valley and foxglove

198:

Instruction: In some institutions, the first 3 doses of muromonab-CD3 are required to be given in Intermediate care when the patient is experiencing acute rejection. The initial doses of muromonab-CD3 can cause violent headaches, fevers, and dyspnea. Dyspnea is more common in fluid overloaded patients and can resemble acute respiratory distress syndrome (ARDS) in severe forms.

199:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO₂) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO₂. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

200:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

201:

Instruction: If a patient is clinically stable, has no other medical reason for an acute level of care, and has tolerated 24 to 48 hours of PO immunosuppressants (after conversion from IV), consider a less intense level of care setting.

202:

An exchange transfusion is the removal of blood or blood components and replacement with donated blood products or IV fluids, usually normal saline. There are 2 main types of exchange transfusions: partial exchange transfusion for polycythemia or hyperviscosity, and red cell exchange for severe sickle cell crisis. Possible side effects may include hypotension, hypocalcemia, infection, emboli, thromboses, and death.

203:

Blood pressure is considered labile when it is extremely variable and difficult to control.

204:

Antifibrinolytic medications reduce blood loss by inhibiting clot degradation. Specific agents include tranexamic acid and aminocaproic acid, and may be indicated in bleeding due to traumatic, surgical, obstetric, gynecological, or gastrointestinal bleeding. Efficacy for these indications is in various stages of investigation, with the strongest

evidence supporting use in bleeding due to trauma and surgery (Steinmetzer et al., J Med Chem 2020, 63: 1445-72; Draxler and Medcalf, Transfus Med Rev 2015, 29: 102-9; Hunt, Anaesthesia 2015, 70 Suppl 1: 50-3, e18).

205:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children:

Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

206:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

207:

Active cooling includes the application of ice to the body surface by ice packs or cooling blankets or the cooling of the central body core by cooled IV fluids or cooled peritoneal lavage.

208:

Thrombolytic agents are medications that act to dissolve culprit clots. Indications for the use of thrombolytic agents include acute myocardial infarction, thrombotic stroke, massive pulmonary embolism, deep vein thrombosis, peripheral arterial thromboembolic occlusion, and vascular graft thrombosis. The most common complication of thrombolytic therapy is hemorrhage. Risk increases with the concomitant use of heparin or other antithrombotic agents. There is also a risk of embolism occurring from a dissolving clot.

209:

Cardiac tamponade is an emergent situation and transthoracic echocardiogram is recommended to evaluate the effusion and subsequent hemodynamic compromise (Doherty et al., J Am Coll Cardiol 2019, 73: 488-516; Adler et al., Eur Heart J 2015, 36: 2921-64; Cosyns et al., Eur Heart J Cardiovasc Imaging 2015, 16: 12-31).

210:

Management of pericardial effusions should be aimed towards treating the underlying disease (e.g., malignant, infectious, connective tissue disorder, tuberculosis, iatrogenic cause) whenever possible. Pericardiocentesis, or cardiac surgery, is indicated for cardiac tamponade or large symptomatic effusions unresponsive to medical therapy (Adler et al., Eur Heart J 2015, 36: 2921-64).

211:

Patients that are symptomatic (dyspneic, orthopneic, or have jugular venous distention) that present with a pericardial effusion may be in cardiac tamponade. Patients with these symptoms require Critical level of care for potential immediate intervention (Mekontso Dessap and Chew, Intensive Care Med 2018, 44: 936-9; Adler et al., Eur

Heart J 2015, 36: 2921-64).

212:

Pericardial drainage and pericardiocentesis are the primary interventions for cardiac tamponade related to pericardial effusion or management of a large chronic pericardial effusion (Adler et al., Eur Heart J 2015, 36: 2921-64).

213:

Prostacyclins stimulate vasodilation while suppressing vascular proliferation and platelet accumulation. Prostacyclin production has been found to be present at low levels in patients with pulmonary arterial hypertension (PAH). The intravenous administration of prostacyclins, epoprostenol, and treprostinil, is the most common treatment for severe PAH. The side effects of these medications include flushing, headache, nausea, diarrhea, rash, jaw, foot pain, and thrombocytopenia. Hemodynamic monitoring may be required due to the potential for severe adverse effects during medication initiation or dosage adjustments. These effects may include systemic vasodilation, bradycardia, hypotension, and marked thrombocytopenia (Vazquez and Klinger, Lung 2020, 198: 581-96; Klinger et al., Chest 2019, 155: 565-86; McLaughlin et al., J Am Coll Cardiol 2015, 65: 1976-97).

214:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous pressure, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

215:

Patients with a ventricular assist device (VAD) are at risk for developing hypovolemia, bleeding, infection, thromboembolism, and arrhythmias. Stability in patients with a VAD is often patient specific. Symptoms of instability in a patient with a VAD include fluctuating vital signs, decreased VAD flows, or decreased level of consciousness.

216:

Evidence of adequate perfusion includes intact mentation, adequate urine output, warm extremities, or the presence of peripheral pulses.

217:

In continuous flow axial pumps, measurement of systolic blood pressure is not possible because there is no audible closure of the aortic valve. The sound heard during auscultation represents the mean arterial pressure (MAP). Use of a Doppler flow probe is helpful in defining blood pressure in patients with a ventricular assist device (VAD). In patients with continuous flow pumps, the goal MAP is between 70 and 90 mmHg. The use of multiple vasoactive or antihypertensive agents may be required to achieve this goal (Mancini and Colombo, Journal of the American College of Cardiology 2015, 65: 2542-55; Wilson et al., J Am Coll Cardiol 2009, 54: 1647-59).

218:

A percutaneous left ventricular assist device (VAD) is used to protect the heart during invasive procedures or temporarily for short-term mechanical circulatory support while treatment decisions are made.

219:

Reference: (Jordan and Caesar, BMJ Qual Improv Rep 2015, 4:)

220:

The "R on T phenomenon" is the superimposition of an ectopic beat on the T wave of a preceding beat. Early observations suggest that R on T is likely to initiate sustained ventricular tachyarrhythmias.

221:

When T and U waves fuse, they form a TU fusion wave on an electrocardiogram. Severe hypokalemia may be

associated with a large U wave created by T and U wave fusion (Zieg et al., Acta Paediatr 2016, 105: 762-72; Levis, Perm J 2012, 16: 53).

222:

The U wave occasionally follows the T wave on an electrocardiogram. A prominent upright U wave is associated with hypokalemia. When the U wave exceeds the T wave amplitude, the serum potassium level is less than 3 mEq/L. Severe hypokalemia may be associated with a giant U wave created by T and U wave fusion (Sandau et al., Circulation 2017, 136: e273-e344; Zieg et al., Acta Paediatr 2016, 105: 762-72).

223:

Symptoms of hyponatremia may include nausea, lethargy, headache, and muscle weakness. If hyponatremia worsens, symptoms may progress to mental status changes (Glasgow Coma Scale (GCS) 9-14). Patients with acute coma, stupor, obtundation, and seizures should be reviewed at a higher level of care due to the need for more frequent neurological monitoring (Weismann et al., Endocr Connect 2016:).

224:

Symptoms of hyponatremia include:

- Coma
- Headache
- Lethargy
- Nausea
- Obtundation
- Seizures
- Stupor
- Vomiting

225:

Patients may present with hypovolemic, euvoletic, or hypervolemic hyponatremia. Those with hypovolemic hyponatremia have decreased total body sodium, total body water and extracellular fluid. Treatment includes isotonic saline and monitoring of serum sodium levels. Patients with euvoletic hyponatremia present with a decreased sodium level caused by excess body water. They do not exhibit the signs of hypovolemic or hypervolemic hyponatremia. Treatment may include isotonic saline and the discontinuation of diuretics. Those with hypervolemic hyponatremia have increased total body sodium, total body water, and markedly increased extracellular fluid with the presence of edema. Treatment includes sodium and water restriction, and administration of vasopressin receptor antagonists.

226:

Glasgow Coma Scale (GCS) is a tool commonly used to measure neurological function. GCS evaluates the level of consciousness in acutely ill patients. A patient's GCS is the combined total score accumulated through the evaluation of the following 3 areas (ranging from 3-15) (Chou et al., In: Glasgow Coma Scale for Field Triage of Trauma: A Systematic Review. 2017):

- Eye opening - 4 = Spontaneous, 3 = To voice, 2 = To pain, 1 = None
- Verbal response - 5 = Oriented, 4 = Confused, 3 = Inappropriate, 2 = Incomprehensible sounds, 1 = None
- Motor response - 6 = Obeys verbal commands, 5 = Localizes to pain, 4 = Withdraws to pain, 3 = Flexion response to pain, 2 = Extension response to pain, 1 = None

Of note, age may impact the relationship between anatomic injury severity and GCS scores such that the same GCS score may correspond to a greater severity of traumatic brain injury (TBI) in adults aged 65 and older than in those younger than 65 (Salottolo et al., JAMA Surg 2014, 149: 727-34). For patients less than 5 years of age, the verbal response component includes a modified scoring as follows:

- 5 = Appropriate word or social smile / Coos and babbles, 4 = Cries but consolable, 3 = Cries, screams, persistently irritable, 2 = Restless, agitated, moans, 1 = None

227:

Status epilepticus is defined as 5 or more minutes of continuous clinical and/or electrographic seizure activity, or recurrent seizure activity without recovery between seizures. Status epilepticus is a neurological emergency with high mortality and a potential for long-term consequences (Strein et al., J Clin Med 2019, 8:). There are many etiologies that may cause status epilepticus (e.g., Alzheimer's disease, multiple sclerosis, post-stroke). A patient with first time seizure that presents as status epilepticus does not imply epilepsy on its own. Refractory seizures, also known as uncontrolled, intractable, or drug resistant seizures, are resistant to treatment and are poorly controlled with medication. Recurrent seizures are seizures that occur more than once and may occur without warning.

228:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

229:

IV fluids used to treat electrolyte imbalances at the Critical level of care include but are not limited to:

- 3% saline
- 5% dextrose
- IV fluid boluses of 0.9% saline

230:

Hypovolemic hyponatremia may be treated with sodium chloride 0.9% (0.009). For severe hyponatremia, treatment with an infusion of sodium chloride 3% (0.03) may be required. Rapid correction of serum sodium can cause spastic quadriparesis or quadriplegia, pseudobulbar palsy, dysarthria, mutism, pupillary or oculomotor abnormalities, stupor, coma, ataxia, dystonia, osmotic demyelinating syndrome, or even death (Weismann et al., Endocr Connect 2016:).

231:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., J Thromb Haemost 2013:).

232:

Hyperosmolar therapy creates an osmotic gradient, facilitating the removal of water across the blood brain barrier, resulting in a decreased brain volume and a decrease in intracranial pressure. While hypertonic saline and mannitol are the 2 most used hyperosmolar agents, there is insufficient evidence to recommend one over the other for improving neurological outcomes in patients with acute ischemic stroke (Cook et al., Neurocrit Care 2020, 32:

647-66). Doses and strengths are generally determined by institutional practice patterns. Common side effects associated with osmotic therapy include dehydration, hypotension, renal failure, and acute respiratory distress.

233:

Venous blood gases (VBG) may be used to assess a patient's ventilation or metabolic status. Venous pH or HCO_3 values closely approximate corresponding values derived from arterial blood gases (ABG) in conditions such as chronic obstructive pulmonary disease (COPD), respiratory distress syndrome, neonatal sepsis, renal failure, pneumonia, diabetic ketoacidosis, and status epilepticus. VBG should not be used to determine oxygenation. Patients with chronic CO_2 retention hospitalized for acute respiratory illness will exhibit increased PCO_2 levels. PCO_2 values should be compared with baseline values when available and other signs and symptoms of respiratory distress should be considered when determining the appropriate level of care (e.g., worsening dyspnea, high respiratory rate, decreased oxygen saturation, confusion, drowsiness) (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; McKeever et al., Thorax 2016, 71: 210-5).

234:

Instruction: Room air assessment may not be clinically appropriate in patients who are dyspneic, tachypneic, cyanotic, or experiencing moderate to severe respiratory distress. If a patient remains hypoxic while on oxygen, room air assessment is not required.

235:

Proper inhaler technique is vital to optimizing the delivery of medication to the lungs and minimizing systemic side effects. Incorrect inhaler usage negatively impacts disease control and quality of life (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

236:

Neuromuscular disorder include:

- Amyotrophic lateral sclerosis
- Guillain-Barre syndrome
- Myasthenia gravis

237:

Myasthenic crisis is characterized by worsening muscle weakness leading to respiratory failure in a patient with myasthenia gravis. However, patients may present for the first time in myasthenic crisis. Bulbar or respiratory weakness suggesting impending respiratory failure warrant admission to the Critical level of care for monitoring (Roper et al., J Emerg Med 2017, 53: 843-53). Guillain-Barre syndrome (GBS) typically presents as distal paresthesia progressing to ascending weakness. Bulbar or respiratory muscle weakness warrants admission to the Critical level of care for close monitoring as up to 30%(.030) of patients with GBS require respiratory support (Shahrizaila et al., Lancet 2021, 397: 1214-28; Ganti and Rastogi, Emerg Med Clin North Am 2016, 34: 795-809). Pulse oximetry and blood gas monitoring may not be useful as hypercapnia and hypoxemia occur later during a myasthenic crisis or with GBS (Roper, The Journal of Emergency Medicine 2017, 53: 843-53). Respiratory failure is common in patients with Amyotrophic lateral sclerosis (ALS). Patients with ALS who are in respiratory failure require management at the Critical level of care (Ganti and Rastogi, Emerg Med Clin North Am 2016, 34: 795-809).

238:

Respiratory interventions for myasthenic crisis, impending myasthenic crisis, or Guillain-Barre syndrome may include:

- Pulmonary function tests (e.g., vital capacity, negative inspiratory force)
- NIPPV/ventilator setting changes

239:

Corrosive substances are any chemical with the potential to cause tissue injury on contact. Alkalis and acids are the two primary types of agents most often responsible for caustic exposures. Common examples of household alkaline

substances include sodium hydroxide, bleach, lye, hair relaxers, dishwasher detergents, and disc batteries. Common household acidic substances include toilet bowl cleaners, swimming pool cleaners, rust removers, metal cleansers, and car battery fluid. Common effects of ingestion of corrosive substances vary based on the strength of the chemical, the amount ingested, and time before treatment. These may include pain, chemical burns, erythema at the site of exposure (e.g., lips, mouth, throat, esophagus, stomach, intestinal lining). Clinical symptoms vary and may include central nervous system depression, cardiac arrhythmias, respiratory depression, nausea, vomiting, diarrhea, fluid and electrolyte imbalance. Patients with a suspected caustic ingestion and symptoms or signs (e.g., oral lesions, vomiting, drooling, dysphagia, hematemesis, dyspnea, abdominal pain) should undergo esophagogastroduodenoscopy (EGD) to identify all consequent digestive tract lesions. Medical management should focus on supportive measures and will vary based on the specific chemical exposure. The following agents may be of value in supportive care:

- Antibiotics - perforations
- Calcium - dysrhythmias related to hypocalcemia
- Proton Pump Inhibitors - esophageal injury, ulceration, ulcerative gastritis
- Narcotics - trauma related pain

The American Association of Poison Control Centers can provide valuable treatment information for hospitalized patients (Tringali et al., Endoscopy 2017, 49: 83-91; Struck et al., Scand J Trauma Resusc Emerg Med 2016, 24: 45; Shah, Crit Care Nurse 2016, 36: 70-5).

240:

Instruction: This criterion may be applied for patients who have diminished cough or gag reflexes or are at high risk for aspiration and require intubation.

241:

Antidotes enhance elimination and/or counteract the effects of a toxin. Emergency toxic ingestions or exposures and their antidotes include the following (Rasimas and Sinclair, Crit Care Clin 2017, 33: 521-41):

Toxins (Antidotes)

- Anticholinergic (physostigmine)
- Benzodiazepines and non-benzodiazepine hypnotics (flumazenil)
- Adrenergic blockers (glucagon)
- Calcium channel blockers (calcium, insulin, or glucose)
- Cyanide or hydrogen sulfide toxicity (preferred-hydroxocobalamin, but also sodium nitrate, or sodium thiosulfate)
- Digitalis glycosides (digoxin immune fab)
- Ethylene glycol, methanol (preferred-fomeprizole, but also ethanol)
- Iron (deferoxamine)
- Isoniazid, hydrazine and monomethylhydrazine (pyridoxine)
- Lead (dimercaptrol (BAL) dose preceding edetate calcium disodium, succimer (DMSA))
- Methemoglobin-forming oxidants (methylene blue)
- Methotrexate (folinic acid)
- Neuroleptics (Bromocriptine or dantrolene)
- Opioids, clonidine, guanfacine, tizanidine (naloxone)
- Organophosphates and carbamates (atropine, pralidoxime)
- Sulfonyleureas (octreotide)
- Tricyclic antidepressants (sodium bicarbonate)
- Valproic acid (I-Carnitine)

242:

A wide complex tachycardia (WCT) is an arrhythmia characterized by a rate greater than 100 beats per minute and a QRS complex greater than 0.12 seconds. The 3 main types of WCT are ventricular tachycardia (VT), torsades de pointes, and supraventricular tachycardia (SVT). When the type of arrhythmia is unable to be determined, the term wide complex arrhythmia is often used. Undetermined wide complex arrhythmias should be treated as VT until proven otherwise.

243:

An overdose of a drug causing sodium channel blockade can result in widened QRS complex leading to ventricular tachycardia or fibrillation. Potassium channel blocker (e.g., amiodarone, dofetilide, sotalol) overdose can result in lengthened QTc duration leading to torsades de pointes (Jang et al., Emerg Med Clin North Am 2014, 32: 79-102).

244:

Long QT syndrome is diagnosed with a QTc interval ≥ 500 ms. Long QT syndrome is also diagnosed with a QTc interval of ≥ 480 to 499 ms when associated with syncope (Shen et al., Journal of the American College of Cardiology 2017, 70: e39-e110).

245:

Prolongation of the QT interval may be caused by a multitude of congenital or acquired factors such as hypokalemia, hypocalcemia, or administration of antiarrhythmic medication. It is often asymptomatic, but places the patient at risk for ventricular arrhythmias, most commonly Torsades de pointes. Congenital long QT syndrome is a common cause of pediatric sudden death during sleep or exercise and may be responsible for a large amount of sudden infant death syndrome (SIDS) (O'Connor et al., Am J Emerg Med 2008, 26: 221-8).

246:

Adenosine receptor agonists, antiarrhythmics, cardiac glycosides, beta blockers, and calcium channel blockers have the potential to affect AV nodal conduction (PR interval) (Nada et al., Am Heart J 2013, 165: 489-500).

247:

Patients with acute antibody-mediated rejection (AMR), or humoral rejection, present with an acute loss of graft function in the weeks or years after transplantation. Clinical symptoms include proteinuria and an elevation in serum creatinine, with a diagnosis requiring documentation of circulating antidonor antibodies. CD4 T cells are required for allograft rejection. Treatment is aimed at reducing inflammation in the allograft, eliminating factors that cause inflammation, and preventing antibody formation while keeping the immune response as intact as possible. Therapies include antibody depletion with plasmapheresis (PLEX), immunoadsorption (IA), immunomodulation with intravenous immunoglobulin (IVIG), and T cell or B cell depleting agents. Compared to PLEX, IA, which is not yet available in the U.S., is more specific and effective in clearing immunoglobulins without the side effects associated with substituting fresh frozen plasma or albumin. With PLEX and IVIG considered standard therapies for AMR, better outcomes have been reported with a combination PLEX, IVIG and the T-cell depleting monoclonal antibody rituximab (Garces et al., Ochsner Journal 2017, 17: 46-55). In addition to rituximab, InterQual® consultants have also used monoclonal antibodies eculizumab and Bortezomib.

248:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

249:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

250:

The patient demonstrates appropriate behavior including no suicidal gestures or threats, or no violent or aggressive behavior.

251:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

252:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

253:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

254:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

255:

Functional impairments may be safely managed in the home environment when the patient is willing and able to perform self-care independently with or without assistive devices, has sufficient caregiver support available to adequately meet the patient's needs or home care services are arranged.

256:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

257:

Neurological deficit(s) may include:

- Altered level of consciousness
- Cognitive deficit
- Motor deficit
- Speech deficit
- Sensory deficit

Neurological disorder may include:

- Amyotrophic lateral sclerosis (ALS)
- Dementia (including Alzheimer's disease)
- Guillain-Barre syndrome (GBS)
- Multiple sclerosis (MS)
- Myasthenia gravis (MG)
- Parkinson's disease (PD)
- Transverse myelitis

258:

Instruction: These criteria refer to a persistent, new neurological deficit(s) that occurs due to the admitting diagnosis or is an unresolved exacerbation from a known neurological disorder.

259:

Instruction: These criteria refer to a functional impairment that develops during the course of this hospitalization.

260:

Inpatient treatment is indicated for dehydration secondary to an eating disorder (American Psychiatric Association, Practice guideline for the treatment of patients with eating disorders, third edition, 2006, p. 1097-222; Pritts and Susman, American Academy of Family Physicians 2003, 67: 297-304).

261:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. These rates should be increased in the presence of dehydration or insensible loss. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

262:

Initial course refers to the initiation of a chronic hemodialysis/peritoneal dialysis program or initiation for acute renal pathology. A repeat initiation for a recurrence or exacerbation of an acute renal problem, provided that there was a gap of more than 14 days between treatments, would also qualify as an initial course.

263:

Hyperviscosity syndrome may be a complication of a hematologic malignancy and is characterized by an extreme elevation of hematocrit, immature white blood cells (WBC), or paraprotein (IgM, IgG, IgA). Symptoms may include neurologic deficits, acute renal failure, and cardiac ischemia.

264:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

265:

Instruction: For this criteria point, repositioning refers to adjustment of the chest tube.

266:

Wound care teaching should include instruction on the following:

- Adjunctive therapies (medications, nutritional plan, and supplements)
- Follow-up plan of care
- Medications and wound management supplies
- When to seek medical care (e.g., signs, symptoms, and complications to report to the health care provider)
- Wound care regimen (including patient-specific factors)

267:

Initial refers to the first time a medication or treatment is utilized. During the period of initiation, if the medication or treatment is temporarily discontinued up to 24 hours (therapeutic pause) it is still considered initial. If a tolerated medication or treatment is discontinued for more than 24 hours and then restarted, it is not considered initial. Medications that were initiated on an outpatient basis and are discontinued prior to hospitalization may be considered initial based on the need for monitoring or duration of discontinuation. Medications that were temporarily held due to concurrent illness and are resumed are not considered initial.

268:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

269:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status;** those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

270:

Resolution of the pericardial effusion is indicated by return of ventricular function to an acceptable level (e.g., cardiac output adequate, blood pressure within normal limits for age, and pulsus paradoxus of 10 mmHg or less).

271:

Determining the cause of acute limb ischemia (ALI) is essential to guide antithrombotic therapy (e.g., antiplatelet agent, anticoagulation) following revascularization. Long-term anticoagulation is recommended after revascularization in patients who may be found to have atrial fibrillation (AF) or intracardiac thrombus as a source; however, anticoagulation therapy may not be required in patients without AF or intracardiac thrombus as the risk for recurrent embolus is lower. Single antiplatelet therapy is recommended in patients with peripheral artery disease to decrease the risk of limb-related and general cardiac complications (Bjorck et al., Eur J Vasc Endovasc Surg 2020, 59: 173-218). However, dual antiplatelet treatment is recommended for patients who require stent placement during revascularization for lower extremity ALI (Aboyans et al., Eur J Vasc Endovasc Surg 2018, 55: 305-

68).

272:

Warfarin dosage is adjusted according to the following international normalized ratio (INR) values (Kernan et al., Stroke 2014, 45: 2160-236; Nishimura et al., Circulation 2014, 129: e521-643; Whitlock et al., Chest 2012, 141: e576S-600S):

- Range of 2.0-3.0 with a target of 2.5 for patients with bioprosthetic valves, mechanical aortic valves, or for patients without implanted mechanical valves
- Range of 2.0-3.0 with a target of 2.5 for patients with ischemic stroke or transient ischemic attack (TIA) with a history of rheumatic mitral valve disease and atrial fibrillation
- Range of 2.5-3.5 with a target of 3.0 for patients with mechanical mitral valves or combination mitral/aortic valve implantation

273:

Reference: (Whitlock et al., Chest 2012, 141: e576S-600S; Monagle et al., Chest 2012, 141: e737S-801S)

274:

Bioprosthetic heart valves, also known as heterografts, are usually porcine or bovine in origin. They have a decreased risk of thromboembolism, but are less durable than mechanical valves (Chikwe et al., JAMA 2015, 313: 1435-42; Chiang et al., JAMA 2014, 312: 1323-9).

275:

Previous practice suggested that patients receive 5 days of unfractionated heparin prior to vitamin K antagonist therapy. This shortened duration is expected to decrease the risk of heparin-induced thrombocytopenia. If the international normalized ratio (INR) becomes therapeutic (greater than 2.0) prior to 5 days of treatment it is acceptable to discontinue parenteral therapy prior to the patient receiving 5 days of treatment (Kearon et al., Chest 2016, 149: 315-52).

276:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

277:

Instruction: This criterion applies to patients with actual, suspected, or a history of heparin-induced thrombocytopenia (HIT).

278:

Patients who develop bleeding while receiving anticoagulation should be evaluated for an underlying source of bleeding. Evaluation may include upper and lower gastrointestinal (GI) studies to rule out underlying GI pathology or an imaging study to rule out intracranial bleeding (Janz, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 9 edn. 2018:1463-79.e3).

279:

Orthostatic hypotension (Juraschek et al., JAMA Intern Med 2017, 177: 1316-23).

280:

Due to the short half-life of non-vitamin K oral anticoagulants (NOACs) (also known as novel oral anticoagulant, direct oral anticoagulant (DOAC), and target-specific oral anticoagulant) such as dabigatran, apixaban, rivaroxaban, and edoxaban, discontinuation of the medication is often all that is required when a patient develops bleeding. When a more rapid response is required, hemodialysis and oral activated charcoal can be used for NOACs without a

specific antidote. Andexanet alfa has received approval for urgent reversal of the anticoagulant effect of apixaban and rivaroxaban (Rogers and Finks, Am J Med 2018).

Idarucizumab has been approved for use as a reversal agent for dabigatran when the episode of bleeding is considered to be life threatening (Hu et al., Vasc Health Risk Manag 2016, 12: 35-44; Abo-Salem and Becker, Curr Opin Pharmacol 2016, 27: 86-91; Vanden Daelen et al., Expert Rev Cardiovasc Ther 2015, 13: 95-103).

281:

Rapid weight gain indicates unhealthy weight gain due to fluids, and possible fluid overload, not a healthy and desired gain in body weight (American Psychiatric Association. Practice Guideline for the Treatment of Patients With Eating Disorders. 2023).

282:

Patients who may require continued stay with ascites include those who have increasing or unchanged abdominal girth or weight, as well as patients who are improved but not clinically stable for discharge because they have had a subtherapeutic response to treatment.

283:

In a patient with mental status changes the basic outpatient workup performed should consist of a computed tomography (CT) scan and electrolyte testing.

284:

The anticoagulation regimen may include low molecular weight heparin (LMWH), unfractionated heparin (UFH), or fondaparinux with or without warfarin, or warfarin alone. For adult patients, non-vitamin K oral anticoagulants (NOACs) (also known as novel oral anticoagulant, direct oral anticoagulant (DOAC), and target-specific oral anticoagulant) such as dabigatran, rivaroxaban, apixaban, or edoxaban, may also be used (Ortel et al., Blood Adv 2020, 4: 4693-738).

285:

Treatment with low molecular weight heparin (LMWH) is recommended, rather than treatment with unfractionated heparin (UFH), in patients with acute deep vein thrombosis (DVT) or pulmonary embolism (PE) (Kearon et al., Chest 2016, 149: 315-52). Both LMWH and fondaparinux levels may be elevated in patients with renal insufficiency. Fondaparinux is contraindicated in patients with a creatinine clearance less than 30 mL/min. LMWH should be used with caution in patients weighing less than 50 kg due to unpredictable pharmacokinetics. In these patients, UFH is preferable (Minichiello and Fogarty, Med Clin North Am 2008, 92: 443-65, x).

286:

Non-vitamin K anticoagulants (NOACs) (also known as novel oral anticoagulant, direct oral anticoagulant (DOAC), and target-specific oral anticoagulant) such as dabigatran, apixaban, rivaroxaban, or edoxaban, may be used instead of warfarin for the long-term treatment of patients with deep vein thrombosis (DVT) or pulmonary embolism (PE) without cancer (Kearon et al., Chest 2016, 149: 315-52).

287:

Dabigatran and edoxaban, both non-vitamin K oral anticoagulants, require at least 5 days of parenteral anticoagulation prior to initiation (Kearon et al., Chest 2016, 149: 315-52; Fontana et al., Swiss Med Wkly 2016, 146: w14286).

288:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

289:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack

of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

290:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

291:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

292:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

293:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk

- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

294:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

295:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

296:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

297:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

298:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

299:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

300:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

301:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs

- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

302:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

303:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

304:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

305:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

306:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.